The Topology and Geometry of Market Regimes

Density Domains, Differential Metrics, Temporal Memory, and Options

Abstract

Today we will talk about the topology and geometry of market regimes through a unified mathematical construction. Weighted logarithmic returns, rolling variances, response estimates, and historical variance references define nonnegative masses and stress coordinates. Two coupled densities determine fragile dominance, antifragile dominance, and their equality set. An explicit boundary and a reversible normalization establish continuity, path-connectedness, and contractibility under regular positive-parameter assumptions. Fixed-state comparisons distinguish movement of an observation from deformation of its classification framework, while channel opening can change the topology of the dominance partition. The analysis then develops the four-dimensional density map, its fibers, differential rank, pullback metric, restricted slices, and singular strata. Temporal confirmation is formulated as a finite-memory system and distinguished from instantaneous dominance and lagged portfolio allocation. Further derivations cover auxiliary stochastic dynamics, invariances, option replication, parity geometry, coordinate shears, exact volatility sensitivities, and derivatives along a volatility smile. Numerical reconstruction connects the algebra to recorded regime states. The complete development preserves the intermediate calculations, mathematical figures, and conditions needed to distinguish geometric classification, temporal memory, pricing transformations, and executed allocation.

Mathematical setting

For nonnegative primitive coordinates (f, n, x, y), define

$$\Phi(f, n, x, y) = (u, v) = \left(\frac{f}{1+x}, \frac{nx}{1+y} \right), \qquad D = u - v.$$

The strict domains are $F = \{D > 0\}$ and $A = \{D < 0\}$; equality is $D = 0$. The topological construction fixes masses and variance references. The differential construction allows all four primitives to vary. Temporal confirmation adds memory, and executed allocation uses the stated lag. Each coordinate system is defined within the construction in which it is used.

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Topology and dominance domains

Assumptions and notation

For the regular two-domain analysis, take $X = [0, \infty)^2$ in variance coordinates and assume $b_{+,t} > 0$, $b_{-,t} > 0$, $f_t > 0$, and $n_t > 0$. Neighborhoods and boundaries are understood relative to X. The change of coordinates $(V_-, V_+) \mapsto (V_-, V_+/C_t(V_-))$ is the reversible transformation used in the topology argument; the clipped variance-to-stress map itself need not be invertible. The separately discussed case $n = 0$, $f > 0$ is outside the regular positive-mass case.

Symbol	Meaning
R_t^+, R_t^-	Weighted logarithmic returns of the two Beta legs.
V_t^+, V_t^-	Annualized rolling variances, using 63 observations in the stated configuration.
$b_{+,t}, b_{-,t}$	Historical variance references: medians of the previous 252 variance observations in the stated configuration.
s_+, s_-	Nonnegative variance stress relative to each reference.
f, n	Admissible fragile and antifragile response masses.
ρ_+, ρ_-, D	Two densities and their signed difference.
F_t, A_t, Γ_t	Strict-dominance domains and their equality boundary.
$C_t(V_-)$	Boundary height in variance coordinates.
G_t, τ_t	Temporally confirmed state and confirmation duration.

1 Introduction: geometry, topology, and market regimes

A market state and the framework that classifies it are distinct mathematical objects. The state can move while the framework is fixed, or the framework can change around a fixed state. The resulting distinction is central to the construction of market regimes.

Geometry describes the shape and position of dominance domains and their boundaries. Topology describes neighborhoods, continuity, paths, connectedness, and separation. A deformation can alter a boundary without changing the connectedness of a domain; opening or removing an entire dominance channel can change the partition itself.

Let F collect states of fragile dominance and A states of antifragile dominance. These sets require an explicit classification rule. The rule is built from two weighted return legs, their variances, historical references, and estimated responses to variance. The two legs are denoted R^+ and R^-; their signs identify roles rather than guarantees about realized returns.

The Beta construction specifies relationships among assets in a trading environment and is distinct from a single conventional market-index beta coefficient. Each constituent asset has a logarithmic return given by

$$r_{i,t} = \ln\left(\frac{P_{i,t}}{P_{i,t-1}}\right) \tag{1}$$

2 Returns and the construction of the two Beta legs

We take the current price of a ticker, divide it by its previous price, and then take the logarithm of the result."

The i simply identifies the ticker, and t identifies the date.

Now, inside my two legs, each ticker also has a weight. One asset might represent 10% of the basket, another 20%, and so on.

The return of a leg can therefore be written as:

$$R_t^+ = \sum_{i \in U^+} w_{i,t}^+ r_{i,t} \tag{2}$$

Look at this equation.

The symbol Σ, which might seem complicated at first, simply means "add everything together."

U^+ is the set of tickers in the first leg. w represents each ticker's weight, and r represents its return.

So the entire equation tells us:

The return of my first leg is equal to the sum of its tickers' returns, each multiplied by its weight

You can check that with a very simple example.

Imagine two tickers. The first represents 60% of the basket and has a logarithmic return of 0.02. The second represents `40%` and has a logarithmic return of -0.01.

$$R^+ = 0.60 \times 0.02 + 0.40 \times (-0.01) \tag{3}$$

$$R^+ = 0.008 \tag{4}$$

See?

We've simply added two weighted contributions.

For the other leg, I apply exactly the same principle:

$$R_t^- = \sum_{j \in U^-} w_{j,t}^- r_{j,t} \tag{5}$$

The $+$ and $-$ signs identify the two legs.

They don't mean their returns must always be positive or negative.

In Beta, the weights depend partly on the assets' historical volatility, with concentration constraints. The relationships between the two baskets are also studied through their cumulative levels and residuals. You can find the detailed construction in the Beta notebook.

Why is this relevant to topology?

Because we're no longer working with a single isolated price trajectory. We have two legs that evolve, and the relationships between them can evolve as well.

The two legs are treated separately.

We take their cumulative logarithmic returns, which I call L^+ and L^-, and transform them into two positive indices:

$$P_t^\pm = \exp\Big(L_t^\pm - L_{t_0}^\pm\Big) \tag{6}$$

We take the cumulative evolution of each leg from the starting date and transform it into a positive index that begins at 1.

Why do this?

Because the Janus operator works with two positive series and calculates their logarithmic returns.

And we can demonstrate that this transformation preserves the returns of the two legs:

$$\ln\left(\frac{P_t^{\pm}}{P_{t-1}^{\pm}}\right) = R_t^{\pm} \tag{7}$$

3 Variance, historical references, and stress

When Janus calculates the logarithmic return of these new indices, it recovers the original return of each Beta leg

That's the connection we're going to use between the two repositories.

Now we have our two legs.

What interests us here isn't developing the entire Beta model. It's understanding how Janus uses these two series to construct the regime domains whose topology we want to demonstrate.

In Janus, We use two quantities to represent the state of our legs: their variances.

Why variances?

Because they measure how spread out their returns are. If the variations stay relatively close to their average, their dispersion is low. If they become much more spread out, it increases.

We can describe this measurement with a first equation:

$$\sigma^2 = \frac{\sum_{i=1}^{N}(x_i - \bar{x})^2}{N-1} \tag{8}$$

In Janus, the stress variance is calculated using the last 63 observations and annualized using a factor of 252:

$$\begin{aligned} V_t^+ &= 252\ \mathrm{Var}_{63}(R^+) \\ V_t^- &= 252\ \mathrm{Var}_{63}(R^-) \end{aligned} \tag{9}$$

Here, this equation is used to measure the recent fluctuations in each of my two legs, then I scale the result to an annual period.

We place the variance of the negative leg on the horizontal axis and the variance of the positive leg on the vertical axis.

A state now becomes a point with two coordinates:

$$z = (V_-, V_+) \tag{10}$$

This equation simply means: "To locate my state, I use the variances of my two legs."

Why two coordinates? Because We study their relationship, not just the dispersion of one isolated leg.

But be careful: variance alone isn't enough.

Imagine a variance of 0.06. Is that a lot? Well, that depends on what the leg has experienced in the past.

If its historical reference is 0.03, the current variance is twice that reference. If the reference is already 0.06, there's no excess at all.

So Janus calculates a historical reference for each leg, which we denote b_+ and b_-.

In the configuration we're using, this reference is the median of the previous 252 variance observations.

Then it calculates what I call variance stress:

$$\begin{aligned} s_+ &= \max\left(\frac{V_+}{b_+} - 1, 0\right) \\ s_- &= \max\left(\frac{V_-}{b_-} - 1, 0\right) \end{aligned} \tag{11}$$

I measure how much the current variance exceeds its reference. If it doesn't exceed it, I keep zero.

The symbol max simply means "take the larger of the two values."

For example:

$$s_+ = \max\left(\frac{0.06}{0.03} - 1, 0\right) = 1 \tag{12}$$

4 Variance response and the two masses

This result means the variance is 100% above its reference.

Now, look at something important.

Even if I keep the exact same variance, its stress can change when its historical reference changes.

The same point on our map can therefore be interpreted differently when the parameters defining the framework change.

And that's only part of the mechanism.

In Janus, I also measure how the excess returns of the two legs respond to variance. The code estimates a relationship of this form:

$$E = \alpha - mU + \varepsilon \tag{13}$$

Excess return is represented by a base level, minus a sensitivity multiplied by variance, plus the part that the relationship doesn't explain.

Here, E represents excess return, U is the variance used in this estimation, m is the sensitivity, α is the base level, and ε is the residual.

I'm using the letter m so we don't confuse this internal Janus coefficient with the relational Beta object.

When this coefficient is positive, the estimated response decreases as variance increases. When it's negative, the estimated response increases with variance.

Janus retains the admissible parts of these responses to construct two masses:

$$\begin{aligned} f &= \max(m_+, 0) \\ n &= \max(-m_-, 0) \end{aligned} \tag{14}$$

5 Fragile density, antifragile pressure, and balance

Read the first one as: "I keep the positive part of the fragile sensitivity of the positive leg."

And the second: "I keep the positive magnitude of the negative response of the negative leg."

These are the masses used to calculate the model's two densities.

The full rolling estimation is developed in the detailed estimation sections.

What matters to us is understanding how these masses help determine the geometry of our regimes.

And now we arrive at an essential equation.

Janus constructs a density for the fragile component:

$$\rho_+ = \frac{f}{1 + s_+} \tag{15}$$

We take the fragile mass and divide it by one plus the stress of the positive leg.

Notice that stress is in the denominator.

The more it increases, the more this density decreases, assuming the mass remains fixed.

For antifragile pressure, Janus constructs:

$$\rho_- = \frac{n\, s_+}{1 + s_-} \tag{16}$$

We take the mass of the antifragile channel, multiply it by the positive stress that activates it, then divide it by one plus the stress of the other leg.

Now we have two densities that depend on the model's conditions.

And to find out which one dominates, all we have to do is compare them.

In words:

$$\text{Balance} = \text{Fragile density} - \text{Antifragile pressure} \tag{17}$$

Using the model's symbols

$$D = \rho_+ - \rho_- \tag{18}$$

Look how simple this is.

If We take a fragile density of 3 and an antifragile pressure of 2, I get:

$$D = 3 - 2 = +1 \tag{19}$$

The balance is positive.

The fragile component dominates.

Now, if antifragile pressure increases to 4:

$$D = 3 - 4 = -1 \tag{20}$$

6 The strict-dominance domains

The balance is negative.

Antifragile pressure dominates.

We've just found a mathematical rule that distinguishes our two domains.

we denote F_t the fragile-dominance domain and A_t the antifragile-dominance domain, at a given date t.

$$F_t = \{z : D_t(z) > 0\} \tag{21}$$

$$A_t = \{z : D_t(z) < 0\} \tag{22}$$

"I gather all states whose balance is positive" and "I gather all states whose balance is negative."

Here, the word regime refers to these dominance domains. Janus also has a temporally confirmed state, which we'll discuss after the geometry.

Now we can start drawing the regime domains, because we know exactly how to determine which side each point belongs to.

Look at this first diagram, constructed using the parameters from the Beta-to-Janus calculation.

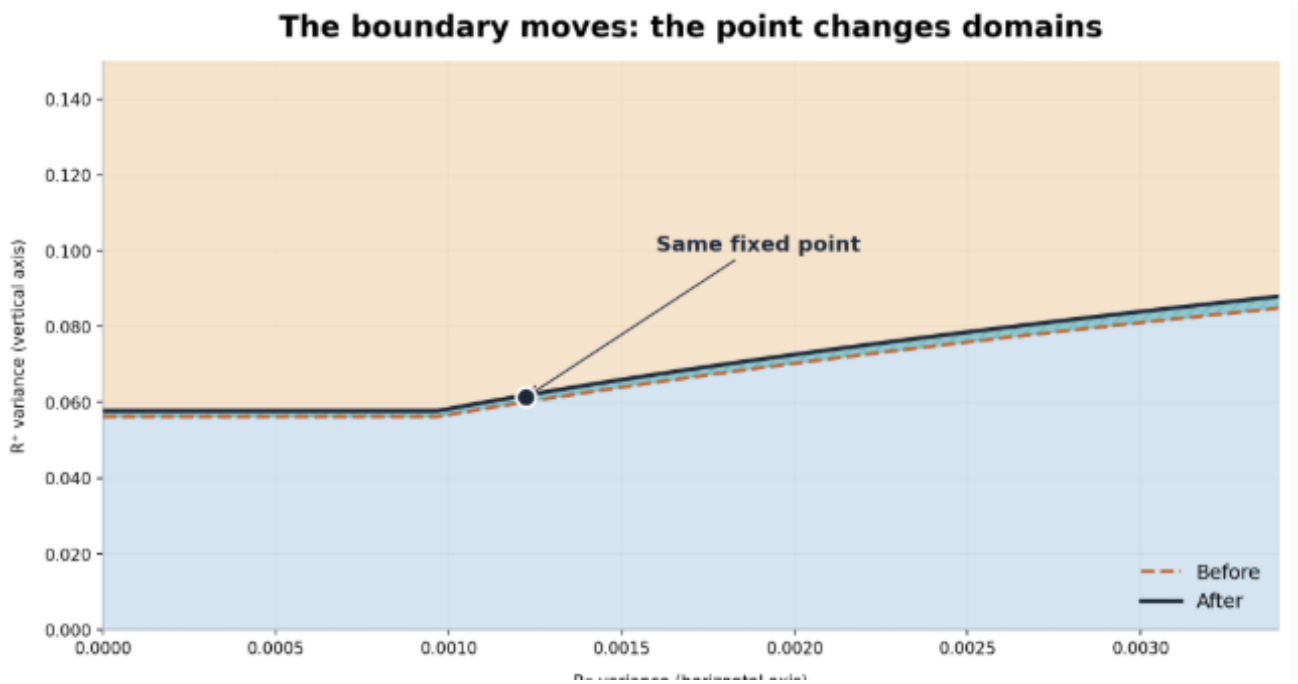


Figure 1: A moving boundary around a fixed state. The blue and orange regions represent fragile and antifragile dominance; dashed and solid curves represent the old and new boundaries.

7 Derivation of the equilibrium boundary

Blue represents the fragile domain.

Orange represents the antifragile domain.

The solid line represents the new boundary, and the dashed line represents the old one.

This diagram is a mathematical map of variance states, not an impenetrable price envelope. The axes represent two annualized variances.

Look at the black point.

It stays in exactly the same place, and yet the boundary moves around it. Under the old framework, that point belongs to the antifragile domain. Under the new one, it belongs to the fragile domain.

a point doesn't necessarily have to move to change domains.

The framework around it can evolve too.

But I'm not going to ask you to believe the drawing. We're going to calculate that boundary and prove why it has this particular shape.

First, I need to find the place where the two densities are in equilibrium.

That happens when:

$$D = 0 \tag{23}$$

Since D is the difference between the two densities, this means

$$\rho_+ = \rho_- \tag{24}$$

At the boundary, fragile density is exactly equal to antifragile pressure.

Now I replace these two densities with the equations we've just learned

Since both stress values are nonnegative, both denominators are positive. So instead of keeping the fractions, We multiply both sides by their denominators straight away.

This gives us:

$$f(1 + s_-) = n\, s_+(1 + s_+) \tag{25}$$

It's exactly the same story, just written in a simpler way using the variables of Janus.

The fragile mass multiplied by one plus negative stress is equal to the antifragile mass multiplied by positive stress and one plus that same positive stress.

And remember, we're still looking for the exact point where our two densities are equal.

That's the boundary between our two regimes.

To find that boundary, I need to isolate positive stress, s_+.

Since both masses are strictly positive, I can divide both sides by n:

$$\frac{f}{n}(1 + s_-) = s_+(1 + s_+) \tag{26}$$

The equation still describes exactly the same equilibrium.

We've simply divided both sides by the same positive quantity to make the next calculation easier.

Then I expand the right-hand side:

$$s_+^2 + s_+ = \frac{f}{n}(1 + s_-) \tag{27}$$

At the boundary, positive stress squared plus that same positive stress must equal the ratio of the two masses multiplied by one plus negative stress.

We've found the equation that determines the critical stress.

To isolate its value, I multiply the equality by 4 and add 1 to both sides:

$$4s_+^2 + 4s_+ + 1 = 1 + 4\frac{f}{n}(1 + s_-) \tag{28}$$

Why?

Because the expression on the left can be written as a square:

$$(2s_+ + 1)^2 = 1 + 4\frac{f}{n}(1 + s_-) \tag{29}$$

We take the square root, then isolate s_+.

Since stress cannot be negative, I keep the nonnegative solution:

$$s_+^\star = \frac{\sqrt{1 + 4\frac{f}{n}(1 + s_-)} - 1}{2} \tag{30}$$

The star simply means "critical value."

We've just demonstrated something important:

for every level of stress on the negative leg, there is a critical level of positive stress at which the two densities are exactly equal.

Below that critical level, the fragile component dominates.

Above it, antifragile pressure dominates.

And now we're going to turn this result into an actual geometric boundary on our variance map.

To do that, I return to an equation you already know:

$$s_+ = \frac{V_+}{b_+} - 1 \tag{31}$$

This relationship is valid at our regular boundary, where positive stress is strictly greater than zero.

We find the value of V_+, because that's the variance we placed on the vertical axis.

So I rearrange the equation:

$$V_+ = b_+(1 + s_+) \tag{32}$$

It tells us: "Positive variance is equal to its reference multiplied by one plus its stress."

Now I replace stress with the critical value we just found.

And I get:

$$C_t(V_-) = \frac{b_{+,t}}{2}\left[1 + \sqrt{1 + 4\frac{f_t}{n_t}\max\left(1, \frac{V_-}{b_{-,t}}\right)}\right] \tag{33}$$

8 Interpretation of the boundary equation

There it is. That's our boundary equation.

It might look a little long, but look closely at what it tells us.

On the left, $C_t(V_-)$ represents the height of our boundary, at a particular date and for a particular value of negative variance.

On the right, we find exactly the quantities we've already explained: the two masses and the two variance references.

In other words:

The height of the boundary is determined by the relationships Janus measures between the two legs, and that height changes when the model's parameters change.

You can even understand the deformations simply by reading the equation.

Look at this ratio:

$$\frac{f_t}{n_t} \tag{34}$$

It's simply fragile mass divided by the intensity of the antifragile channel.

When the variance references are held constant, increasing this ratio also increases the critical height.

That means the boundary moves upward on our chart.

Now look at:

$$\max\left(1, \frac{V_-}{b_-}\right) \tag{35}$$

As long as negative variance doesn't exceed its reference, this expression stays equal to 1. So the height of the boundary doesn't yet depend on V_-.

That's why our boundary starts with a horizontal section.

But once negative variance exceeds its reference, the ratio V_-/b_- becomes greater than 1.

The expression starts increasing with that variance, and the boundary curves upward.

So the geometry we're looking at isn't a drawing chosen at random.

IT'S THE SHAPE IMPOSED BY THE EQUATIONS OF JANUS.

Here is that shape using the two successive frameworks of March 16 and 17, 2016, calculated from the Beta legs connected to Janus.

9 Regime geometry and neighborhoods

Look at the two images.

Blue represents all the states in which the fragile component dominates. Orange represents the states in which antifragile pressure dominates.

The black line separates the two domains.

And now I can write these two regions mathematically.

The fragile domain:

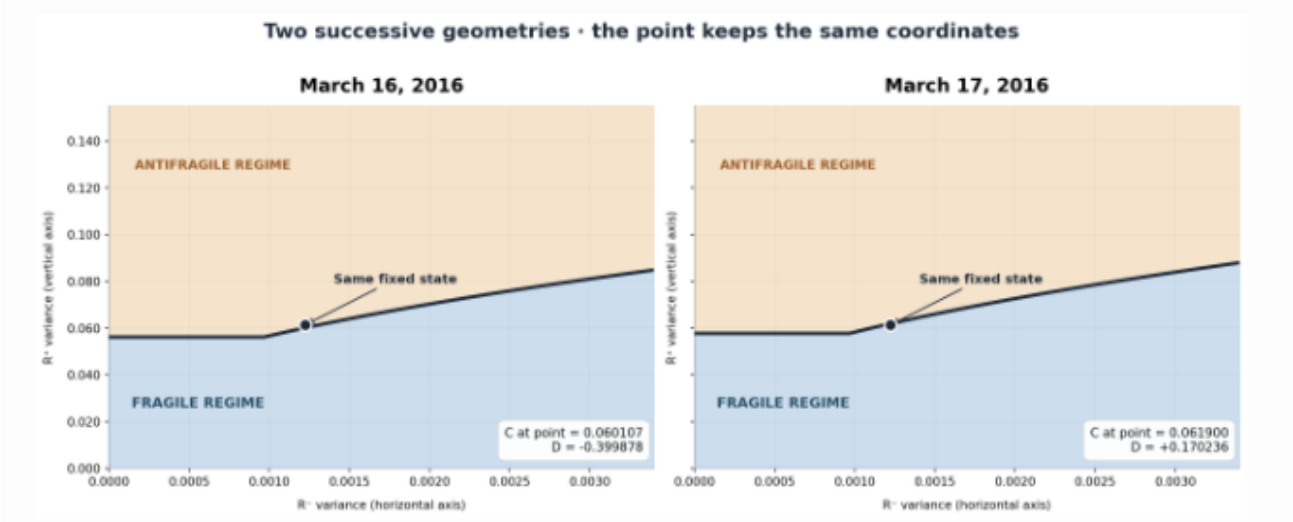


Figure 2: Two successive geometries for March 16 and March 17, 2016. The state keeps the same variance coordinates while the framework and classification change.

$$F_t = \{(V_-, V_+) : V_+ < C_t(V_-)\} \tag{36}$$

We take every pair of variances whose vertical coordinate is lower than the height of the boundary.

The antifragile domain:

$$A_t = \{(V_-, V_+) : V_+ > C_t(V_-)\} \tag{37}$$

We take every pair of variances located above the boundary.

And finally, the boundary itself:

$$\Gamma_t = \{(V_-, V_+) : V_+ = C_t(V_-)\} \tag{38}$$

Γ, simply gives our boundary a name.

We've now written down the two domains and the boundary separating them.

And that's fundamental for topology.

Why?

Because now I can answer mathematically the questions we asked at the beginning of the construction.

Does each domain form a single connected region? Can a state move slightly without leaving its domain? Can we connect two states with a path that stays inside? What happens when the boundary deforms?

Let's start with neighborhoods.

Imagine a point inside the fragile domain, sufficiently far from its boundary.

If I change its two variances very slightly, there is a small region around it in which the balance remains positive.

I can call that small region a neighborhood.

To write it mathematically, We take a point z and a small positive number ε.

$$B_\varepsilon(z) = \{z' : \|z' - z\| < \varepsilon\} \tag{39}$$

I gather all the points whose distance from z is smaller than ε.

The symbol $\|\cdot\|$ measures the usual distance between two points on the chart.

Why introduce distance here?

Simply because it lets us describe and draw a neighborhood.

Topology then studies the neighborhood relationships and the properties they define.

If our point lies strictly inside the fragile domain, we can find a neighborhood small enough to stay entirely within that domain.

Mathematically:

$$z \in F_t \implies \exists \varepsilon > 0 : B_\varepsilon(z) \cap X \subset F_t \tag{40}$$

This expression might look complicated, but all it says is:

"If my point lies inside the fragile domain, then there is a small region around it that remains entirely inside the same domain."

X represents the set of possible pairs of variances, while $\subset$ means "is contained in."

Why is this property true?

Because our balance D is a continuous function of the two variances when the parameters defining the framework are fixed and the variance references are strictly positive.

A point whose balance is strictly positive cannot become negative under an arbitrarily small perturbation.

To change its sign, the state must encounter the boundary, or the conditions defining the balance must change.

We've just demonstrated a first property of our domains: every strictly interior point has a neighborhood that preserves its dominance.

But now imagine moving closer to the boundary.

The margin becomes smaller and smaller.

At the boundary, the balance is exactly zero. You can find fragile and antifragile states as close as you like to that separation.

And that's what allows us to demonstrate that, when both masses are positive:

$$\partial F_t = \partial A_t = \Gamma_t \tag{41}$$

The symbol ∂ represents the boundary of a set, in our state space X.

So this equality tells us:

The fragile domain and the antifragile domain share exactly the same boundary.

Now let's move on to connectedness.

Can we demonstrate the same property for each of our two Janus domains?

Yes, in regular frameworks where both masses are strictly positive.

And We show you a very simple method.

We have a curved boundary whose height changes depending on the horizontal position.

Instead of measuring the point's height directly, We measure its height relative to the height of the boundary.

I write:

$$u = \frac{V_+}{C_t(V_-)} \tag{42}$$

We take the height of the point and divide it by the height of the boundary at exactly the same horizontal position.

Imagine the boundary has a height of 10.

If your point is at 5:

$$u = 5/10 = 0.5 \tag{43}$$

The point is halfway to the critical height.

If it's at 10:

$$u = 10/10 = 1 \tag{44}$$

It's exactly on the boundary.

And if it's at 15:

$$u = 15/10 = 1.5 \tag{45}$$

It's above the boundary.

Do you see what we've just done?

We've replaced a curved boundary with a reference that is always equal to 1.

Here's what that transformation produces.

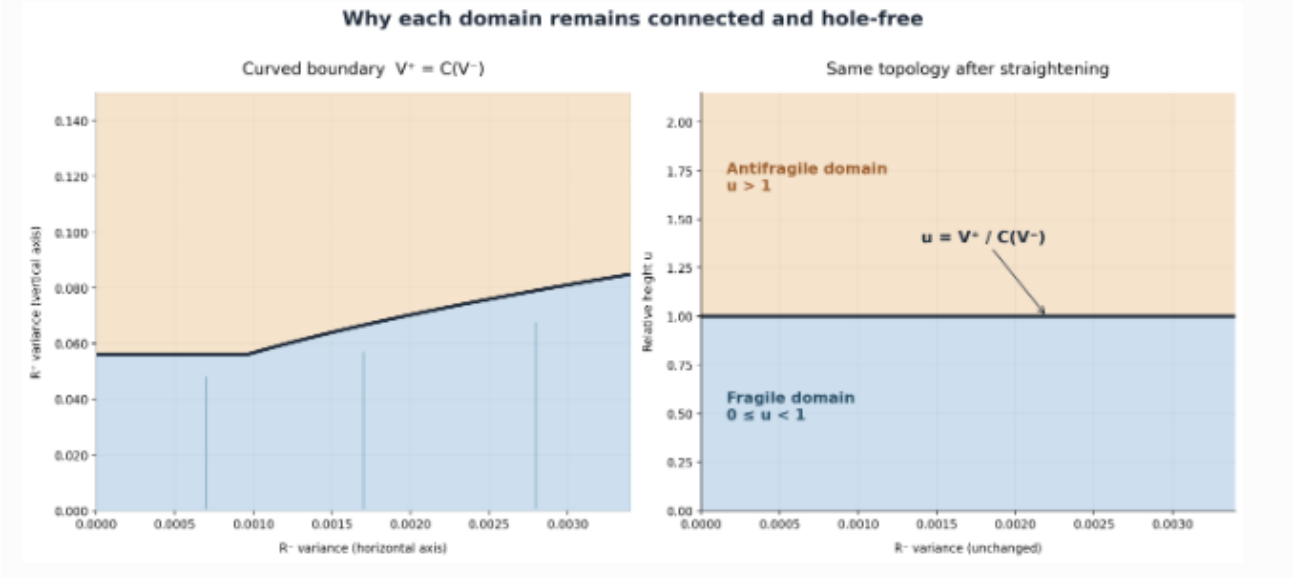


Figure 3: Straightening the boundary using $u = V_+/C_t(V_-)$. The transformed strict-dominance domains lie below and above $u = 1$.

On the left, you can see the original geometry of Janus.

On the right, we've straightened the boundary.

Now look at the two domains.

The fragile domain becomes:

$$0 \leq u < 1 \tag{46}$$

And the antifragile domain:

$$u > 1 \tag{47}$$

The boundary becomes:

$$u = 1 \tag{48}$$

Much easier to understand, isn't it?

You now have two regions separated by a horizontal line.

Take any two points in the blue region.

Can you connect them with a straight line without leaving that region?

Yes.

You can draw a segment between them, and every point on that segment remains inside the blue region.

We can demonstrate that with an equation:

$$\gamma(\lambda) = (1 - \lambda)z_1 + \lambda z_2, \qquad 0 \leq \lambda \leq 1 \tag{49}$$

10 Paths, contractibility, and homeomorphism

This equation tells us: "I start at the first point, then progressively move toward the second until I reach it."

When $\lambda = 0$, I'm at the first point.

When $\lambda = 1$, I'm at the second.

Between those two values, I'm moving along the segment.

If both starting heights are below 1, every intermediate weighted combination remains below 1.

The same reasoning works in the antifragile domain, where both heights are greater than 1.

Each domain is therefore path-connected: I can connect any two of its states with a continuous path that remains inside the domain.

And we can go a little further.

A domain can be a single connected piece and still contain a hole. Imagine a ring: you can walk all the way around it, but you can't shrink a loop surrounding its hole into a point without leaving the ring.

Do our two regions contain holes?

Look at our straightened regions.

I can take a loop inside either one and gradually shrink it toward a point chosen inside that same domain.

Mathematically, I can write that deformation as:

$$H(z, \lambda) = (1 - \lambda)z + \lambda z_0 \tag{50}$$

Every point of my loop progressively moves toward the point z_0

Because our straightened regions are convex, these movements remain entirely inside their respective regions.

So we can contract any loop into a point.

This property is called contractibility.

And because we can return from our straightened regions to the original curved regions through a continuous, reversible transformation:

$$V_+ = u\, C_t(V_-) \tag{51}$$

the two representations have the same topological organization.

This is what we call a homeomorphism.

The word sounds complicated, but it describes a simple idea:

"I can continuously and reversibly transform a shape without losing its connections or artificially creating holes."

We've therefore demonstrated that our two regular domains are both path-connected and contractible.

Their geometry can change while these topological properties remain the same.

Go back to our boundary equation:

$$C_t(V_-) = \frac{b_{+,t}}{2}\left[1 + \sqrt{1 + 4\frac{f_t}{n_t}\max\left(1, \frac{V_-}{b_{-,t}}\right)}\right] \tag{52}$$

11 A changing framework: March 16-17, 2016

Notice that little t?

It represents time.

The masses and references defining the framework are recalculated using the available historical data.

So our boundary can change from one date to another.

I can write that more simply:

$$C_{t_1}(V_-) \neq C_{t_2}(V_-) \tag{53}$$

At the same horizontal position, the boundary can have a different height at two different dates.

And if that boundary moves far enough, it can end up on the other side of a state that we keep fixed.

We've actually found this mechanism in the data from the two repositories.

Let's take March 16, 2016.

The calculated parameters include:

$$\begin{aligned} f_{16} &\approx 10.292917 \\ n_{16} &\approx 10.115890 \end{aligned} \tag{54}$$

The positive variance reference is approximately:

$$b_{+,16} = 0.0345594225 \tag{55}$$

And the negative variance reference is:

$$b_{-,16} = 0.0009679545 \tag{56}$$

We take the state observed on that date:

$$z = (0.0012231272,\ 0.0613259825) \tag{57}$$

The first coordinate is negative-leg variance. The second is positive-leg variance.

By replacing the parameters in our boundary equation, I get the following critical height at that horizontal position:

$$C_{16}(V_-) \approx 0.060107381 \tag{58}$$

The height of the point is 0.0613259825.

That's higher than the boundary:

$$0.0613259825 > 0.060107381 \tag{59}$$

So the point belongs to the antifragile domain.

To verify this another way, I calculate the two Janus densities at the same point.

They are approximately:

$$\begin{aligned} \rho_{+,16} &= 5.800433 \\ \rho_{-,16} &= 6.200311 \end{aligned} \tag{60}$$

The balance becomes:

$$D_{16} = 5.800433 - 6.200311 \approx -0.399878 \tag{61}$$

It's negative.

We get exactly the same dominance.

Now watch what We do.

I keep both coordinates of my point exactly the same. I don't change them.

I simply replace the March 16 framework with the parameters calculated for March 17.

The masses become approximately:

$$\begin{aligned} f_{17} &= 9.996732 \\ n_{17} &= 8.985637 \end{aligned} \tag{62}$$

The positive variance reference becomes:

$$b_{+,17} = 0.0346433700 \tag{63}$$

The negative reference remains practically unchanged in this comparison.

And when I recalculate the boundary using these new parameters, I get:

$$C_{17}(V_-) \approx 0.061900145 \tag{64}$$

Now look at this relationship:

$$0.060107381 < 0.0613259825 < 0.061900145 \tag{65}$$

It tells us that the height of our fixed point lies between the old boundary and the new boundary.

Under the first framework, the point is above the boundary.

Under the second, it's below.

So its new classification is fragile.

And if I calculate the two densities directly under the new framework:

$$\begin{aligned} \rho_{+,17} &\approx 5.647207 \\ \rho_{-,17} &\approx 5.476971 \end{aligned} \tag{66}$$

The balance becomes:

$$D_{17} = 5.647207 - 5.476971 \approx +0.170236 \tag{67}$$

It's positive.

Now look at the enlarged chart.

The dashed line represents the old boundary. The solid line represents the new one.

The black point is exactly the same in this comparison.

The hatched region represents all the states that change domains because of this modification to the framework.

We can even write that region mathematically:

$$Z = \{(V_-, V_+) : C_{16}(V_-) < V_+ < C_{17}(V_-)\} \tag{68}$$

12 Reclassification through boundary deformation

I gather all the states located above the old boundary but below the new one.

So it's not just the coordinates of our particular point that matter.

An entire region of the state space changes domains when we update the framework.

And in our calculation, we can distinguish the two effects.

If I change only the observed variances while keeping the March 16 framework, the balance remains negative in this episode.

But if I keep the old variances and change only the parameters defining the framework, the balance becomes positive.

So we've isolated the effect of the boundary's deformation.

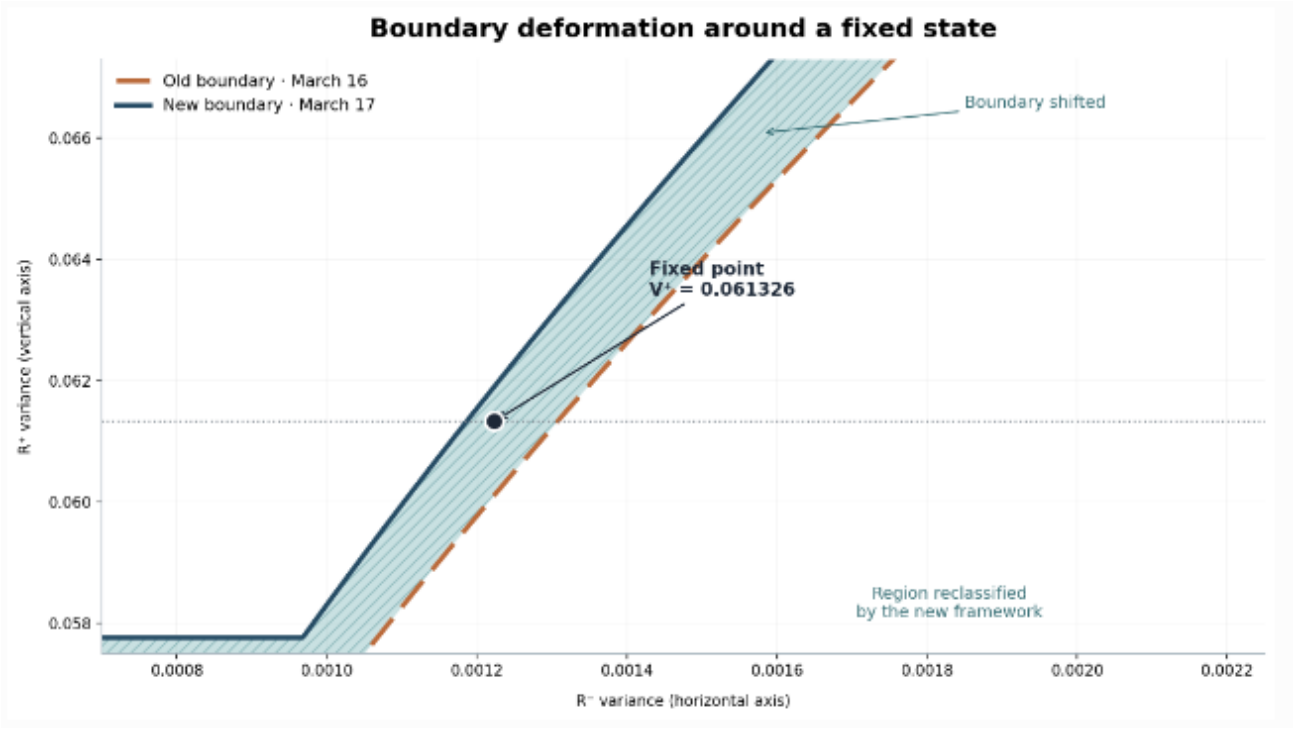


Figure 4: Boundary deformation around a fixed state. The hatched region contains states between the old and new boundaries and corresponds to the reclassification set Z.

This is a mathematical comparison using parameters derived from historical data, not an observation in which market parameters physically changed independently of the variances used to estimate them.

That distinction is precisely what allows us to understand the mechanism.

We've demonstrated that the geometry of the domains can evolve enough to reclassify a fixed state.

And now let's return to topology.

During this deformation, the fragile domain still exists, and so does the antifragile domain.

Each remains a single connected region.

Their shared boundary changes its shape and position, but their topological type is preserved.

The domains deform while preserving their connectedness.

But are there also situations where more than just the geometry changes?

Yes. And we're going to demonstrate that directly using an equation from Janus.

Go back to antifragile pressure:

$$\rho_- = \frac{n\, s_+}{1 + s_-} \tag{69}$$

13 The appearance of a second domain

Now ask yourself a very simple question.

What happens if $n = 0$?

The answer is right there in the equation:

$$\rho_- = \frac{0 \times s_+}{1 + s_-} = 0 \tag{70}$$

Antifragile pressure is zero everywhere in our stress space.

And if fragile mass remains strictly positive, then:

$$D = \frac{f}{1 + s_+} > 0 \tag{71}$$

The balance remains strictly positive for every finite level of stress.

In other words, the fragile domain occupies the entire state space, and the antifragile domain doesn't exist under this framework.

We can write:

$$F = X, \qquad A = \varnothing \tag{72}$$

The symbol X represents the set of possible states. The symbol $\varnothing$ simply means "empty set."

Now imagine that n becomes positive while f remains positive.

Antifragile pressure can then become dominant once positive stress exceeds a certain level.

A finite boundary appears, separating the two domains.

So we move from a configuration in which only one dominance domain is present to a configuration in which two domains are present.

And we've also found this in the Beta data connected to Janus.

On April 9, 2020, the calculated masses were:

$$\begin{aligned} f_9 &\approx 4.835370 \\ n_9 &= 0 \end{aligned} \tag{73}$$

The antifragile domain was absent.

On April 10:

$$\begin{aligned} f_{10} &\approx 4.713758 \\ n_{10} &\approx 0.254082 \end{aligned} \tag{74}$$

Both masses are now positive.

Here's the transformation.

For this particular drawing, I chose stress as our coordinates: positive stress on the horizontal axis and negative stress on the vertical axis.

This allows us to show the role of n directly, without bringing the variance references into the drawing.

Look at the image on the left.

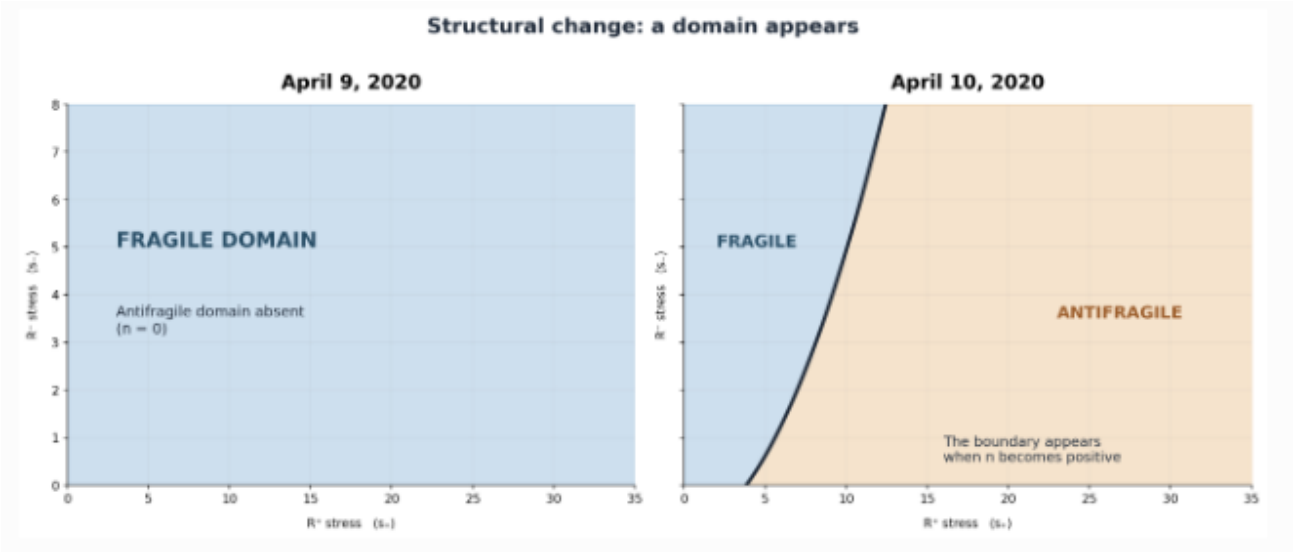


Figure 5: The appearance of an antifragile domain between April 9 and April 10, 2020. Here the axes are the two stress coordinates. The absence of the antifragile domain on April 9 follows from $n = 0$, rather than from the limits of the plotted window.

The fragile domain occupies the entire region represented.

The antifragile domain is absent because its pressure is zero.

In the image on the right, a boundary has appeared.

Both domains now exist.

It's important to understand that you cannot conclude a domain is empty simply because you don't see it within the window of a chart.

Here, its absence is demonstrated by the equation with $n = 0$.

When $n > 0$, the antifragile domain exists beyond a finite boundary, even if that boundary is too high to appear within a small plotting window.

We can now count the connected components present in our partition of strict-dominance domains.

I write:

$$S = F \cup A \tag{75}$$

14 Connected components and topological change

I combine the two dominance domains into one set.

When $n = 0$ and $f > 0$, only one domain is present.

So S has one connected component.

When $f > 0$ and $n > 0$, we have two domains separated by their boundary, which does not belong to S.

The set S then has two connected components.

$$\text{Number of connected components:} \quad 1 \longrightarrow 2 \tag{76}$$

15 Instantaneous dominance and temporal confirmation

The organization of the partition has changed from one dominance domain to two distinct domains.

This time, we're not dealing with geometric deformation alone. The number of connected components of the partition has changed.

So we've found two different mathematical phenomena in Janus.

In the first, both domains are present and their boundary deforms. Their geometry changes, but their topological properties remain the same.

In the second, a previously absent domain appears. The organization of the partition itself changes.

And this is where we can speak precisely about a topological change in the regime partition.

But there's one last question.

Does crossing to the other side of the boundary mean Janus has immediately confirmed a regime change?

No, because Janus distinguishes between instantaneous dominance and its temporally confirmed state.

we denote that state G_t.

The balance D_t answers one question: "Which component dominates right now?"

The state G_t answers another: "Have the conditions required to confirm the new regime been met?"

In the Janus code, entering the confirmed pressure state requires the balance to remain negative for a duration calculated by the model.

That duration is written as:

$$\tau_t = \text{round}\left[\text{clip}\left(\frac{1}{\sqrt{\max(|D_t|, 10^{-6})}}, 3, 30\right)\right] \tag{77}$$

Let me translate that.

First, We take the absolute value of the balance, meaning its magnitude regardless of whether it's positive or negative.

If that magnitude is large, the quantity

$$\frac{1}{\sqrt{|D_t|}} \tag{78}$$

is small. If the magnitude is small, this quantity increases.

The clip function then limits the result to between 3 and 30 observations, and round rounds it to an integer.

So the complete equation tells us:

"The model adjusts the confirmation period according to the strength of the imbalance between the two densities, while imposing a minimum and maximum duration."

When Janus is waiting to enter the confirmed pressure state, it counts consecutive observations where the balance is negative.

Confirmation occurs when that counter reaches the calculated duration.

To exit that state, its default configuration uses an acceleration signal from the positive leg. So that's a different rule from simply reversing the sign of the balance.

In the April 2020 Beta-to-Janus example, the new domain appears under the April 10 framework, and the temporal Janus state then confirms its transition on April 14.

That distinction matters.

The boundary determines geometric dominance. The temporal rule determines operational confirmation.

Price movements feed the variations of the two legs.

Those variations feed the variances, historical references, and estimated responses.

Those quantities determine the two Janus densities.

The densities define the domains and their boundary.

And when the parameters change, the geometry of those domains can change with them.

Topology becomes useful because it allows us to say precisely what is preserved and what is not: neighborhoods, paths, connectedness, the presence of domains, and their separation.

Topology doesn't automatically tell us tomorrow's price. It allows us to construct and study the organization of the states in which our model can evolve.

And that's exactly what we wanted to understand.

16 Computational reconstruction

The construction can be reconstructed numerically from the two return series.

The numerical procedure reconstructs the two dominance domains from the return data and the stated equations.

Start with the two series R^+ and R^-.

Transform the two return series into positive indices, then calculate the response masses, historical references, and diagnostic quantities.

Use the rolling estimation conventions stated in the detailed derivation.

The goal is to understand how the quantities produced by the algorithm become a geometry of market regimes.

Then you return to our equations:

$$\begin{aligned} s_+ &= \max\left(\frac{V_+}{b_+} - 1, 0\right) \\ s_- &= \max\left(\frac{V_-}{b_-} - 1, 0\right) \end{aligned} \tag{79}$$

Then:

$$\begin{aligned} \rho_+ &= \frac{f}{1 + s_+} \\ \rho_- &= \frac{n\, s_+}{1 + s_-} \end{aligned} \tag{80}$$

And finally:

$$D = \rho_+ - \rho_- \tag{81}$$

17 Numerical implementation and reconstruction protocol

For positive references and nonnegative masses, the following function evaluates the stress and density equations on scalar inputs or arrays.

The implementation is:

```
import numpy as np

def calculate_topology(V_plus, V_minus, f, n, b_plus, b_minus):
    V_plus = np.asarray(V_plus, dtype=float)
    V_minus = np.asarray(V_minus, dtype=float)
    stress_plus = np.maximum(V_plus / b_plus - 1.0, 0.0)
    stress_minus = np.maximum(V_minus / b_minus - 1.0, 0.0)
    rho_plus = f / (1.0 + stress_plus)
    rho_minus = n * stress_plus / (1.0 + stress_minus)
    balance = rho_plus - rho_minus
    fragile = balance > 0
    antifragile = balance < 0
    return balance, fragile, antifragile
```

Next, you calculate this balance across a grid of variance pairs, coloring fragile states in blue and antifragile states in orange.

You draw the boundary wherever the two densities are equal, when both masses are positive.

We reconstruct the map for March 16, 2016, and then the one for March 17.

You keep exactly the same point on both drawings.

And you should recover this result:

$$D_{16} < 0 \qquad \text{and} \qquad D_{17} > 0 \tag{82}$$

In this fixed-point comparison, dominance shifts from antifragile to fragile.

But most importantly, We explain why.

You need to be able to show that the parameters defining the framework changed, causing the boundary to move around the point.

Then We straighten that boundary using:

$$u = \frac{V_+}{C(V_-)} \tag{83}$$

You should recover the two regions separated by $u = 1$ and explain why they are connected.

Finally, go back to the case where $n = 0$.

We demonstrate, directly from the equation for ρ_-, why the antifragile domain is absent when $f > 0$.

Then explain what changes when both masses become positive.

All that's left is to translate them into calculations and reproduce the diagrams.

The following sections develop estimation, differential geometry, temporal confirmation, pricing transformations, and allocation.

You now have what you need to understand why the boundary isn't just a line added to a chart.

It's calculated from the model's equations. Its shape depends on the estimated conditions. Its deformation changes the domains, and certain changes even alter the topological organization of their partition.

PRICES EVOLVE, MARKET CONDITIONS EVOLVE, AND THE DOMAINS OF YOUR MODEL CAN EVOLVE WITH THEM. THAT'S WHY YOU CAN'T JUST STUDY WHAT MOVES INSIDE THE FRAMEWORK. YOU ALSO NEED TO UNDERSTAND HOW THE FRAMEWORK ITSELF TRANSFORMS.

18 Consequences of the topological construction

The construction separates three questions. First, the instantaneous balance determines which density dominates at a state. Second, the boundary geometry describes how the classification changes across variance states and across parameter settings. Third, the temporal rule determines when the algorithm confirms an operational transition. These objects should remain distinct.

For positive masses and positive references, the explicit boundary is continuous and strictly positive. Its normalized coordinate places the two strict-dominance domains in convex regions separated by $u = 1$. Each domain is therefore path-connected and contractible, even as its boundary bends or moves. By contrast, the transition from $n = 0$ to $n > 0$ with $f > 0$ changes the number of connected components in the union of strict-dominance domains, because the equality boundary is excluded from that union. The ambient state space X itself has not been split or changed.

The March 2016 example is a fixed-state comparison across two estimated frameworks. It illustrates the mathematical effect of changing parameters; it does not establish an intervention that changes those parameters independently of the data used to estimate them. The April 2020 example separates the emergence of a geometric domain from the later confirmation of an operational state. Both examples are numerical illustrations of the classification equations.

The algebra establishes classification and topological properties under the stated assumptions. It does not, by itself, establish predictive accuracy, profitability, robustness to estimation error, or out-of-sample performance. Those questions require separately specified data, estimation procedures, execution assumptions, and empirical validation. Within that scope, the central conclusion is precise: a market state may be reclassified either because it moves within a fixed framework or because the framework's boundary changes around it.

Detailed estimation of primitive coordinates

19 Market states and classification frameworks

The object of study is a classification framework for the responses of assets to changing market conditions. Observed price movements, the parameters of a valuation model, and the geometry used to classify a market state are distinct objects. A random price component may be represented through Brownian motion without assuming that observed financial prices follow a pure Brownian process.

The construction distinguishes fragile and antifragile contributions. Fragility describes an adverse response to shocks; robustness describes resistance; antifragility describes a favorable response. Here these terms receive operational definitions through variance-response estimates and a specified density operator. They are not permanent labels assigned to individual assets.

The two Beta legs provide weighted return series. A rolling regression measures the direction and intensity of each leg's response to variance; its signed coefficient is called financial mass. This coefficient is distinct from market capitalization or trading volume. A second calculation compares recent variance with a historical reference to define excess stress.

Masses and stresses enter two nonnegative density contributions. Their difference defines fragile dominance, antifragile dominance, or equality. The topological construction describes the organization and connectivity of these domains. The detailed analysis derives their shape, the effect of parameter changes, and the geometry of the transformations used by the model.

20 Returns, levels, and positive leg indices

A fixed state can change classification when the framework changes. In the historical comparisons developed below, holding variances and reference values fixed while replacing the masses is sufficient to reverse dominance. This separates a change in the observed point from a change in the map that classifies it.

The mathematical objects are kept distinct throughout: prices are observed; returns measure price changes; variance measures dispersion; mass estimates a response; stress compares variance with its reference; densities define dominance; temporal confirmation adds memory; and allocation is applied afterward. An analogy between layers does not establish an identity of their equations.

Write m for signed mass, f for retained fragile mass, n for the nonnegative magnitude of the antifragile channel, and D for the balance. The earlier notation a, b, Δ corresponds to f, n, D. A zero balance does not by itself establish robustness or introduce a third allocation regime.

The two Beta legs consist of asset sets with specified weights. Their returns are

$$R_t^+ = \sum_{i \in U^+} w_{i,t}^+ r_{i,t}, \qquad R_t^- = \sum_{j \in U^-} w_{j,t}^- r_{j,t}. \tag{84}$$

We read this equation properly.

The big symbol $\sum$, called sigma, simply means "add everything together."

U^+ is the set of assets in the first leg.

The notation $i \in U^+$ means that we take each asset belonging to that set.

w(i,t)$^+$ is that asset's weight at date t.

And r(i,t) is its logarithmic return.

The equation says

"We take each asset's return, multiply it by its weight, and add the contributions to obtain the return of the first leg."

The second leg follows exactly the same principle.

And pay attention to the signs.

The + and − identify the roles of the two legs. They don't mean the first leg must generate positive returns or the second must generate negative returns.

Next, define the individual logarithmic return

$$r_{i,t} = \ln\left(\frac{P_{i,t}}{P_{i,t-1}}\right). \tag{85}$$

P(i,t) is the current price of asset i.

P(i,t−1) is its price at the previous observation.

We divide the current price by the previous one, then apply the natural logarithm.

Why use logarithms?

Because consecutive logarithmic returns add together

$$\ln\left(\frac{P_t}{P_{t-2}}\right) = \ln\left(\frac{P_t}{P_{t-1}}\right) + \ln\left(\frac{P_{t-1}}{P_{t-2}}\right). \tag{86}$$

The intermediate prices cancel when we multiply the two price ratios.

So logarithmic returns give us a convenient additive representation of successive price changes.

Now take a simple example with two assets.

The first asset has weight 0.60 and logarithmic return 0.02.

The second has weight 0.40 and return −0.01.

The basket return is

$$R^+ = 0.60(0.02) + 0.40(-0.01). \tag{87}$$

The first contribution is 0.012, and the second is −0.004.

So

$$R^+ = 0.008. \tag{88}$$

That's a weighted basket return.

But Beta is not limited to these two returns.

The complete construction also includes admissible asset universes, weights, hedges, cumulative levels, centered residuals, and state diagnostics.

For example, one relationship is

$$\texttt{sb_alpha_excess} = R^+ - \texttt{sb_beta}\, R^-. \tag{89}$$

Here, sb_beta is a hedge coefficient.

The equation says:

"Take the return of the first leg, subtract the second leg's return multiplied by the hedge coefficient, and retain the residual."

Another relationship is

$$\texttt{sb_S} = L^{+} - \texttt{sb_k}\, L^{-} - \texttt{sb_S_center}. \tag{90}$$

This one operates on cumulative logarithmic levels.

sb_k is used in the relation between those levels, while sb_S_center removes a centering reference.

The first equation concerns returns.

The second concerns cumulative levels.

And sb_beta, sb_fragility_beta, the complete Beta object, and the rolling financial mass of Janus are not interchangeable.

Now, to connect the Beta legs to Janus, We construct positive indices

$$P_t^{\pm} = \exp\left(L_t^{\pm} - L_{t_0}^{\pm}\right). \tag{91}$$

The function exp is the exponential function, the inverse of the natural logarithm.

$L_t\pm$ is the cumulative logarithmic level at date t.

$L(t_0)\pm$ is its level at the starting date.

At the starting date

$$L_{t_0}^{\pm} - L_{t_0}^{\pm} = 0. \tag{92}$$

And since $\exp(0) = 1$, each index starts at 1.

Now, did we preserve the original Beta returns?

We prove it.

$$\begin{aligned} \ln\left(\frac{P_t^{\pm}}{P_{t-1}^{\pm}}\right) &= (L_t^{\pm} - L_{t_0}^{\pm}) - (L_{t-1}^{\pm} - L_{t_0}^{\pm}) \\ &= L_t^{\pm} - L_{t-1}^{\pm} \\ &= R_t^{\pm}. \end{aligned} \tag{93}$$

Look at the starting-level terms.

They cancel!

We're left with the difference between two consecutive cumulative logarithmic levels, which is exactly the original leg return.

We've proved that the constructed indices preserve the logarithmic returns of the two beta legs!

But that doesn't automatically make these indices the executable prices of self-financing portfolios, and it doesn't replace Beta's other coordinates.

Beta's Q1–Q4 states also don't automatically become Janus's fragile and antifragile domains.

21 Sample variance and annualization

Now we have the two series we need.

Before estimating their financial masses, however, we need to understand exactly what we're measuring with variance.

Imagine five returns, expressed in percentage points

1,1,1,1,1.

Their average is 1, and every observation equals the average.

There's no dispersion.

Now consider -1,0,1,2,3.

The average is still 1.

But the returns are far more spread out.

That's what variance measures.

For a sample of N returns

$$\widehat{\mathrm{Var}}_N(r) = \frac{1}{N-1}\sum_{i=1}^{N}(r_i - \bar{r})^2. \tag{94}$$

Here, $\bar{r}$ is the average return.

For each observation, we subtract that average.

We square the difference, add the squared differences, then divide by N − 1.

Why squares?

Because positive and negative deviations shouldn't cancel.

And why N − 1?

Because this is the sample-variance convention used when the mean is also estimated from the same observations.

In our second example, the deviations are

-2,-1,0,1,2.

Their squares are

4,1,0,1,4.

The sum is 10.

With five observations

$$\widehat{\mathrm{Var}}_5 = \frac{10}{5-1} = 2. \tag{95}$$

The first example has variance zero.

Variance doesn't tell us whether returns are rising or falling. it measures how dispersed they are around their mean!

And keep the units straight: variance is measured in squared return units. A variance of 0.05 is not a volatility of 5%. Volatility is the square root of variance.

Now, Janus uses variance for two different calculations.

The first measures the financial mass.

The second measures recent stress.

They use different windows, and we must keep them separate.

For the mass, Janus uses

$$U_t = 252\,\widehat{\mathrm{Var}}_{252}(r)_t. \tag{96}$$

It calculates the variance of the last 252 returns and annualizes it using a factor of 252.

The corresponding excess-return measure is

$$E_t = \ln\left(\frac{P_t}{P_{t-252}}\right) - 0.02. \tag{97}$$

This is the logarithmic return over 252 observations minus the constant 2% reference in the studied configuration.

22 Variance response and least-squares mass estimation

Now, we're going to estimate how the two quantities move together.

The relationship is

$$E = \alpha - mU + \varepsilon. \tag{98}$$

E is the excess return.

α is the baseline of the fitted relationship.

U is the annualized variance.

m is the signed sensitivity we're trying to find.

ε is the unexplained residual.

This equation says:

"Excess return is represented by a baseline, minus a sensitivity multiplied by variance, plus the residual."

That sensitivity m is what we call financial mass in Janus!

But I don't choose it arbitrarily.

I estimate it from historical observations.

The program fits 252 historical pairs (U_i, E_i).

It searches for the line that minimizes the sum of squared errors

$$Q(\alpha, m) = \sum_{i=1}^{N} (E_i - \alpha + mU_i)^2. \tag{99}$$

Why is the term $+\mathrm{mU}_i$ inside the error when the fitted relationship contains $-\mathrm{mU}_i$?

Because

$$E_i - (\alpha - mU_i) = E_i - \alpha + mU_i. \tag{100}$$

We're subtracting the fitted value.

Next, minimize that error.

First, differentiate with respect to α

$$\frac{\partial Q}{\partial \alpha} = -2\sum_i (E_i - \alpha + mU_i). \tag{101}$$

At the minimum, this derivative is zero

$$\sum_i (E_i - \alpha + mU_i) = 0. \tag{102}$$

Expand

$$\sum_i E_i - N\alpha + m\sum_i U_i = 0. \tag{103}$$

Isolate α and divide by N

$$\alpha = \bar{E} + m\bar{U}. \tag{104}$$

The bars represent averages.

Now substitute this expression back into the error

$$\begin{aligned} E_i - \alpha + mU_i &= E_i - (\bar{E} + m\bar{U}) + mU_i \\ &= (E_i - \bar{E}) + m(U_i - \bar{U}). \end{aligned} \tag{105}$$

We're now working with deviations from the means.

Differentiate the sum of squares with respect to m.

The minimum condition is

$$\sum_i (U_i - \bar{U})\big[(E_i - \bar{E}) + m(U_i - \bar{U})\big] = 0. \tag{106}$$

Expand

$$\sum_i (U_i - \bar{U})(E_i - \bar{E}) + m\sum_i (U_i - \bar{U})^2 = 0. \tag{107}$$

Move the first term to the other side and divide

$$m = -\frac{\sum_i (U_i - \bar{U})(E_i - \bar{E})}{\sum_i (U_i - \bar{U})^2}. \tag{108}$$

The numerator measures how the deviations of U and E move together.

The denominator measures the dispersion of the successive U values inside the regression window.

Using the standard statistical terminology

$$m = -\frac{\mathrm{Cov}(U, E)}{\mathrm{Var}(U)}. \tag{109}$$

And be careful!

Var(U) is the variance of the successive calculated U values, not simply U itself.

Now we need to demonstrate that this really is a minimum.

Call the calculated mass m⋆.

After choosing the optimal intercept for each mass value

$$Q(m) = Q(m^\star) + (m - m^\star)^2 \sum_i (U_i - \bar{U})^2. \tag{110}$$

The error obtained with another mass equals the minimum error plus a nonnegative quantity."

When U varies inside the window, the sum of squared deviations is strictly positive.

The extra term vanishes only when m = m⋆.

Therefore

$$Q(m) \geq Q(m^\star). \tag{111}$$

We've demonstrated that the estimated mass minimizes the squared error!

If U is constant, the denominator is zero and the slope is not identifiable.

The code handles unavailable estimates according to its initialization and carry-forward conventions.

An initialization zero, an unavailable estimate, and a genuinely estimated zero mass are not the same thing.

Now, there's already a geometric meaning behind this regression.

Collect the centered observations into vectors Uc and Ec.

Our formula becomes

$$m = -\frac{\langle U_c, E_c \rangle}{\|U_c\|^2}. \tag{112}$$

The angle brackets denote the dot product.

The double bars denote the vector norm.

Using the angle ϑ between the centered vectors

$$\langle U_c, E_c \rangle = \|U_c\| \|E_c\| \cos\vartheta. \tag{113}$$

Substituting and simplifying

$$m = -\frac{\|E_c\|}{\|U_c\|} \cos\vartheta. \tag{114}$$

If the angle is obtuse, its cosine is negative, so the mass is positive.

If the angle is acute, the cosine is positive, so the mass is negative.

If the projection is zero and Uc $\neq 0$, the mass is zero.

Financial mass already has a precise geometric interpretation: it's a projection coefficient calculated from the historical observations!

23 Interpretation, estimation checks, and retained masses

Now, what does its sign mean economically?
Our fitted response is:

$$\widehat{E}(U) = \alpha - mU. \tag{115}$$

The hat means we're talking about the value predicted by the fitted line, not an observed future return.
Increase variance by δU > 0, while keeping the coefficients fixed:

$$\widehat{E}(U + \delta U) = \alpha - m(U + \delta U). \tag{116}$$

Subtract the original fitted response

$$\widehat{E}(U + \delta U) - \widehat{E}(U) = -m\delta U. \tag{117}$$

If m > 0, the change is negative.
The fitted response deteriorates when variance increases.
That's the fragile response in this construction.
If m < 0, the change is positive.
The fitted response improves when variance increases.
That's the antifragile response in this statistical construction.
If m $= 0$, the fitted linear response has no sensitivity to U.
The mass measures the direction and intensity of the estimated response to variance!
It doesn't guarantee the asset will make money during the next shock.
Next, examine the actual data.
For SPY, over the 252 historical pairs from 26 February 2019 to 25 February 2020

$$m_{\mathrm{SPY}} \simeq +15.258823, \qquad \alpha_{\mathrm{SPY}} \simeq 0.441035. \tag{118}$$

The fitted response is

$$\widehat{E}_{\mathrm{SPY}}(U) \simeq 0.441035 - 15.258823U. \tag{119}$$

The slope is negative.
In this window, higher variance is associated with a lower fitted excess return.
Now look at GLD over the same window

$$m_{\mathrm{GLD}} \simeq -42.523791, \qquad \alpha_{\mathrm{GLD}} \simeq -0.398160. \tag{120}$$

Its fitted response becomes

$$\widehat{E}_{\mathrm{GLD}}(U) \simeq -0.398160 + 42.523791U. \tag{121}$$

The slope is positive

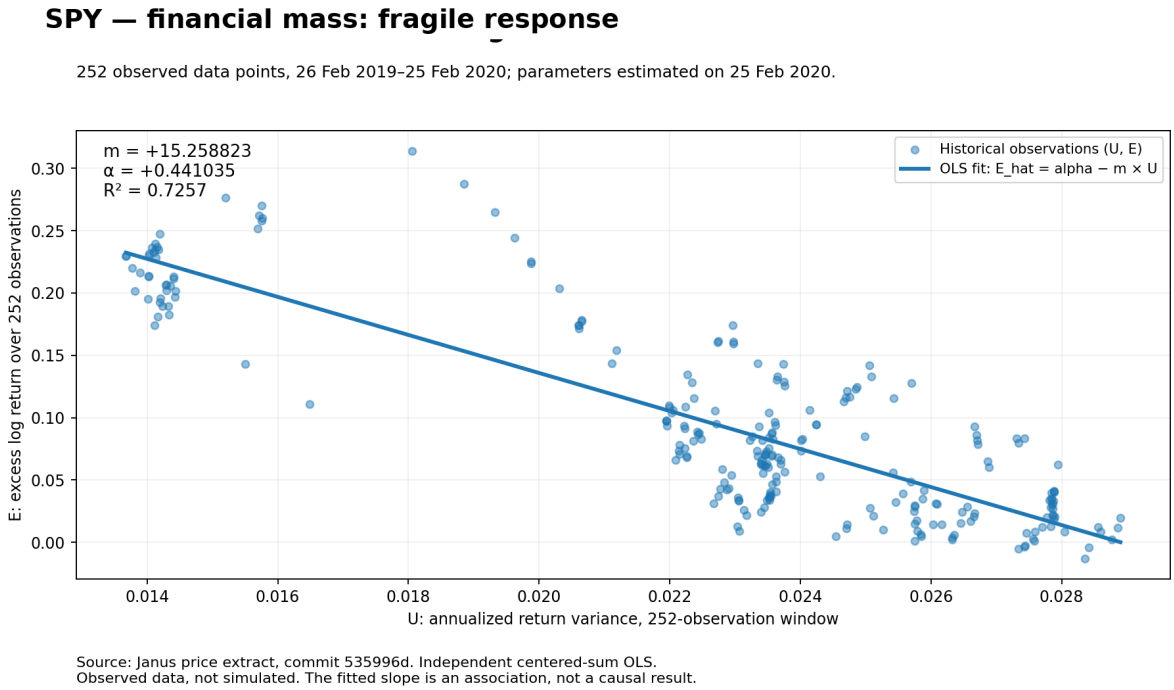


Figure 6: Rolling variance-response estimation for the fragile leg. The fitted slope determines the signed response mass.

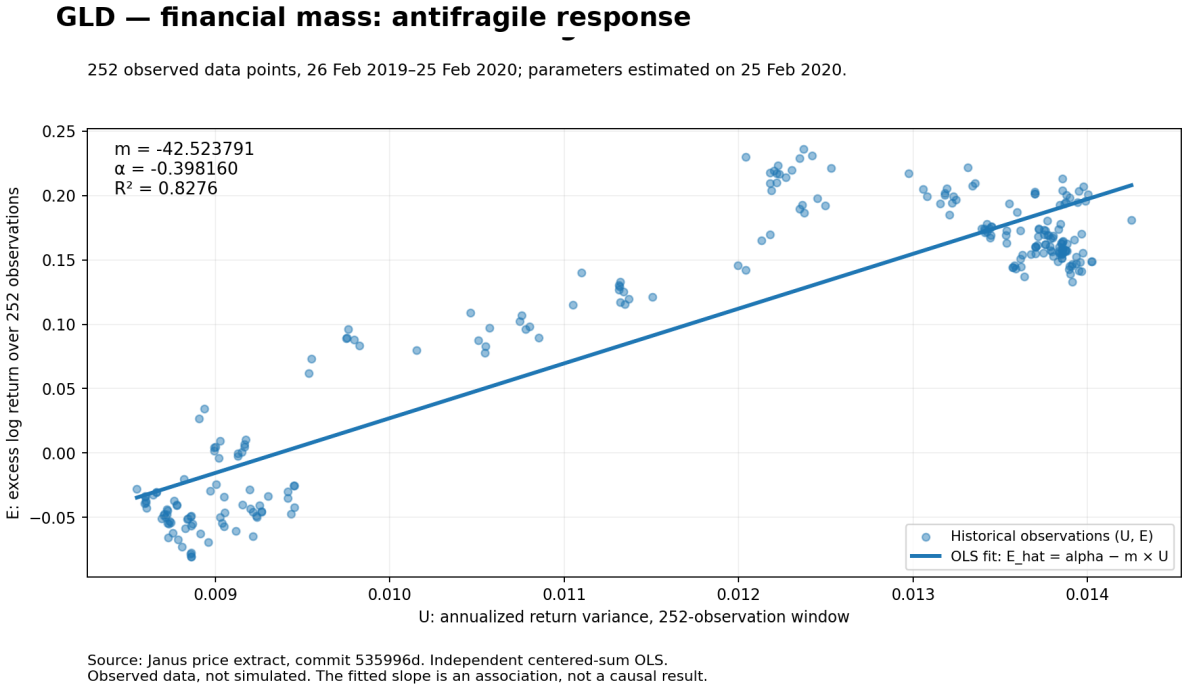


Figure 7: Rolling variance-response estimation for the second leg, with the negative response retained as a nonnegative antifragile-channel magnitude.

Now apply the same small variance increase to each fitted relationship

$$\delta U = 0.001. \tag{122}$$

For SPY

$$\delta \widehat{E}_{\mathrm{SPY}} \simeq -0.015259. \tag{123}$$

For GLD

$$\delta \widehat{E}_{\mathrm{GLD}} \simeq +0.042524. \tag{124}$$

The two fitted relationships have opposite slopes.

But this is a comparison of sensitivities within each asset's estimated response, not necessarily the same external shock hitting both assets.

Their R^2 values are approximately 0.725731 and 0.827619.

Those figures measure the historical fit, not the probability of correctly forecasting the next session.

The fitted residuals also satisfy

$$\sum_i \varepsilon_i \simeq 0, \qquad \sum_i (U_i - \bar{U})\varepsilon_i \simeq 0. \tag{125}$$

The numerical discrepancies are on the order of 10^{-14} or less.

So we've measured the responses and verified the estimation.

But knowing each asset's mass doesn't yet tell us which domain the pair belongs to!

That's where the Janus coupling operator comes in.

Now, Janus doesn't directly use every signed mass.

It first retains two admissible components

$$f = \max(m_+, 0), \qquad n = \max(-m_-, 0). \tag{126}$$

We read the first equation.

m_+ is the signed mass estimated for the first leg.

If it's positive, we keep it. If it's negative, we retain zero.

For example

$$m_+ = 4 \quad \Longrightarrow \quad f = 4. \tag{127}$$

But

$$m_+ = -4 \quad \Longrightarrow \quad f = 0. \tag{128}$$

For the second leg, we first reverse the sign of its mass, then retain the positive part.

If

$$m_- = -4, \tag{129}$$

Then

$$n = \max(4, 0) = 4. \tag{130}$$

But if

$$m_- = +4, \tag{131}$$

Then

$$n = \max(-4, 0) = 0. \tag{132}$$

Why treat the two masses differently?

Because f represents the retained positive fragile mass of the first leg, while n represents the positive magnitude of the second leg's negative response.

Both f and n are nonnegative, even though the original signed mass m_- can be negative.

And if $n = 0$, the second leg hasn't disappeared from the market. Its antifragile-response channel simply isn't retained by this operator at that date.

24 Historical variance references and stress coordinates

Now we have the masses.

But that's still not enough.

Imagine two assets with exactly the same financial mass. Do they necessarily make the same contribution to the regime?

No!

Because their recent variances may be very different relative to their own historical behavior.

That's where stress comes in.

For each leg, Janus calculates

$$V_t^{\pm} = 252\,\widehat{\mathrm{Var}}_{63}(R^{\pm})_t. \tag{133}$$

This is the variance of the last 63 leg returns, annualized using the factor 252.

Notice the difference from the mass regression

$$U_t = 252\,\widehat{\mathrm{Var}}_{252}(r)_t. \tag{134}$$

The mass uses U, based on a 252-return variance.

The stress uses V, based on 63 returns.

These are two different variance measurements, with two different roles!

Now suppose the recent variance is V=0.06

Is that high?

Well, compared with what?

If the asset's usual reference is 0.03, the recent variance is twice that reference.

But if the reference is 0.10, the same 0.06 is actually below it.

That's why Janus constructs a historical baseline.

For each leg

$$b_t^{\pm} = \text{median}\left(V_{t-252}^{\pm}, \ldots, V_{t-1}^{\pm}\right). \tag{135}$$

The median is the middle value after ordering the observations.

For example, in 1,2,3,4,100,

the median is 3.

Unlike the mean, it isn't pulled as strongly by the extreme value 100.

And look at the final index in our historical window: t − 1.

The current variance doesn't enter the reference used to evaluate itself.

Now we calculate the relative excess

$$\widetilde{s}_{\pm} = \frac{V_{\pm}}{b_{\pm}} - 1. \tag{136}$$

The tilde indicates stress before clipping.

If V = b, their ratio is 1, and the raw stress is zero.

If V > b, it is positive.

If V < b, it is negative.

But Janus keeps only the non negative part

$$s_{\pm} = \max(\widetilde{s}_{\pm}, 0). \tag{137}$$

We calculate it.

For V=0.06, b=0.03,

We get

$$s = \max\left(\frac{0.06}{0.03} - 1, 0\right) = \max(1, 0) = 1. \tag{138}$$

The recent variance is 100% above its reference.

Now keep the same V = 0.06, but use b = 0.10

$$s = \max(0.6 - 1, 0) = \max(-0.4, 0) = 0. \tag{139}$$

The variance hasn't changed. Only the reference has.

And the measured stress is completely different.

Variance alone doesn't define stress. stress depends on the relationship between current variance and its historical reference!

We can also write

$$\frac{1}{1 + s_{\pm}} = \frac{b_{\pm}}{\max(b_{\pm}, V_{\pm})}. \tag{140}$$

We prove it.

If V ≤ b, stress is zero

$$\frac{1}{1+s} = 1 \tag{141}$$

And because max(b, V) = b

$$\frac{b}{\max(b, V)} = 1 \tag{142}$$

Both sides agree.

If V > b, stress is

$$s = \frac{V}{b} - 1 \tag{143}$$

Add 1

$$1 + s = \frac{V}{b} \tag{144}$$

Take the reciprocal

$$\frac{1}{1+s} = \frac{b}{V}. \tag{145}$$

And since V > b, the maximum in the denominator is V.

So the identity holds in both cases.

Notice how the functions change their behavior at particular conditions.

The median depends on the ordering of historical observations.

The clipping operator changes its expression when the raw stress crosses zero.

The transformation is already piecewise-defined before we even construct the densities!

Now, to simplify the notation, I'll write

$$x = s_+, \qquad y = s_-. \tag{146}$$

These are simply shorter names for the two stresses.

Our four primitive variables are

$$p = (f, n, x, y) \in \mathcal{K} = [0, +\infty)^4. \tag{147}$$

The symbol ∈ means "belongs to."

The set $[0, +\infty)^4$ contains four-dimensional points whose coordinates are all nonnegative.

That describes the admissible mathematical inputs to the coupling operator. It doesn't say the four variables are statistically independent in historical data.

Density geometry

25 The four-dimensional primitive state and density map

Now look at the two Janus densities

$$u = \rho_+ = \frac{f}{1+x}, \qquad v = \rho_- = \frac{nx}{1+y}. \tag{148}$$

The Greek letter ρ is pronounced "rho."

u and v are simply shorter names for these densities.

We examine the first

$$u = \frac{f}{1+x} \tag{149}$$

The numerator contains the retained fragile mass.

The denominator contains one plus the first leg's stress.

If x = 0, the denominator is 1

$$u = f. \tag{150}$$

If x = 1

$$u = \frac{f}{2} \tag{151}$$

If x = 3

$$u = \frac{f}{4} \tag{152}$$

So at fixed mass, increasing the first leg's stress reduces its fragile contribution.

Now look at the second density

$$v = \frac{nx}{1+y} \tag{153}$$

The numerator contains the retained antifragile magnitude n, multiplied by the stress x.

If x = 0

$$v = 0 \tag{154}$$

Even if n > 0, the channel is not activated.

If x increases, the second contribution increases, provided n > 0.

But y, the second leg's own stress, is in the denominator.

Increasing y attenuates that contribution.

So the same stress x weakens the first contribution and activates the second. the stress y attenuates the second contribution!

That's the coupling.

And this is an important scientific distinction: these density formulas are the operator defined by Janus. The regression provides the masses, but doesn't uniquely impose this particular coupling on every possible financial model.

26 Coupling, balance, and the dominance domains

Now, the two densities are non negative.

To determine which one dominates, Janus calculates

$$\boxed{D = u - v} \tag{155}$$

If u = 4 and v = 2

$$D = +2 \tag{156}$$

The fragile contribution dominates.

If u = 2 and v = 4

$$\text{D=-2.} \tag{157}$$

The antifragile contribution dominates.

And if u = v

$$D = 0 \tag{158}$$

The contributions are equal.

The antifragile density itself isn't negative. It's the second contribution that is subtracted in the balance.

Now we can define the three sets

$$\boxed{F = \{p : D(p) > 0\}, \qquad A = \{p : D(p) < 0\}, \qquad Z = \{p : D(p) = 0\}.} \tag{159}$$

The braces mean "the set of states satisfying the condition."

F contains positive-balance states.

A contains negative-balance states.

And Z contains zero-balance states.

A zero balance doesn't automatically imply robustness.

Both contributions might simply be zero.

And the domain of a pair isn't a permanent label attached to every individual asset inside that pair.

Now, the geometry becomes a very concrete question.

Where exactly are both contributions equal?

At the boundary

$$D = 0 \tag{160}$$

Since D = u − v

$$\mathrm{u} = \mathrm{v} \tag{161}$$

Substitute the density formulas

$$\frac{f}{1+x} = \frac{nx}{1+y} \tag{162}$$

Multiply both sides by (1 + x)(1 + y).

We can do that without changing the equality because x, y ≥ 0, so both denominators are positive.

The fractions simplify

$$\boxed{f(1+y) = nx(1+x).} \tag{163}$$

This is the exact equilibrium condition.

On the left, fragile mass is multiplied by one plus the second leg's stress.

On the right, antifragile magnitude is multiplied by the activating stress and by one plus that same stress.

27 The equilibrium graph and its stress-plane boundary

Now put the balance over a common denominator

$$\boxed{D = \frac{f(1+y) - nx(1+x)}{(1+x)(1+y)}.} \tag{164}$$

We name the numerator and denominator locally

$$\mathrm{N}_D = f(1+y) - nx(1+x), \tag{165}$$

$$\mathrm{H}_D = (1+x)(1+y) > 0. \tag{166}$$

So

$$\boxed{f = \frac{nx(1+x)}{1+y}.} \tag{167}$$

Because H_D is strictly positive, the sign of D is precisely the sign of N_D.

Now solve the boundary equation for f

$$\boxed{f = \frac{nx(1+x)}{1+y}.} \tag{168}$$

That means for any choice of n, x, y, the equation gives the value of f that balances the two contributions.

Three coordinates remain free.

We can parameterize the zero-balance level

$$\text{(n,x,y)} \longmapsto \left(\frac{nx(1+x)}{1+y}, n, x, y \right). \tag{169}$$

So in our four-dimensional primitive space, the global zero level is a three-dimensional graph, with boundaries and corners inherited from the nonnegative constraints.

A curve drawn on a two-dimensional chart is only a slice of this larger zero-balance set!

Now look at the shape in different coordinate systems.

In the density plane

$$\text{D=0} \iff v = u. \tag{170}$$

That's a diagonal.

At fixed stresses, in the mass plane

$$f = \frac{x(1+x)}{1+y}\, n \tag{171}$$

That's a straight line when x > 0.

But now fix the masses f, n > 0, and let the stresses vary.

Starting from

$$f(1+y) = nx(1+x) \tag{172}$$

divide by f

$$1 + y = \frac{n}{f}\, x(1+x) \tag{173}$$

Subtract 1

$$\boxed{y = \frac{n}{f}\, x(1+x) - 1} \tag{174}$$

Expand

$$y = \frac{n}{f}\, x^2 + \frac{n}{f}\, x - 1 \tag{175}$$

That's a quadratic function of x.

Our boundary is a parabola in the stress plane!

Why was it a straight line in the density plane?

Because we changed the coordinates.

The equilibrium condition didn't change. Only its representation did.

Now, which side of the parabola is fragile, and which side is antifragile?

We prove it.

At fixed masses and fixed y

$$D(x, y) = \frac{f}{1+x} - \frac{nx}{1+y} \tag{176}$$

Differentiate with respect to x

$$\boxed{\frac{\partial D}{\partial x} = -\frac{f}{(1+x)^2} - \frac{n}{1+y} < 0} \tag{177}$$

Both terms are negative for f, n > 0.

So the balance decreases strictly as x increases.

At x = 0

$$D(0, y) = f > 0 \tag{178}$$

And as x $\to +\infty$, the first term approaches zero while the subtracted second term grows without bound

$$\lim_{x \to +\infty} D(x, y) = -\infty \tag{179}$$

The balance is continuous, starts positive, decreases strictly, and eventually becomes negative.

So it crosses zero exactly once!

There is a unique critical stress x$\star$ for every fixed y.

Next, calculate it.

We have

$$f(1+y) = nx(1+x) \tag{180}$$

Divide by n > 0

$$x^2 + x = \frac{f}{n}(1+y) \tag{181}$$

Multiply both sides by 4 and add 1

$$(2x+1)^2 = 1 + 4\frac{f}{n}(1+y) \tag{182}$$

Take the non negative root

$$\boxed{x^\star = \frac{\sqrt{1 + 4(f/n)(1+y)} - 1}{2}} \tag{183}$$

The star means "critical value."

Since the balance is strictly decreasing

$$x < x^\star \iff D > 0 \tag{184}$$

$$x = x^\star \iff D = 0 \tag{185}$$

$$x > x^\star \iff D < 0 \tag{186}$$

The symbol $\iff$ means "if and only if."

At y = 0, the admissible parabola begins at

$$x_0 = \frac{\sqrt{1 + 4f/n} - 1}{2} \tag{187}$$

Next, apply this to the historical SPY/GLD state of 25 February 2020.

The retained masses are:

$$f \simeq 15.258823, \qquad n \simeq 42.523791 \tag{188}$$

And for that observation

$$y = 0 \tag{189}$$

The threshold is

$$\boxed{x^\star \simeq 0.280276} \tag{190}$$

But the observed activating stress is

$$x_{\text{observed}} \simeq 0.315577 \tag{191}$$

Since

$$0.315577 > 0.280276 \tag{192}$$

the observed state lies beyond the critical stress.

So D < 0.

We verify it directly

$$u = \frac{15.258823}{1 + 0.315577} \simeq 11.598583 \tag{193}$$

$$v = \frac{42.523791(0.315577)}{1+0} \simeq 13.419512 \tag{194}$$

Therefore

$$\boxed{D \simeq -1.820929} \tag{195}$$

The analytical threshold, the density calculation, and the geometric map agree.

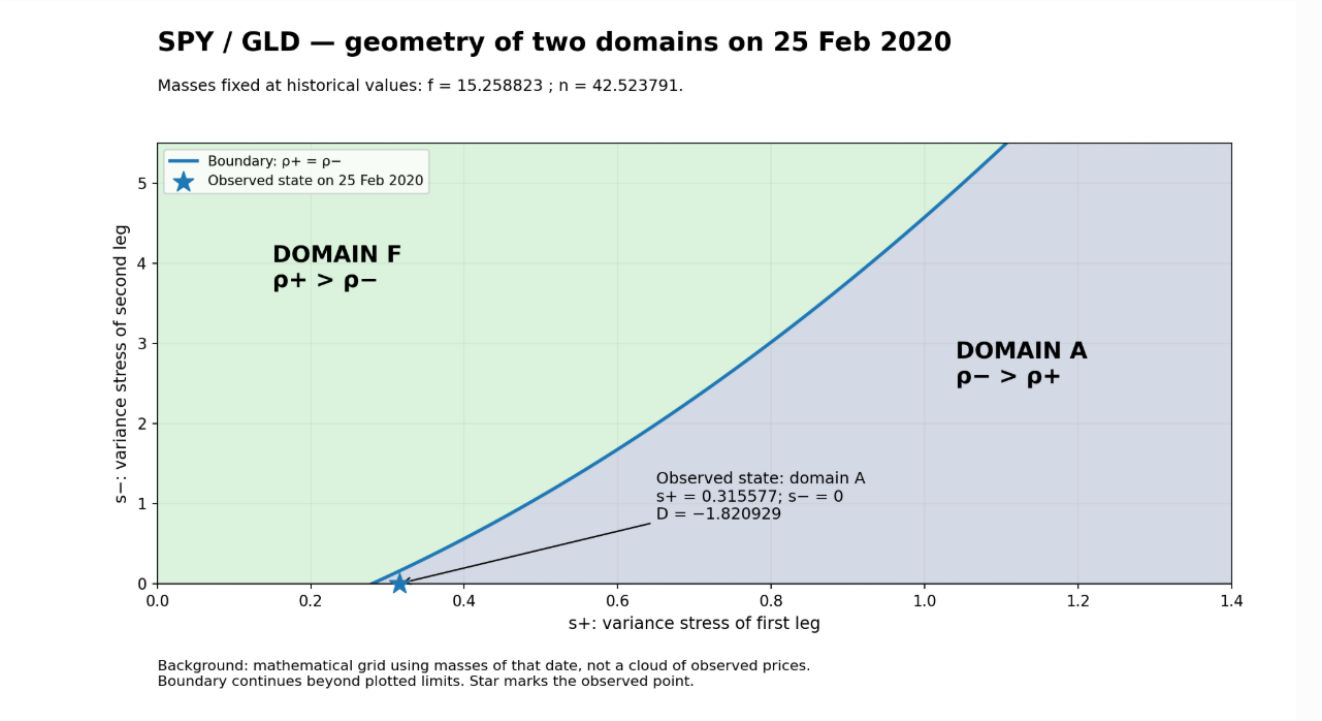


Figure 8: Dominance in the two stress coordinates. The observed state is separate from the curve determined by the framework parameters.

In the figure, the colored regions are a mathematical grid of possible states calculated under the masses of that date.

The star marks the observed state.

Those background points are not thousands of observed prices.

28 Boundary sensitivity and invariant mass scaling

Next, understand how the masses deform this boundary.

Define their ratio

$$q = \frac{f}{n} \tag{196}$$

The critical stress becomes

$$x^\star(q,y) = \frac{\sqrt{1+4q(1+y)}-1}{2} \tag{197}$$

Differentiate with respect to q

$$\boxed{\frac{\partial x^\star}{\partial q} = \frac{1+y}{\sqrt{1+4q(1+y)}} > 0} \tag{198}$$

The derivative of the square root is combined with the derivative of its inner expression. The numerical factors cancel, leaving this strictly positive result.

What does that mean?

When fragile mass increases relative to antifragile mass, more positive stress is required to cross the boundary!

And when the antifragile channel strengthens relatively, the critical stress falls.

We can also isolate the effects of the four primitives on the balance

$$\frac{\partial D}{\partial f} = \frac{1}{1+x} > 0 \tag{199}$$

$$\frac{\partial D}{\partial n} = -\frac{x}{1+y} \leq 0 \tag{200}$$

$$\frac{\partial D}{\partial x} = -\frac{f}{(1+x)^2} - \frac{n}{1+y} \leq 0 \tag{201}$$

$$\frac{\partial D}{\partial y} = \frac{nx}{(1+y)^2} \geq 0 \tag{202}$$

At fixed stresses, increasing f raises the balance.

Increasing n lowers it when x > 0.

Increasing x reduces the fragile contribution and activates the antifragile one.

Increasing y attenuates the subtracted antifragile contribution.

Now, in real historical data, these variables can change simultaneously.

The full differential is

$$\boxed{\begin{aligned} dD = {} & \frac{df}{1+x} - \frac{x\,dn}{1+y} \\ & - \left[\frac{f}{(1+x)^2} + \frac{n}{1+y}\right] dx \\ & + \frac{nx}{(1+y)^2}\,dy. \end{aligned}} \tag{203}$$

Each term represents a local sensitivity multiplied by a small change in its corresponding variable.

But remember: this is a local differential.

For finite changes between two dates, we must recompute the formulas or use an explicitly defined finite-change decomposition.

We cannot automatically turn a local derivative into a complete causal explanation of a historical episode.

Next, examine another property.

Suppose both masses are multiplied by the same positive constant λ

$$(f, n) \mapsto (\lambda f, \lambda n) \tag{204}$$

Then

$$\begin{aligned} D_{\text{new}} &= \frac{\lambda f}{1+x} - \frac{\lambda n x}{1+y} \\ &= \lambda \left(\frac{f}{1+x} - \frac{nx}{1+y} \right). \end{aligned} \tag{205}$$

So

$$\boxed{D_{\text{new}} = \lambda D} \tag{206}$$

The ratio f/n hasn't changed.

The zero-balance boundary stays in the same place.

But the magnitude of the imbalance changes.

That can affect the confirmation duration and the exit condition.

The mass ratio and the common scale of the masses do not play the same role!

Now We return to the variance map.

Remember the point that stays in exactly the same place while the framework moves around it?

We now have everything we need to calculate that framework.

I'll represent the observed state by

$$\boxed{z = (V_-, V_+)} \tag{207}$$

The first coordinate V_- is horizontal.

The second V_+ is vertical.

These are the two measured annualized variances.

Now I'll collect the parameters defining the classification framework

$$\boxed{\theta = (f, n, b_+, b_-)} \tag{208}$$

29 The boundary in variance coordinates

The letter θ, groups the two retained masses and the two historical variance references.

Don't confuse it with the angle ϑ from our earlier geometric interpretation of the regression.

They are different mathematical objects.

Why separate z and θ?

Because z tells us where our observed point is in the variance plane.

The parameters θ, together with the model equations, tell us how that plane is divided between the domains.

Both are calculated from related histories, so we're not claiming they're economically independent.

We're separating them mathematically to examine their roles.

Now start with the positive stress

$$x = \max\left(\frac{V_+}{b_+} - 1, 0\right) \tag{209}$$

On the regular boundary, with f, n > 0, the critical stress x⋆ is strictly positive.

So on that boundary

$$x = \frac{V_+}{b_+} - 1 \tag{210}$$

Add 1

$$1 + x = \frac{V_+}{b_+} \tag{211}$$

Multiply by b_+

$$\boxed{V_+ = b_+(1 + x)} \tag{212}$$

Now, for the second leg

$$y = \max\left(\frac{V_-}{b_-} - 1, 0\right) \tag{213}$$

Therefore

$$\boxed{1 + y = \max\left(1, \frac{V_-}{b_-}\right)} \tag{214}$$

Now take our critical stress

$$x^\star = \frac{\sqrt{1 + 4(f/n)(1 + y)} - 1}{2} \tag{215}$$

Substitute it into

$$V_+ = b_+(1 + x^\star) \tag{216}$$

We obtain

$$V_+ = b_+ \left[1 + \frac{\sqrt{1 + 4(f/n)(1 + y)} - 1}{2}\right] \tag{217}$$

Put the terms in the brackets over the same denominator

$$1 + x^\star = \frac{1 + \sqrt{1 + 4(f/n)(1 + y)}}{2} \tag{218}$$

Finally, substitute the expression for 1 + y

$$\boxed{C_\theta(V_-) = \frac{b_+}{2}\left[1 + \sqrt{1 + 4\frac{f}{n}\max\left(1, \frac{V_-}{b_-}\right)}\right]} \tag{219}$$

That's the exact regular Janus boundary in the variance plane!

C_θ(V_) is the critical boundary height for a chosen horizontal coordinate V_ under fixed parameters θ.

The fragile domain consists of points below that height

$$\boxed{F_\theta = \{(V_-, V_+) : V_+ < C_\theta(V_-)\}} \tag{220}$$

The antifragile domain consists of points above it

$$\boxed{A_\theta = \{(V_-, V_+) : V_+ > C_\theta(V_-)\}} \tag{221}$$

And the common regular boundary is

$$\boxed{\Gamma_\theta = \{(V_-, V_+) : V_+ = C_\theta(V_-)\}} \tag{222}$$

The Greek letter Γ, is simply the name assigned to this boundary.

Now look at the maximum inside the square root.

As long as

$$V_- \leq b_- \tag{223}$$

the maximum is 1.

So the critical height doesn't yet depend on V_.

The boundary begins with a horizontal section!

When

$$V_- > b_- \tag{224}$$

the ratio V_/b_becomes larger than 1.

The boundary starts rising with the second leg's variance.

It remains continuous where the formula changes, but its slope can change at that point because the maximum switches between its two branches.

That shape is not an arbitrary curve drawn by hand.

It follows from the equations defining the masses, references, and stress coupling.

And remember what this map represents.

Its axes are variances, not price levels.

It isn't an impenetrable tunnel that forces the price to remain inside physical walls.

It's a map of states used to classify the model's two dominance conditions.

30 Fixed states under changing frameworks

Next, return to the historical example where the point remains fixed.

We use the Beta-to-Janus data from 16 and 17 March 2016.

The point observed on 16 March is

$$\boxed{z_{16} = (0.001223127171,\ 0.061325982523)} \tag{225}$$

The point observed on 17 March is

$$\boxed{z_{17} = (0.001218962154,\ 0.061554572102)} \tag{226}$$

The 16 March parameters are

$$f_{16} \simeq 10.292917212, \qquad n_{16} \simeq 10.115889759 \tag{227}$$

$$b_{+,16} \simeq 0.034559422519, \qquad b_{-,16} \simeq 0.000967954482 \tag{228}$$

The 17 March parameters are

$$f_{17} \simeq 9.996732251, \qquad n_{17} \simeq 8.985637079 \tag{229}$$

$$b_{+,17} \simeq 0.034643370010, \qquad b_{-,17} \simeq 0.000967954482 \tag{230}$$

Now, under the framework of 16 March, the critical height at the horizontal coordinate of z_{16} is approximately

$$C_{16}(V_-) \simeq 0.0601073817 \tag{231}$$

But the observed point has vertical coordinate

$$V_+ \simeq 0.0613259825 \tag{232}$$

The point is above the boundary

$$0.0613259825 > 0.0601073817 \tag{233}$$

So it belongs to the antifragile-dominance domain.

The balance is

$$D_{16}(z_{16}) \simeq -0.399878 \tag{234}$$

Now here's what We notice.

I keep exactly the same point!

I don't change V_-.

I don't change V_+.

I only replace the framework parameters with those of 17 March.

The new boundary height becomes

$$C_{17}(V_-) \simeq 0.0619001448 \tag{235}$$

Now compare the three numbers

$$\boxed{0.0601073817 < 0.0613259825 < 0.0619001448} \tag{236}$$

The first is the old boundary height.

The second is the fixed point's height.

The third is the new boundary height.

The point lies between the two boundaries.

Under the old framework, it was above the boundary.

Under the new framework, it is below it.

Its balance becomes

$$\boxed{D_{17}(z_{16}) \simeq +0.170236} \tag{237}$$

The point didn't move. the boundary moved around it!

But now we can ask a more precise question.

The framework contains the masses and the variance references.

Which part of the parameter change was sufficient to reverse the classification?

We separate them.

I keep the point from 16 March fixed and calculate four combinations

Parameters at the fixed point	Balance D	Domain
16 March masses and references	-0.399878	Antifragile
17 March masses, 16 March references	$+0.125975$	Fragile
16 March masses, 17 March references	-0.351365	Antifragile
17 March masses and references	$+0.170236$	Fragile

Look at the second row.

I changed only the masses.

The reference values remained those of 16 March.

And the balance became positive!

Now look at the third row.

I changed only the references.

The balance remained negative.

Therefore, in this example, changing the masses alone is sufficient to reclassify the fixed point, while changing only the references is not.

Why did the masses matter so much?

Look at their ratio.

On 16 March

$$\frac{f_{16}}{n_{16}} \simeq 1.017500 \tag{238}$$

On 17 March

$$\frac{f_{17}}{n_{17}} \simeq 1.112523 \tag{239}$$

The ratio increased.

Earlier, we proved that a higher f/n increases the critical stress threshold.

The antifragile magnitude decreased proportionally more than the fragile mass.

So the boundary moved in exactly the direction predicted by the derivative.

31 Reclassification regions and moving observations

Next, separate something else.

The actual observed point on 17 March is not identical to the one on 16 March.

So we should distinguish changing the observed point from changing the framework.

Here are all four point–framework combinations

Framework	Observed point	Balance D
16 March	16 March	-0.399878
16 March	17 March	-0.495736
17 March	16 March	$+0.170236$
17 March	17 March	$+0.083469$

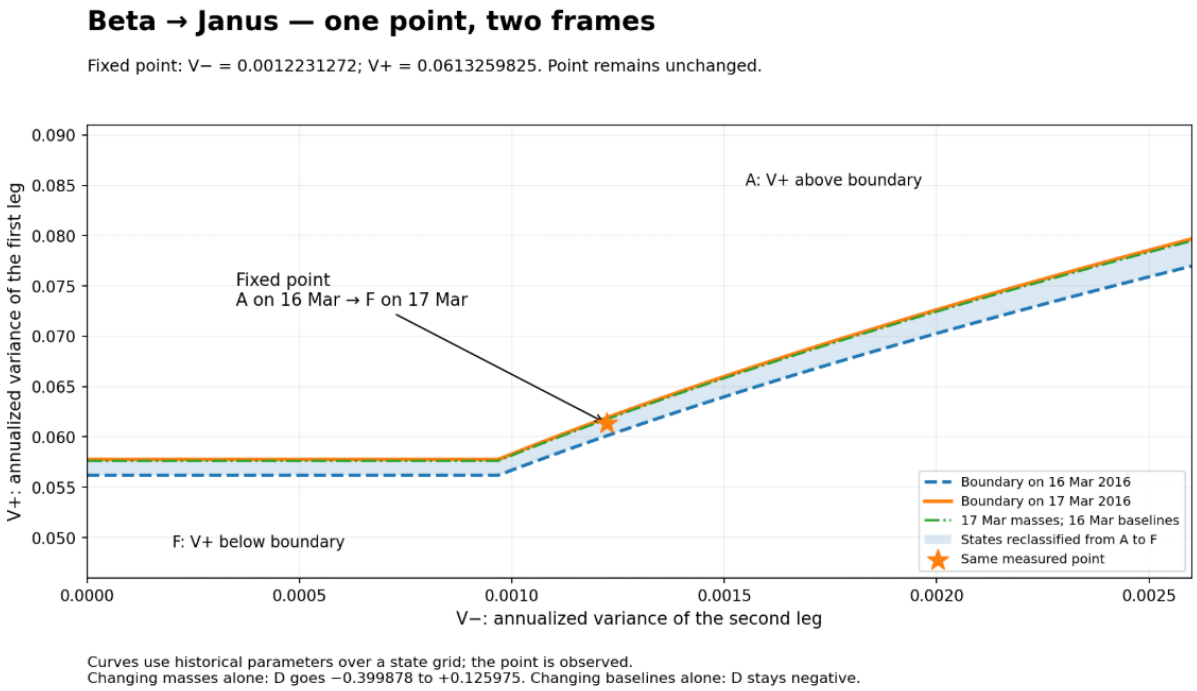


Figure 9: A fixed state evaluated under two successive frameworks. Changing the masses and reference values can reverse dominance without moving the state.

Changing only the point under the old framework preserves a negative balance.

Changing only the framework while keeping the old point produces a positive balance.

In this comparison, the framework change is sufficient to reverse dominance. the point movement alone isn't!

Now we can describe not only the historical point but the whole region reclassified by the movement of the boundary.

Define

$$\boxed{\mathcal{Z}_{16\to 17} = \{(V_-, V_+) : C_{16}(V_-) < V_+ < C_{17}(V_-)\}} \tag{240}$$

This is the region above the old boundary and below the new one.

So the change doesn't concern just one historical observation.

An entire area of the state space can change domains when the framework is deformed!

However, this comparison is algebraic.

The parameters come from historical observations, and they are not independent of the market conditions used to estimate them.

Holding a point fixed lets us isolate the effect of changing the parameters in the model.

It doesn't establish that the masses physically changed independently of the variances in the real market.

Next, verify that the opposite type of transition can occur.

For AAPL/GLD on 14 and 15 August 2024, the retained masses change from

$$(f_{14}, n_{14}) \simeq (0.513172,\ 0.590107) \tag{241}$$

to

$$(f_{15}, n_{15}) \simeq (0.444246,\ 1.166296) \tag{242}$$

The variance references change from

$$(b_{+,14}, b_{-,14}) \simeq (0.043593100,\ 0.015372422) \tag{243}$$

to

$$(b_{+,15}, b_{-,15}) \simeq (0.043749298,\ 0.015424691) \tag{244}$$

Now keep the 14 August point fixed.
Its positive variance is approximately

$$V_+ \simeq 0.072992448 \tag{245}$$

Under the 14 August framework, the critical height is

$$C_{14} \simeq 0.082575242 \tag{246}$$

The point lies below the boundary.
It belongs to the fragile domain.
Under the 15 August framework, the critical height drops to

$$C_{15} \simeq 0.065391764 \tag{247}$$

Now the same point lies above the boundary.
It belongs to the antifragile domain.
The four balance calculations are

$$D_{\theta_{14}}(z_{14}) \simeq +0.102164 \tag{248}$$

$$D_{\theta_{14}}(z_{15}) \simeq +0.096913 \tag{249}$$

$$D_{\theta_{15}}(z_{14}) \simeq -0.135329 \tag{250}$$

$$D_{\theta_{15}}(z_{15}) \simeq -0.145169 \tag{251}$$

Under the old framework, changing the point alone preserves fragile dominance.
Under the new framework, the fixed old point becomes antifragile.
The two historical examples show opposite transitions.

For the Beta-to-Janus example

$$A \longrightarrow F \tag{252}$$

For AAPL/GLD

$$F \longrightarrow A \tag{253}$$

And in the AAPL/GLD episode, the balance becomes negative on 15 August, but the model confirms the change only on 19 August, after the candidate observations on the 15th, 16th, and 19th.

We'll explain that delay when we study the confirmation mechanism.

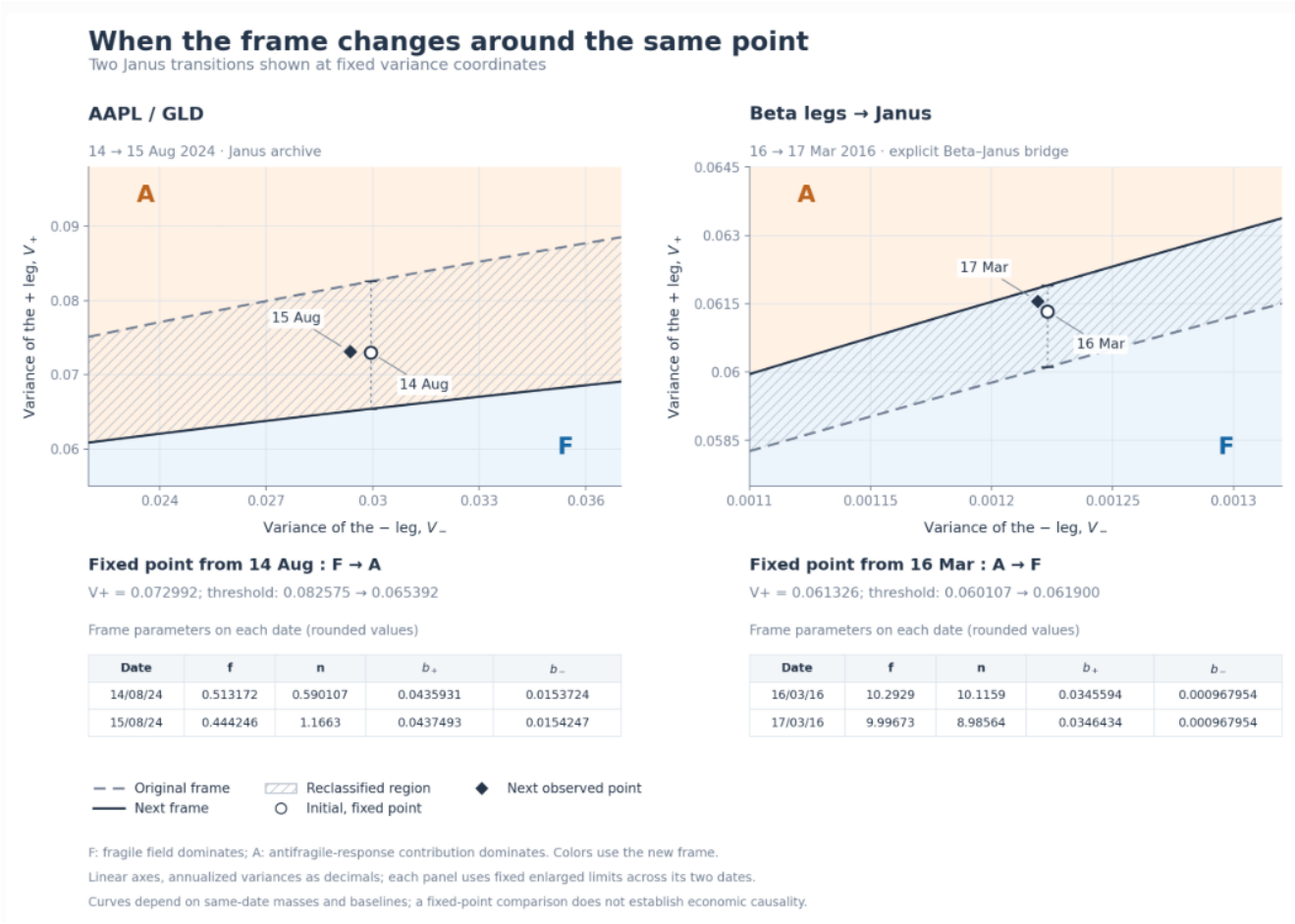


Figure 10: Two fixed-point comparisons illustrating reclassification by a change in the framework.

Differentials, fibers, and metrics

32 The density map and its nonunique inverse

But first, we've got another important geometric question.
We began with four variables

$$(f, n, x, y) \tag{254}$$

The density map returns only two

$$(u, v) \tag{255}$$

What information does this transformation keep, and what does it lose?
That's what we're going to demonstrate with fibers, Jacobians, and transported metrics.
We have the transformation

$$\boxed{\Phi(f, n, x, y) = \left(\frac{f}{1+x}, \frac{nx}{1+y}\right) = (u, v)} \tag{256}$$

The Greek letter Φ, is the name of the map.
It takes four inputs and produces two outputs.
Now imagine I give you

$$u = 1, \qquad v = 2 \tag{257}$$

Can you tell me exactly which masses and stresses produced those two densities?
No! Not from those two numbers alone.
We prove it.
From the first density

$$u = \frac{f}{1+x} \tag{258}$$

multiply by 1 + x

$$\boxed{f = u(1+x)} \tag{259}$$

To recover f, I still need to know x.
From the second

$$v = \frac{nx}{1+y} \tag{260}$$

multiply by 1 + y

$$v(1+y) = nx \tag{261}$$

If x > 0, divide by x

$$\boxed{n = \frac{v(1+y)}{x}} \tag{262}$$

To recover n, I still need x and y.

So even when u and v are fixed, I can choose different values of x, y and adjust f, n to obtain exactly the same densities.

We check it with two states.

First

$$(f, n, x, y) = (2, 2, 1, 0) \tag{263}$$

The densities are

$$u = \frac{2}{1+1} = 1 \tag{264}$$

$$v = \frac{2(1)}{1+0} = 2 \tag{265}$$

Now take

$$(f, n, x, y) = (3, 2, 2, 1) \tag{266}$$

The densities are

$$u = \frac{3}{1+2} = 1 \tag{267}$$

$$v = \frac{2(2)}{1+1} = 2 \tag{268}$$

So both states produce

$$\boxed{(u, v) = (1, 2)} \tag{269}$$

Their primitive variables differ, but their balance is identical

$$D = 1 - 2 = -1 \tag{270}$$

The density map doesn't preserve every detail of the primitive state!

The collection of all primitive states that produce the same output is called a fiber.

For fixed positive densities u, v the complete regular fiber is

$$\boxed{\Phi^{-1}(u,v) = \left\{ \left(u(1+x), \frac{v(1+y)}{x}, x, y \right) : x > 0,\ y \geq 0 \right\}} \tag{271}$$

Here Φ^{-1}(u, v) means the preimage of the point (u, v).

It isn't one unique inverse function.

The braces collect all the primitive states producing those densities.

And x, y remain free parameters, subject to their admissible conditions.

There are two free coordinates.

The regular fiber is two-dimensional!

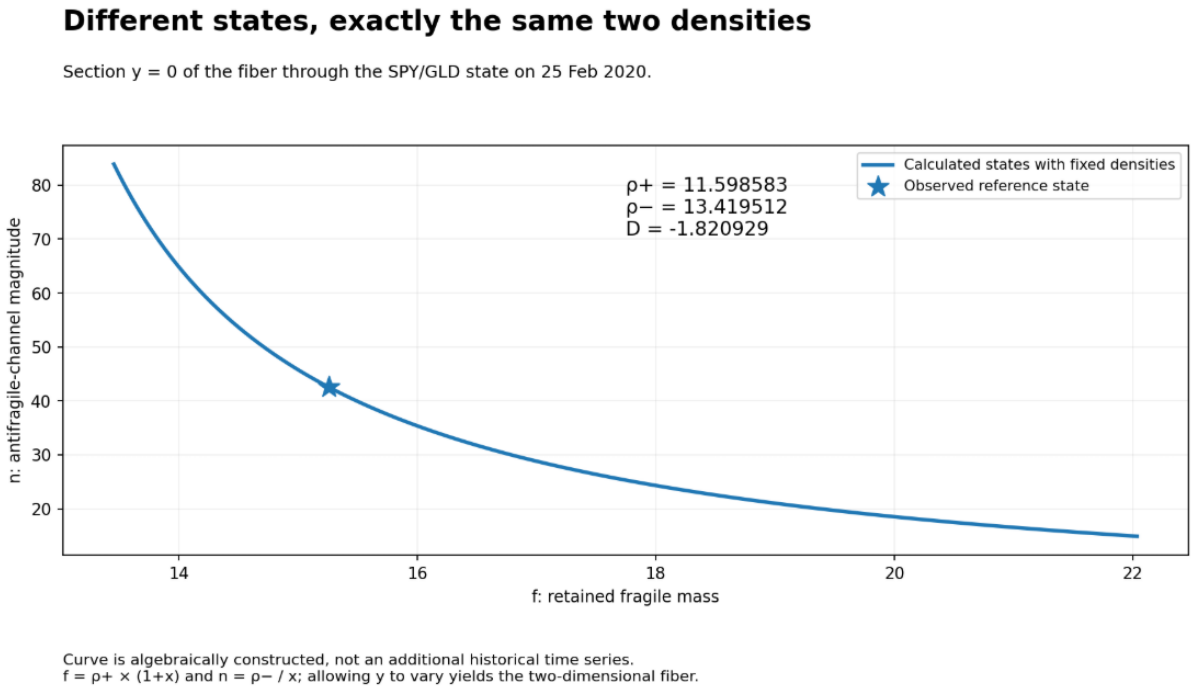


Figure 11: A two-dimensional slice through a density fiber: distinct primitive states can have the same two density coordinates.

This figure shows a section of a fiber constructed from the SPY/GLD densities on 25 February 2020.

It's an algebraic construction, not another historical time series.

33 Jacobian, rank, and invisible tangent directions

Now We measure this information loss locally.

For that, we need a Jacobian.

A Jacobian collects the partial derivatives of several outputs with respect to several inputs.

A partial derivative describes what happens to a function when one input changes while the others remain fixed.

Our map has two outputs

$$u = \frac{f}{1+x} \tag{272}$$

$$v = \frac{nx}{1+y} \tag{273}$$

And four inputs f, n, x, y.

We calculate the derivatives of u.

With respect to f

$$\frac{\partial u}{\partial f} = \frac{1}{1+x} \tag{274}$$

With respect to n

$$\frac{\partial u}{\partial n} = 0 \tag{275}$$

Because n doesn't appear in the first density.

With respect to x

$$\frac{\partial u}{\partial x} = -\frac{f}{(1+x)^2} \tag{276}$$

And with respect to y

$$\frac{\partial u}{\partial y} = 0 \tag{277}$$

Now for v

$$\frac{\partial v}{\partial f} = 0 \tag{278}$$

$$\frac{\partial v}{\partial n} = \frac{x}{1+y} \tag{279}$$

$$\frac{\partial v}{\partial x} = \frac{n}{1+y} \tag{280}$$

$$\frac{\partial v}{\partial y} = -\frac{nx}{(1+y)^2} \tag{281}$$

Put these derivatives into a matrix

$$\boxed{J = \begin{pmatrix} \dfrac{1}{1+x} & 0 & -\dfrac{f}{(1+x)^2} & 0 \\ 0 & \dfrac{x}{1+y} & \dfrac{n}{1+y} & -\dfrac{nx}{(1+y)^2} \end{pmatrix}} \tag{282}$$

A matrix is simply an organized table of numbers or expressions.

Each row corresponds to an output.

Each column corresponds to an input.

The first row shows how u responds to f, n, x, y.

The second shows how v responds.

Now, what does the rank tell us?

It tells us how many independent output directions the map can produce locally.

The Jacobian has two rows, so its rank cannot exceed 2.

But when is it actually 2?

We examine two 2×2 minors.

The first uses the columns corresponding to f, n

$$\boxed{\det J_{(f,n)} = \frac{x}{(1+x)(1+y)}} \tag{283}$$

For a matrix

$$\begin{pmatrix} ab \\ cd \end{pmatrix} \tag{284}$$

the determinant is ad − bc.

Our first minor has nonzero determinant when $x > 0$.

Now use the columns corresponding to f, x

$$\boxed{\det J_{(f,x)} = \frac{n}{(1+x)(1+y)}} \tag{285}$$

This one has nonzero determinant when $n > 0$.

So whenever $n > 0$ or $x > 0$, at least one of these minors is nonzero.

The rank is 2.

But if

$$n = x = 0 \tag{286}$$

the second row of the Jacobian vanishes.

The first row doesn't vanish because

$$\frac{1}{1+x} > 0 \tag{287}$$

So the rank is 1.

We obtain

$$\boxed{\operatorname{rang} J = \begin{cases} 2, & n > 0 \text{ or } x > 0, \\ 1, & n = x = 0. \end{cases}} \tag{288}$$

The notation rang is retained from the original mathematics; it means rank.

Our map has four inputs, but can transmit at most two independent output directions!

Next, find the directions it cannot see.

Start with the total differentials.

For u

$$\boxed{du = \frac{df}{1+x} - \frac{f\,dx}{(1+x)^2}} \tag{289}$$

For v

$$\boxed{dv = \frac{x\,dn + n\,dx}{1+y} - \frac{nx\,dy}{(1+y)^2}} \tag{290}$$

The symbols df, dn, dx, dy represent infinitesimal changes in the four primitive variables.

The numeratorx dn + n dx appears because the derivative of the product nxis

$$d(nx) = x\,dn + n\,dx \tag{291}$$

Now, an invisible direction is a change that satisfies

$$du = 0, \qquad dv = 0 \tag{292}$$

Start with du = 0

$$\frac{df}{1+x} - \frac{f\,dx}{(1+x)^2} = 0 \tag{293}$$

Multiply by $(1 + x)^2$

$$(1+x)df - f\,dx = 0 \tag{294}$$

Therefore

$$\boxed{df = \frac{f}{1+x}\,dx} \tag{295}$$

So a change in x can be compensated by a suitable change in f, keeping u fixed.

Now impose dv = 0

$$\frac{x\,dn + n\,dx}{1+y} - \frac{nx\,dy}{(1+y)^2} = 0 \tag{296}$$

Multiply by $(1 + y)^2$

$$(1+y)(x\,dn + n\,dx) - nx\,dy = 0 \tag{297}$$

Expand

$$x(1+y)dn + n(1+y)dx - nx\,dy = 0 \tag{298}$$

Solve for dn, assuming $x > 0$

$$\boxed{dn = -\frac{n}{x}\,dx + \frac{n}{1+y}\,dy} \tag{299}$$

Those two equations tell us how the masses can compensate changes in the stresses while preserving the densities.

Now choose a first invisible direction by letting x vary and keeping y fixed

$$\boxed{k_x = \begin{pmatrix} \dfrac{f}{1+x} \\ -\dfrac{n}{x} \\ 1 \\ 0 \end{pmatrix}} \tag{300}$$

Choose a second direction by letting y vary while keeping x fixed

$$\boxed{k_y = \begin{pmatrix} 0 \\ \dfrac{n}{1+y} \\ 0 \\ 1 \end{pmatrix}} \tag{301}$$

The four entries of each vector correspond to the variations of f, n, x, y.

And when we multiply by the Jacobian

$$\boxed{Jk_x = 0, \qquad Jk_y = 0} \tag{302}$$

So both directions are invisible to the density outputs, to first order.

They are tangent to the regular fibers.

34 The pullback metric and quotient coordinates

Next, talk about a metric.

A metric is a rule for measuring lengths, distances, and angles.

In the ordinary Euclidean plane, moving 3 units horizontally and 4 units vertically gives a displacement of length 5.

That's Pythagoras

$$3^2 + 4^2 = 5^2 \tag{303}$$

In the density plane, we'll use the same Euclidean rule.

For a small displacement (du, dv)

$$\boxed{ds_\rho^2 = du^2 + dv^2} \tag{304}$$

It says:

"The squared length of the displacement is the sum of the squares of its two components."

Now substitute the differentials of the densities

$$\boxed{\begin{aligned} ds_\rho^2 = &\left(\frac{df}{1+x} - \frac{f\,dx}{(1+x)^2}\right)^2 \\ &+ \left(\frac{x\,dn + n\,dx}{1+y} - \frac{nx\,dy}{(1+y)^2}\right)^2 \end{aligned}} \tag{305}$$

Look at what this means.

The primitive state can move in four dimensions while producing

$$du = dv = 0 \tag{306}$$

In that case

$$ds_\rho^2 = 0 \tag{307}$$

Even though the primitive variables changed!

That is exactly what happens along a fiber.

In matrix notation, the transported metric is

$$\boxed{g_4 = J^\top J} \tag{308}$$

The symbol J^T means the transpose: rows become columns and columns become rows.

Because J has two rows and four columns, J^TJ is a 4×4 matrix.

Why this product?

For a primitive displacement w, the induced density displacement is Jw.

Its squared Euclidean length is

$$\|Jw\|^2 \tag{309}$$

And

$$\boxed{w^\top g_4 w = \|Jw\|^2 \geq 0} \tag{310}$$

The result cannot be negative.

But it can be zero for a nonzero w, whenever Jw = 0.

Therefore

$$\boxed{\ker g_4 = \ker J, \qquad \operatorname{rang} g_4 = \operatorname{rang} J} \tag{311}$$

The symbol ker denotes the kernel: all vectors sent to zero.

Since the rank is at most 2 in a 4×4 matrix

$$\boxed{\det g_4 = 0 \quad \text{everywhere}} \tag{312}$$

The metric is degenerate.

In the regular sector, two directions are measured and two are null.

At n = x = 0, the ambient rank drops to 1, leaving three null directions.

This transported metric is not a nondegenerate four-dimensional riemannian or lorentzian metric!

It has no ordinary four-dimensional matrix inverse, so it does not directly provide the usual nondegenerate four-dimensional Levi-Civita connection.

But that doesn't mean the construction has no geometry.

It means this metric measures the two density directions and ignores movements inside their fibers.

In the regular sector x > 0, we can use

$$(u, v, x, y) \tag{313}$$

instead of

$$(f, n, x, y) \tag{314}$$

The inverse relations are

$$f = u(1 + x), \qquad n = \frac{v(1 + y)}{x} \tag{315}$$

In these coordinates

$$ds^2_\rho = du^2 + dv^2 \tag{316}$$

So

$$\boxed{[g_4] = \mathrm{diag}(1, 1, 0, 0)} \tag{317}$$

The two 1s correspond to measured directions.

The two zeros correspond to invisible directions.

This is the geometry of the quotient: primitive states giving the same densities are identified.

Now there's a singular case we must preserve.

When

$$v = 0 \tag{318}$$

the equation

$$v = \frac{nx}{1+y} \tag{319}$$

can vanish because n = 0, because x = 0, or both.

These are different branches meeting at n = x = 0.

So we cannot pretend that the complete map forms one globally regular fiber bundle.

A three-dimensional kernel at a singular point also does not automatically mean the corresponding fiber is a regular three-dimensional manifold.

Differential rank and singular fiber geometry are two different questions!

35 Fixed-mass slices

Next, change the problem.

Instead of allowing the masses to vary, I'll fix them

$$f, n > 0 \tag{320}$$

Only the stresses x > 0, y ≥ 0 can move.

The transformation is still

$$u = \frac{f}{1+x}, \qquad v = \frac{nx}{1+y} \tag{321}$$

But now it has two inputs and two outputs.

The Jacobian restricted to the stresses is

$$\boxed{J_{xy} = \begin{pmatrix} -\dfrac{f}{(1+x)^2} & 0 \\ \dfrac{n}{1+y} & -\dfrac{nx}{(1+y)^2} \end{pmatrix}} \tag{322}$$

Its determinant is

$$\boxed{\det J_{xy} = \frac{fnx}{(1+x)^2(1+y)^2} > 0} \tag{323}$$

Every factor in the numerator is positive in the active sector.

So this restricted map is locally invertible.

We can even find the inverse explicitly.

From

$$u = \frac{f}{1+x} \tag{324}$$

solve for x

$$\boxed{x = \frac{f}{u} - 1} \tag{325}$$

Then from

$$v = \frac{nx}{1+y} \tag{326}$$

solve for y

$$\boxed{y = \frac{n(f/u - 1)}{v} - 1} \tag{327}$$

Now check the admissible values.

The condition x > 0 gives

$$0 < u < f \tag{328}$$

And y ≥ 0 gives

$$0 < v \leq n\left(\frac{f}{u} - 1\right) \tag{329}$$

So the exact image of this fixed-mass slice is

$$\boxed{\Omega_{f,n} = \left\{0 < u < f, \quad 0 < v \leq n\left(\frac{f}{u} - 1\right)\right\}} \tag{330}$$

The equality corresponds to the boundary y = 0.

The image of the fixed-mass stress slice is not the entire positive density quadrant!

Next, transport the Euclidean density metric onto this two-dimensional stress slice.

Since the masses are fixed

$$du = -\frac{f}{(1+x)^2}\,dx \tag{331}$$

$$dv = \frac{n}{1+y}\,dx - \frac{nx}{(1+y)^2}\,dy \tag{332}$$

Substitute into

$$ds^2 = du^2 + dv^2 \tag{333}$$

And expand the squares using

$$(a-b)^2 = a^2 - 2ab + b^2 \tag{334}$$

After collecting the coefficients, we obtain

$$\boxed{g_{xx} = \frac{f^2}{(1+x)^4} + \frac{n^2}{(1+y)^2}} \tag{335}$$

$$\boxed{g_{xy} = g_{yx} = -\frac{n^2 x}{(1+y)^3}} \tag{336}$$

$$\boxed{g_{yy} = \frac{n^2 x^2}{(1+y)^4}} \tag{337}$$

The metric is

$$ds^2 = g_{xx}\,dx^2 + 2g_{xy}\,dx\,dy + g_{yy}\,dy^2 \tag{338}$$

The cross term cannot be discarded. It represents the combined effect of changing the two stresses.
Now calculate its determinant

$$\begin{aligned} \det g_2 &= g_{xx}g_{yy} - g_{xy}^2 \\ &= \left[\frac{f^2}{(1+x)^4} + \frac{n^2}{(1+y)^2}\right]\frac{n^2x^2}{(1+y)^4} - \frac{n^4x^2}{(1+y)^6} \\ &= \boxed{\frac{f^2n^2x^2}{(1+x)^4(1+y)^4} > 0} \end{aligned} \tag{339}$$

The two terms involving n^4x^2 cancel.
So this two-dimensional metric is nondegenerate for f, n, $x > 0$.
It remains regular at $y = 0$.
But when $x = 0$, the determinant vanishes.
That's because all values of y then produce

$$(u, v) = (f, 0) \tag{340}$$

One direction has become invisible.

Now, does crossing D = 0 make the metric degenerate?

No!

The determinant remains positive whenever f, n, x > 0, including on the regular dominance boundary.

Crossing a dominance boundary is not a collapse of the metric!

36 Flatness, boundary distance, and local stability

Next, study curvature.

The boundary in the Euclidean stress plane is

$$y = \frac{n}{f}\, x(1 + x) - 1 \tag{341}$$

Its first derivative is

$$y' = \frac{n}{f}(1 + 2x) \tag{342}$$

Its second derivative is

$$y'' = \frac{2n}{f} \tag{343}$$

For a Euclidean curve y = Y(x), curvature is

$$\kappa(x) = \frac{|Y''(x)|}{[1 + Y'(x)^2]^{3/2}} \tag{344}$$

Substituting our derivatives

$$\boxed{\kappa_{\text{stress}}(x) = \frac{2n/f}{[1 + (n/f)^2(1 + 2x)^2]^{3/2}} > 0} \tag{345}$$

The boundary has positive Euclidean curvature in stress coordinates.

But in the metric transported from the density plane, the same boundary becomes

$$u = v \tag{346}$$

That's a straight line with zero geodesic curvature.

The transported metric is flat because it comes from a locally invertible transformation into the Euclidean plane.

Are those results contradictory?

Not at all.

We're using different metrics to measure curvature!

Positive Euclidean curvature in the stress plane and zero geodesic curvature under the transported metric can both be correct.

Next, measure the distance to the density boundary.

Take a point

$$(u, v) \tag{347}$$

The boundary is

$$u = v \tag{348}$$

The orthogonal projection onto the diagonal is

$$\left(\frac{u+v}{2}, \frac{u+v}{2}\right) \tag{349}$$

Why?

Because the point (a, a) minimizing

$$(u-a)^2 + (v-a)^2 \tag{350}$$

satisfies

$$a = \frac{u+v}{2} \tag{351}$$

Now calculate the squared distance

$$\left(\frac{u-v}{2}\right)^2 + \left(\frac{v-u}{2}\right)^2 = \frac{(u-v)^2}{2} \tag{352}$$

Taking the square root

$$\boxed{d_\rho = \frac{|u-v|}{\sqrt{2}} = \frac{|D|}{\sqrt{2}}} \tag{353}$$

This is an exact Euclidean distance in density coordinates.

It's not a monetary distance or a probability of loss.

Now define two new coordinates

$$\boxed{\xi = \frac{u-v}{\sqrt{2}}, \qquad \chi = \frac{u+v}{\sqrt{2}}} \tag{354}$$

ξ is the normal coordinate: it measures signed dominance.

χ is the tangential coordinate: it measures the common level of the contributions.

The inverse is

$$u = \frac{\chi + \xi}{\sqrt{2}}, \qquad v = \frac{\chi - \xi}{\sqrt{2}} \tag{355}$$

Because u, v ≥ 0

$$\boxed{\chi \geq |\xi|} \tag{356}$$

And the Euclidean metric is preserved

$$du^2 + dv^2 = d\xi^2 + d\chi^2 \tag{357}$$

We've rotated the axes without changing lengths.

Now consider a perturbation (δu, δv).

The balance changes by

$$\delta D = \delta u - \delta v \tag{358}$$

By the Cauchy–Schwarz inequality

$$\boxed{|\delta D| \leq \sqrt{2}\sqrt{\delta u^2 + \delta v^2}} \tag{359}$$

So if

$$\boxed{\sqrt{\delta u^2 + \delta v^2} < d_\rho} \tag{360}$$

the perturbation is too small to reverse the balance sign.

That's a sufficient geometric stability condition, not a probability of loss.

And because Ω_(f,n) isn't globally convex, the distance to the diagonal in the full quadrant isn't automatically the intrinsic geodesic distance to the reachable boundary inside the constrained fixed-mass slice.

37 Log-density coordinates and stratification

Next, simplify the boundary another way.

When both densities are positive, consider their ratio

$$\frac{u}{v} \tag{361}$$

If it's above 1, the fragile contribution dominates.

If it's below 1, the antifragile contribution dominates.

Now take the logarithm

$$\boxed{\zeta = \ln\left(\frac{u}{v}\right) = \ln\left(\frac{f(1+y)}{nx(1+x)}\right)} \tag{362}$$

The Greek letter ζ because $\ln(1) = 0$, the boundary becomes

$$\zeta = 0 \tag{363}$$

The logarithm is increasing, so the sign of ζ is the sign of dominance.
Since

$$e^{\zeta} = \frac{u}{v} \tag{364}$$

we get

$$u = ve^{\zeta} \tag{365}$$

The balance becomes

$$\boxed{D = v(e^{\zeta} - 1)} \tag{366}$$

We can also solve for f

$$\boxed{f = \frac{nx(1+x)}{1+y}e^{\zeta}} \tag{367}$$

So (ζ, n, x, y) gives another coordinate system for the active sector f, n, x > 0, y ≥ 0.
The boundary becomes $\zeta = 0$.
The two strict domains become the regions $\zeta > 0$ and $\zeta < 0$.
They are path-connected and contractible in this regular representation.
But this logarithmic coordinate is undefined when one of the densities is zero.
We can't take the logarithm of zero.
So we have to keep the zero-mass and zero-stress cases.
And that's where the stratified geometry comes in.
A stratum is a portion of the state space where a particular set of constraints holds.
For example, one stratum may contain states where all the relevant masses and stresses are positive.
Another may contain states where a mass is exactly zero.
Another may contain states where the activating stress is zero.
When one of these variables vanishes, an entire density contribution may disappear.
And that changes the number of independent directions available within the stratum.
We begin with f, n, x.

Each one can be zero or strictly positive.

Two possibilities for each of three variables gives:

$$2 \times 2 \times 2 = 8 \tag{368}$$

So we have eight families.

Each family can also be subdivided according to whether y = 0 or y > 0.

That gives 16 strata in total.

In the table, + means strictly positive.

The reported rank is the rank of the map restricted to that stratum.

A variable fixed at zero isn't allowed to move away from that face when calculating the restricted rank.

(f, n, x)	Densities	Balance D	Rank
$(+,+,+)$	$u > 0, v > 0$	Any sign	2
$(+,+,0)$	$u = f, v = 0$	Positive	1
$(+,0,+)$	$u = f/(1+x), v = 0$	Positive	1
$(+,0,0)$	$u = f, v = 0$	Positive	1
$(0,+,+)$	$u = 0, v > 0$	Negative	1
$(0,+,0)$	$u = v = 0$	Zero	0
$(0,0,+)$	$u = v = 0$	Zero	0
$(0,0,0)$	$u = v = 0$	Zero	0

We read the first row.

Both masses and the activating stress are positive.

Therefore

$$u = \frac{f}{1+x} > 0 \tag{369}$$

$$v = \frac{nx}{1+y} > 0 \tag{370}$$

Both contributions exist.

The balance can be positive, negative, or zero depending on their relative magnitudes.

The restricted map has rank 2.

Now look at the second row.

We have

$$f > 0, \qquad n > 0, \qquad x = 0 \tag{371}$$

The densities become

$$u = f, \qquad v = 0 \tag{372}$$

The balance is

$$D = f > 0 \tag{373}$$

38 Restricted rank, ambient rank, and zero-balance faces

The antifragile contribution cannot dominate in this stratum.

And because v remains identically zero when we stay on the face x = 0, it contributes no independent variation direction.

The restricted rank is 1.

The same reasoning applies when n = 0, even if x > 0.

Now consider

$$f = 0, \qquad n > 0, \qquad x > 0 \tag{374}$$

Then

$$u = 0 \tag{375}$$

$$v = \frac{nx}{1+y} > 0 \tag{376}$$

$$v = \frac{nx}{1+y} > 0 \tag{377}$$

So

$$D = -v < 0 \tag{378}$$

The antifragile contribution necessarily dominates.

Finally, in the last three families

$$f = 0, \qquad nx = 0 \tag{379}$$

Both densities vanish

$$u = v = 0 \tag{380}$$

Therefore

$$\boxed{D \equiv 0} \tag{381}$$

The symbol ≡ indicates that the equality holds throughout the entire stratum.

It isn't merely an equality at one isolated point.

Now we can summarize the restricted rank

$$\boxed{\operatorname{rang} D(\Phi|_{\text{strate}}) = \mathbf{1}_{\{f>0\}} + \mathbf{1}_{\{n>0,\ x>0\}}} \tag{382}$$

The notation Φ|_strate means the transformation restricted to the stratum.

The symbol **1** is an indicator: it equals 1 when its condition holds and 0 otherwise.

If both density channels can vary, the rank is 2.

If only one can vary, it's 1.

If both are identically zero on the stratum, the restricted rank is zero.

But this is not the same thing as the ambient rank of the Jacobian.

In the ambient space, we can vary all four primitive coordinates.

On a stratum where f = 0 is imposed, changing f is no longer an admissible tangent direction.

That's why the two ranks can differ.

Now, the global zero-balance equation remains

$$\boxed{f = \frac{nx(1+x)}{1+y}} \tag{383}$$

But in a zero-density stratum, that equality holds everywhere.

It doesn't divide that stratum into a positive side and a negative side.

Next, examine the corner

$$f = n = x = 0 \tag{384}$$

Take a small positive number ϵ.

Approach the corner using

$$(f, n, x) = (\epsilon, 0, 0) \tag{385}$$

The balance is

$$D = \epsilon > 0 \tag{386}$$

So positive-balance states can approach the corner.

Now approach it using

$$(f, n, x) = (0, \epsilon, \epsilon) \tag{387}$$

The balance becomes

$$\boxed{D = -\frac{\epsilon^2}{1+y} < 0} \tag{388}$$

So negative-balance states can approach the same corner.

Notice the square ϵ^2.

The negative contribution appears at second order.

If we examined only the gradient tangent to the zero stratum, we'd miss that behavior.

That's why corners and approach directions matter.

Next, distinguish variables before clipping from the nonnegative variables after clipping.

Recall

$$f = \max(m_+, 0) \tag{389}$$

$$n = \max(-m_-, 0) \tag{390}$$

$$x = \max(\widetilde{s}_+, 0) \tag{391}$$

Suppose

$$m_+ < 0 \tag{392}$$

Then f = 0.
Now suppose either

$$m_- > 0 \tag{393}$$

so n = 0, or

$$\widetilde{s}_+ < 0 \tag{394}$$

so x = 0.
Then nx = 0.
The combined conditions

$$\boxed{m_+ < 0 \quad \text{and} \quad (m_- > 0 \text{ or } \widetilde{s}_+ < 0)} \tag{395}$$

therefore imply

$$f = 0, \qquad nx = 0, \qquad D = 0 \tag{396}$$

39 Clipping plateaus and empirical strata

These are strict inequalities.
We can change the raw variables slightly without necessarily changing the truth of those inequalities.
Entire open neighborhoods of raw inputs can therefore produce the same zero balance after clipping.
These are open zero-balance plateaus in raw-variable space.
But in the already-transformed nonnegative quadrant, the zero strata are lower-dimensional faces.
They aren't open subsets of the full quadrant.
Clipping can erase entire directions of variation!
Next, verify these configurations in the historical data.
After excluding the initialization period, the three pairs contain 7,008 estimable observations.
Their distribution across the eight families is

(f, n, x)	SPY/GLD	QQQ/GLD	AAPL/GLD	Total
$(0, 0, 0)$	35	40	39	114
$(0, 0, +)$	25	29	107	161
$(0, +, 0)$	50	155	231	436
$(0, +, +)$	96	211	285	592
$(+, 0, 0)$	361	373	247	981
$(+, 0, +)$	326	305	354	985
$(+, +, 0)$	852	653	556	2061
$(+, +, +)$	591	570	517	1678

The three families where both densities vanish contain

$$114, \qquad 161, \qquad 436 \tag{397}$$

observations.

Add them

$$\boxed{711 = 114 + 161 + 436} \tag{398}$$

So there are 711 estimated zero balances after initialization.

Now count the restricted ranks.

Rank 2 occurs in the fully active family

$$1,678 \text{ observations} \tag{399}$$

Rank 1 occurs in the families where one positive density direction remains

$$4,619 \text{ observations} \tag{400}$$

Rank 0 occurs in the zero-density strata

$$711 \text{ observations} \tag{401}$$

Check the total

$$1,678 + 4,619 + 711 = 7,008 \tag{402}$$

Now calculate the ambient Jacobian rank.

We proved it equals 1 when

$$n = x = 0 \tag{403}$$

That condition occurs in two families

$$(0, 0, 0) \qquad \text{and} \qquad (+, 0, 0) \tag{404}$$

They contain 1095 observations.

The remaining 5913 have ambient rank 2.

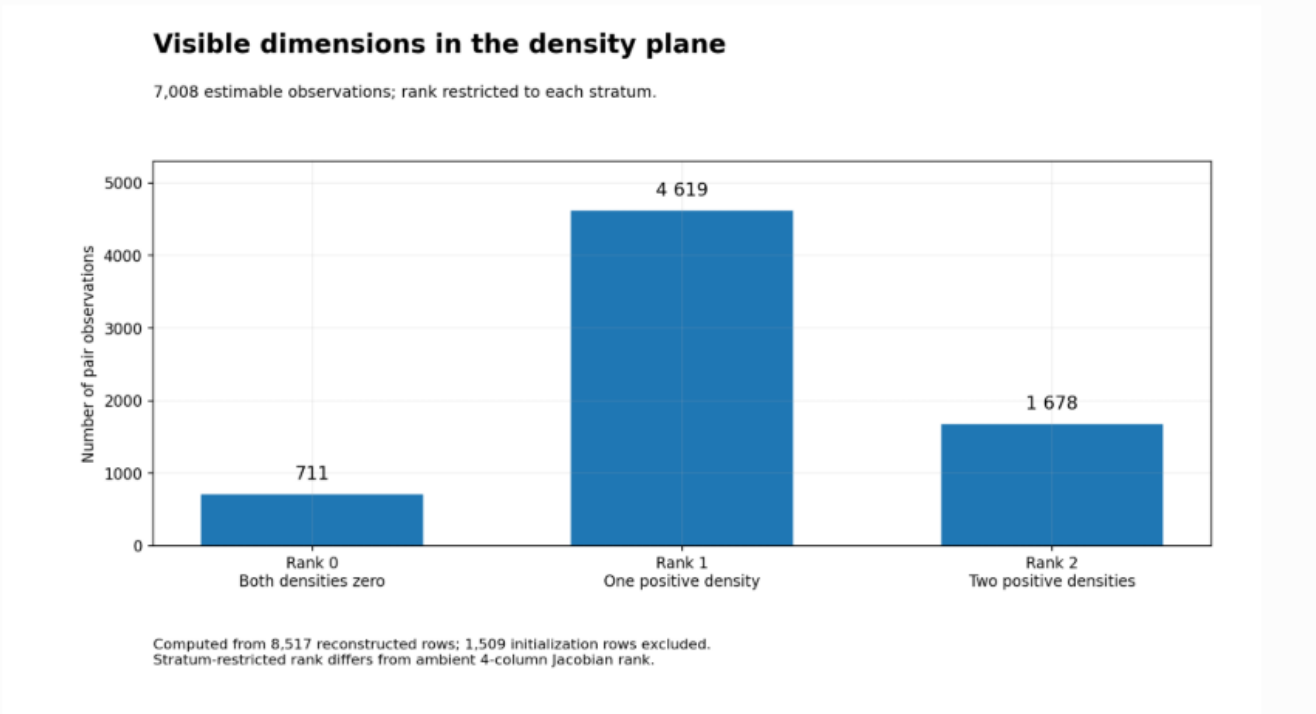


Figure 12: Counts of the three restricted-rank families after initialization: 711 observations of rank zero, 4,619 of rank one, and 1,678 of rank two.

The two counts differ because we're studying different sets of permitted directions.

And here's another distinction.

The 1,678 observations where both densities are positive contain 595 observations with $y = 0$ and 1,083 with $y > 0$.

But there are also observations where f, $n > 0$ while $x = 0$.

In that case

$$v = \frac{n(0)}{1+y} = 0 \tag{405}$$

The antifragile mass is available, but the channel isn't activated.

Two positive masses and two positive densities are not the same event!

40 Opening and disappearance of dominance channels

Next, study what happens when an entire domain disappears.

Suppose

$$f > 0, \qquad n = 0 \tag{406}$$

The balance becomes

$$D = \frac{f}{1+x} - 0 \tag{407}$$

So

$$\boxed{D = \frac{f}{1+x} > 0} \tag{408}$$

This is true for every finite stress x $\geq$ 0.

Therefore

$$\boxed{F = X, \qquad A = \varnothing} \tag{409}$$

The symbol $\varnothing$ means the empty set.

The fragile domain occupies the entire admissible state space, while the strict antifragile-dominance domain is absent.

Now suppose n becomes strictly positive, with f $>$ 0.

Our critical stress

$$x^\star = \frac{\sqrt{1 + 4(f/n)(1+y)} - 1}{2} \tag{410}$$

becomes finite.

The antifragile domain can now exist for sufficiently high activating stress.

What happens as n $\to 0^+$ while f remains fixed?

The ratio f/n grows without bound.

Therefore

$$\boxed{n \to 0^+ \quad \Longrightarrow \quad x^\star \to +\infty} \tag{411}$$

The boundary is pushed beyond every finite stress.

At n = 0, there's no finite antifragile-dominance threshold.

That's demonstrated by the equation, not by the absence of a curve in a limited plot.

We must also preserve the other cases.

If

$$f = 0, \qquad n > 0 \tag{412}$$

then

$$D = -\frac{nx}{1+y} \tag{413}$$

The balance is negative when x $>$ 0, but zero when x = 0.

In variance coordinates, x = 0 means

$$V_+ \leq b_+ \tag{414}$$

So the zero set can occupy an entire band rather than acting as a regular boundary separating two signs.
If

$$f = n = 0 \tag{415}$$

then

$$D = 0 \tag{416}$$

everywhere.
Both strict-dominance domains are empty.
Next, check a historical episode where the antifragile channel becomes available.
On 9 April 2020, in the Beta-to-Janus bridge

$$f_9 \simeq 4.835370 \tag{417}$$

$$n_9 = 0 \tag{418}$$

$$x_9 \simeq 42.712857 \tag{419}$$

That's an extremely high positive stress.
But because n = 0, the antifragile density is zero.
The balance is

$$\boxed{D_9 \simeq +0.110617} \tag{420}$$

Now look at 10 April 2020.
The masses become

$$f_{10} \simeq 4.713758 \tag{421}$$

$$n_{10} \simeq 0.254082 \tag{422}$$

And the activating stress becomes

$$x_{10} \simeq 42.564050 \tag{423}$$

The stress has actually decreased slightly.
But the antifragile channel is now available.
The densities are

$$u_{10} \simeq 0.108203 \tag{424}$$

$$v_{10} \simeq 1.981944 \tag{425}$$

And the balance is

$$\boxed{D_{10} \simeq -1.873742} \tag{426}$$

Stress decreases, but antifragile dominance appears!

Why?

Because the second channel has opened and its contribution now exceeds the first.

This is a numerical counterexample to the idea that high variance alone is sufficient to create an antifragile domain.

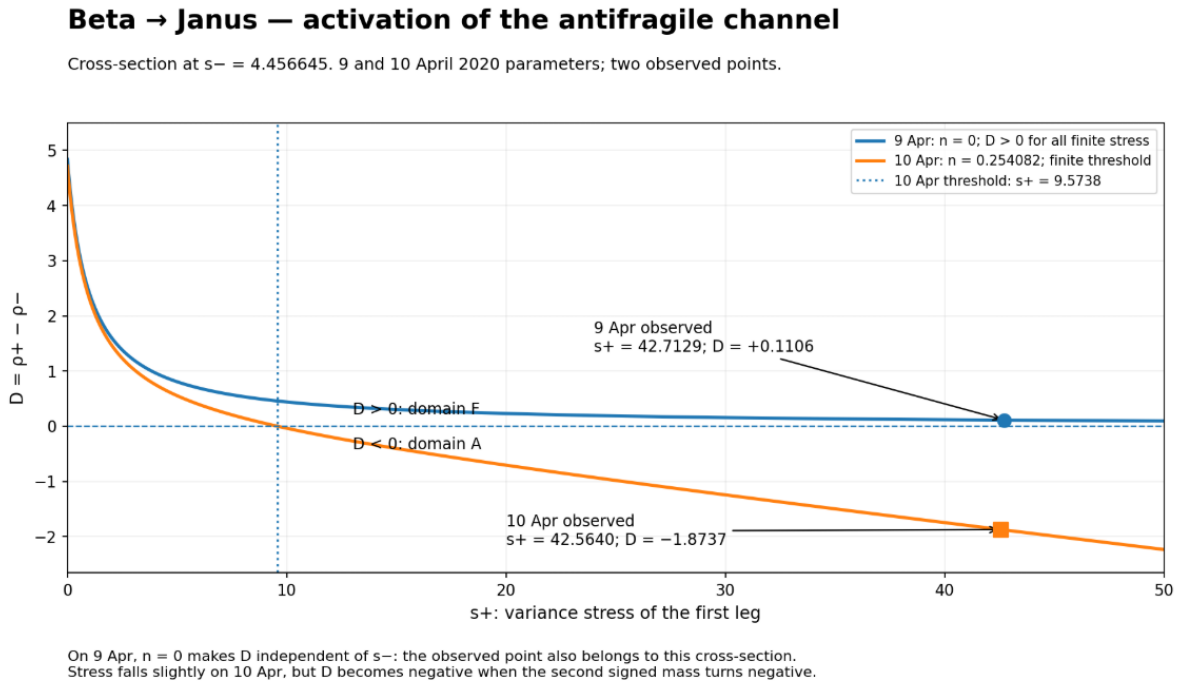


Figure 13: Opening of the antifragile channel between 9 and 10 April 2020, expressed in stress coordinates.

Now, we can separate the observed-point movement from the framework change again.

If I keep the point from 9 April fixed and apply the parameters from 10 April, the balance becomes approximately

$$\mathrm{D} \simeq -1.875473. \tag{427}$$

41 Topology of regular domains and changing partitions

That's not the balance of the point actually observed on 10 April.

It's a fixed-point calculation isolating the new framework.

On 9 April, under that framework, there is no antifragile-dominance region.

On 10 April, there is one.

The geometric appearance takes place under the new framework, while temporal confirmation occurs on 14 April, after the candidate observations on 10, 13, and 14 April.

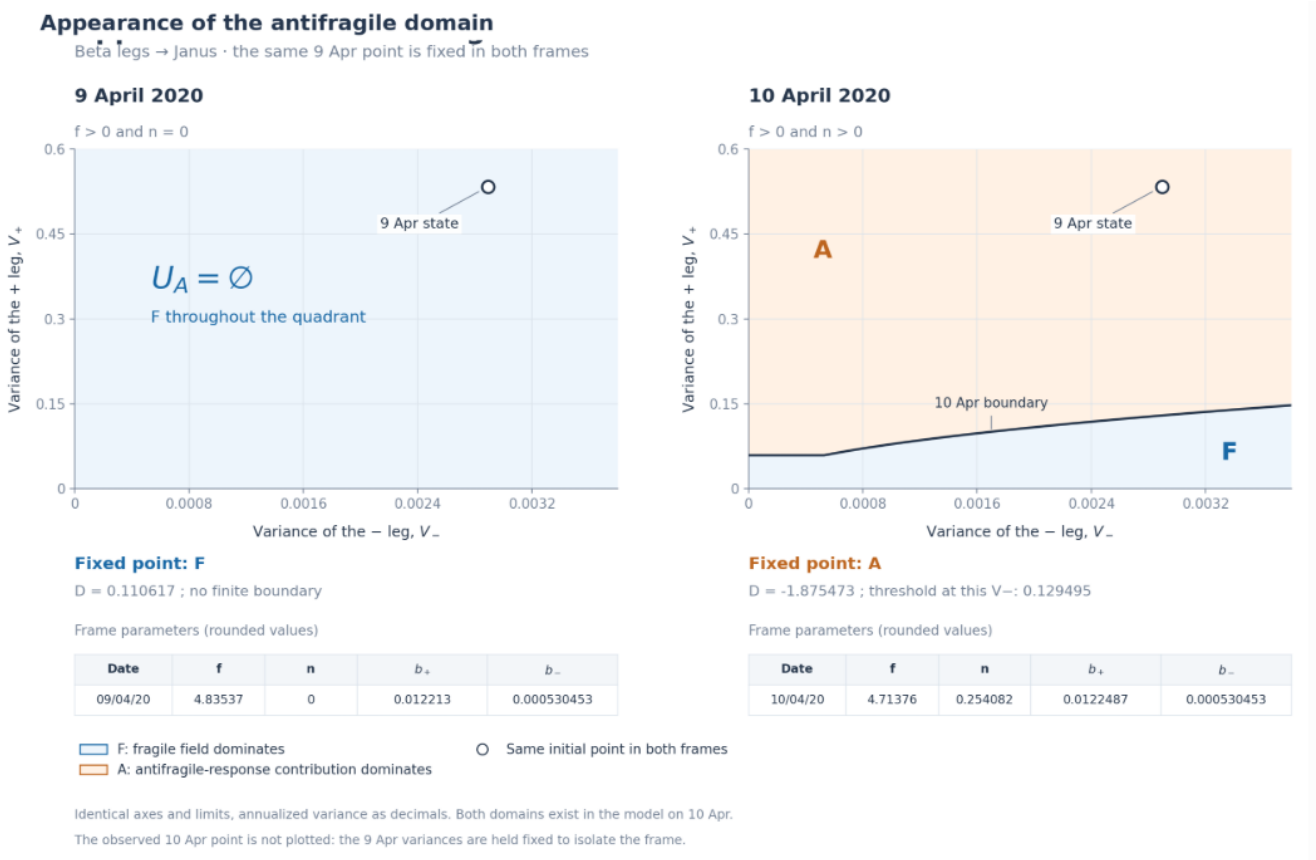


Figure 14: The same channel-opening comparison in variance coordinates. Channel availability and the level of stress are distinct conditions.

Now we're back to topology.

A boundary deforming isn't necessarily a topological change. but the appearance or disappearance of a domain can change the topology of the strict-dominance partition!

We prove this with the exact boundary equation.

Assume

$$f, n, b_+, b_- > 0 \tag{428}$$

The function Cθ(V$_-$) is continuous and strictly positive.

Now We straighten our curved boundary.

I'll define a transformation that measures a point's height relative to the boundary height at the same horizontal coordinate:

$$\boxed{T_\theta(V_-, V_+) = \left(V_-, \frac{V_+}{C_\theta(V_-)}\right)} \tag{429}$$

We call the new vertical coordinate η

$$\eta = \frac{V_+}{C_\theta(V_-)} \tag{430}$$

Why introduce η?

To avoid confusing this relative height with the density u.

Imagine the boundary height is 10.

If the point has height 5

$$\eta = \frac{5}{10} = 0.5 \tag{431}$$

It's below the boundary.

If its height is 10

$$\eta = 1 \tag{432}$$

It's on the boundary.

If its height is 15

$$\eta = 1.5 \tag{433}$$

It's above it.

So our curved boundary becomes

$$\boxed{\eta = 1} \tag{434}$$

We've straightened the boundary!

Now, does this transformation preserve the relevant topological organization?

We need an inverse.

From

$$\eta = \frac{V_+}{C_\theta(V_-)} \tag{435}$$

multiply by Cθ(V$_-$)

$$V_+ = \eta C_\theta(V_-) \tag{436}$$

So

$$\boxed{T_\theta^{-1}(V_-, \eta) = (V_-, \eta C_\theta(V_-))} \tag{437}$$

Because Cθ is continuous and strictly positive, the transformation and its inverse are continuous.

This is a homeomorphism.

A homeomorphism is a continuous transformation with a continuous inverse. It preserves the topological organization of the spaces being compared.

Our domains become

$$\boxed{T_\theta(F_\theta) = [0, +\infty) \times [0, 1)} \tag{438}$$

And

$$\boxed{T_\theta(A_\theta) = [0, +\infty) \times (1, +\infty)} \tag{439}$$

The symbol $\times$ here means we form every pair of coordinates from the two intervals.

Both regions are convex.

What does convex mean?

Take two points inside the same region and draw the straight segment connecting them.

The entire segment stays inside that region.

For points z_1, z_2, the segment is

$$\gamma(\lambda) = (1 - \lambda)z_1 + \lambda z_2, \qquad 0 \leq \lambda \leq 1 \tag{440}$$

At $\lambda = 0$, we're at the first point.

At $\lambda = 1$, we're at the second.

Between them, we move continuously along the straight segment.

To construct a path in the original curved domain, we transform both endpoints, draw the segment in the straightened region, then apply the inverse

$$\boxed{\gamma(\lambda) = T_\theta^{-1}\left[(1 - \lambda)T_\theta(z_1) + \lambda T_\theta(z_2)\right]} \tag{441}$$

The straight segment remains in the relevant straightened domain.

Its inverse image therefore remains in the original domain.

We've proved that each regular domain is path-connected!

Next, prove something stronger.

Choose a fixed point z_0 inside one domain.

We move every other point continuously toward it without ever leaving the domain.

Define

$$\boxed{H(z, \lambda) = T_\theta^{-1}\left[(1 - \lambda)T_\theta(z) + \lambda T_\theta(z_0)\right]} \tag{442}$$

At $\lambda = 0$, every point remains in its original position.

At $\lambda = 1$, every point reaches z_0.

And throughout the movement, everything stays inside the domain.

This property is called contractibility.

We've shown that the entire domain can be continuously contracted to a single point.

Next, establish the common boundary.

Since D is continuous, a point with $D > 0$ has a sufficiently small neighborhood where the balance stays positive.

The same is true for $D < 0$.

But on the regular boundary, where:

$$D = 0 \tag{443}$$

small vertical changes can reach either strict domain.

In the admissible variance space

$$X = [0, +\infty)^2 \tag{444}$$

small vertical changes can reach either strict domain.

In the admissible variance space

$$X = [0, +\infty)^2 \tag{445}$$

we obtain

$$\boxed{\partial F_\theta = \partial A_\theta = \Gamma_\theta} \tag{446}$$

The symbol ∂ denotes the boundary of a set.

The coordinate axes don't become additional artificial boundaries between regimes.

We're considering boundaries relative to the admissible space X.

Next, examine the case where $n = 0$.

We proved

$$F = X, \qquad A = \varnothing \tag{447}$$

Define the union of strict-dominance domains

$$\boxed{S = F \cup A} \tag{448}$$

The symbol $\cup$ means union.

When $n = 0$, $f > 0$, $S = X$ has one connected component.

When $f, n > 0$, both domains exist, and their zero-balance boundary is excluded from S.

So S has two connected components.

We can write

$$\boxed{\#\pi_0(S) : 1 \longrightarrow 2} \tag{449}$$

This notation counts the connected components of the strict-dominance union.

The topology of the partition has changed. the ambient variance space X has not!

That's the precise distinction from the topological construction.

A boundary can change its position or shape while the two regular domains remain path-connected and contractible.

But when a channel opens or disappears, the number of strict-dominance domains can change.

Temporal confirmation

42 Geometric dominance and local kinematics

Now, does crossing a geometric boundary mean that Janus immediately confirms a change of trading state?

NO!

Because the detector also has kinematics and temporal memory.

And we have to distinguish three things:

The current geometric dominance.

The confirmed regime.

And the actual portfolio allocation.

That's what we're going to reconstruct next.

The dominance calculated at a particular date is

$$D = u - v \tag{450}$$

But Janus doesn't simply look at the sign of D and immediately execute every transition.

It also calculates local kinematic information from the price series.

The program starts by normalizing prices

$$\boxed{Z_t = \frac{P_t}{P_0}} \tag{451}$$

P_t is the current price.

P_0 is the price at the starting observation.

If the price doubles relative to its starting level, $Z_t = 2$.

If it stays at the starting level, $Z_t = 1$.

Now the program fits a local polynomial to the normalized series.

What does that mean?

Imagine taking a short section of a curve and fitting a polynomial to describe its recent shape.

The polynomial value describes its local estimated level.

The first derivative describes its local slope.

The second derivative describes how that slope is changing.

Janus extracts

$$\boxed{\mathcal{R}_t = \widehat{p}_t(0)} \tag{452}$$

$$\boxed{\mathcal{R}_{1,t} = \widehat{p}'_t(0)} \tag{453}$$

$$\boxed{\mathcal{R}_{2,t} = \widehat{p}_t''(0)} \tag{454}$$

The hat indicates an estimated polynomial.

One prime means first derivative.

Two primes mean second derivative.

These are local fitted quantities, not exact global derivatives of a smooth observed price trajectory.

And there's an implementation detail we have to preserve.

In the configuration studied, the nominal smooth_window=21 setting actually uses 11 observations, at positions from -10 to 0, with initial padding to the left.

The polynomial is fitted again at each date.

To reproduce the kinematics exactly, we have to follow the window actually used by the code!

Now define a numerical floor

$$\boxed{M_\epsilon(R) = \max(R, 10^{-6})} \tag{455}$$

The value 10^{-6} is 0.000001.

It prevents division by zero or by an extremely small denominator.

For each leg, define

$$\boxed{\ell_\pm = \frac{\mathcal{R}_{2,\pm}}{\max(\mathcal{R}_\pm, 10^{-6})}} \tag{456}$$

The symbol ℓ represents the fitted second derivative normalized by the scale level.

It isn't the financial mass m.

It isn't the stress.

It's an additional kinematic quantity.

Now Janus combines it with our balance

$$\begin{aligned} a_+ + a_- &= \ell_+ + \frac{D}{2} + \ell_- - \frac{D}{2} \\ &= \ell_+ + \ell_- \end{aligned} \tag{457}$$

So

$$\boxed{a_+ + a_- = \ell_+ + \ell_-} \tag{458}$$

The two complete accelerations are not generally opposites!

Now, in the main configuration, the default exit condition is

$$\boxed{a_+ > 0} \tag{459}$$

The corresponding exit boundary is where

$$a_+ = 0 \tag{460}$$

Substitute

$$\ell_+ + \frac{D}{2} = 0 \tag{461}$$

Then substitute the definition of ℓ_+

$$\frac{\mathcal{R}_{2,+}}{\max(\mathcal{R}_+, 10^{-6})} + \frac{D}{2} = 0 \tag{462}$$

Move the balance term to the other side and multiply by the strictly positive denominator

$$\boxed{\mathcal{R}_{2,+} = -\frac{D}{2}\max(\mathcal{R}_+, 10^{-6})} \tag{463}$$

That's the actual default exit boundary.

It isn't simply the condition that the fitted second derivative becomes positive.

And because the denominator contains a maximum, the expression is piecewise-defined at the numerical floor.

The floor isn't activated in the archived case being studied, but it still belongs to the full implementation.

43 Duration thresholds, clipping, and rounding

Now we know how the model can exit.

How does it confirm entry?

That's where the confirmation duration enters.

The rule is

$$\boxed{\tau = \text{arrondi}_{\text{pair}}\left[\text{clip}_{[3,30]}\left(\frac{1}{\sqrt{\max(|D|, 10^{-6})}}\right)\right]} \tag{464}$$

We unpack every part.

First

$$|D| \tag{465}$$

The vertical bars mean absolute value.

They measure the magnitude of the balance regardless of its sign.

A balance of $+4$ and one of -4 both have absolute value 4.

Then

$$\max(|D|, 10^{-6}) \tag{466}$$

This applies a numerical floor so we don't divide by zero.

Next, take the square root and its reciprocal

$$\frac{1}{\sqrt{\max(|D|, 10^{-6})}} \tag{467}$$

What happens as |D| increases?

The square root increases, so its reciprocal decreases.

What happens when |D| becomes small?

The reciprocal increases.

So a strong imbalance tends to require fewer confirmation observations before the bounds are applied, while a weak imbalance tends to require more.

Now clip limits the result to the interval from 3 to 30.

If the value is below 3, the program retains 3.

If it's above 30, it retains 30.

Finally, the program rounds to an integer.

And the rounding rule matters: exact half-integers are rounded to the even integer.

For example

$$3.5 \longmapsto 4, \qquad 4.5 \longmapsto 4 \tag{468}$$

The result τ is an integer between 3 and 30 observations.

Now remember the distance to the density diagonal

$$d_\rho = \frac{|D|}{\sqrt{2}} \tag{469}$$

We can rearrange it

$$|D| = \sqrt{2}\, d_\rho \tag{470}$$

Then, away from the floor

$$\frac{1}{\sqrt{|D|}} = 2^{-1/4} d_\rho^{-1/2} \tag{471}$$

Including the floor and the bounds gives

$$\boxed{\tau = \text{arrondi}_{\text{pair}} \left[\text{clip}_{[3,30]} \left(2^{-1/4} \max \left(d_\rho, \frac{10^{-6}}{\sqrt{2}} \right)^{-1/2} \right) \right]} \tag{472}$$

This equation expresses the same rule using the Euclidean density-plane distance.

The distance to the zero-balance diagonal enters the confirmation duration!

But that distance isn't a probability of loss.

At D = 0, the duration is 30 after applying the floor, bounds, and rounding.

Now, how is the duration used?

When the model is outside the confirmed refuge state and D < 0, it maintains a counter of consecutive negative-balance observations.

If the previous counter is c, the new candidate counter is

$$\boxed{k = c + 1} \tag{473}$$

Entry requires

$$\boxed{k \geq \tau} \tag{474}$$

If the balance becomes nonnegative while outside refuge, the counter resets.

44 Exact entry thresholds and finite waiting time

Now here's a subtlety.

Since τ depends on the magnitude of the current balance, the actual entry condition isn't determined by the sign of D alone.

For each candidate counter, there is a negative balance threshold.

For $3 \leq k \leq 29$, define

$$\boxed{h_k = \frac{4}{(2k+1)^2}} \tag{475}$$

Where does this expression come from?

Look at the value of |D| making the unrounded duration exactly halfway between k and k + 1

$$\frac{1}{\sqrt{|D|}} = k + \frac{1}{2} \tag{476}$$

Invert

$$\sqrt{|D|} = \frac{1}{k + \frac{1}{2}} \tag{477}$$

Square

$$|D| = \frac{1}{(k + \frac{1}{2})^2} \tag{478}$$

Now

$$k + \frac{1}{2} = \frac{2k+1}{2} \tag{479}$$

Therefore

$$\boxed{|D| = \frac{4}{(2k+1)^2} = h_k} \tag{480}$$

Now remember the round-to-even rule.

If k is even, equality at the half-integer rounds toward k, so the equality is accepted.

If k is odd, equality rounds toward k + 1, so the inequality must be strict.

Thus

$$\boxed{k \text{ even}: \quad \text{entry} \iff D \leq -h_k} \tag{481}$$

And

$$\boxed{k \text{ odd}: \quad \text{entry} \iff D < -h_k} \tag{482}$$

For three candidate observations

$$h_3 = \frac{4}{49} \tag{483}$$

Since 3 is odd

$$\boxed{D < -\frac{4}{49}} \tag{484}$$

For four

$$\boxed{D \leq -\frac{4}{81}} \tag{485}$$

For five

$$\boxed{D < -\frac{4}{121}} \tag{486}$$

Before three candidate observations, entry is impossible under the main configuration.

At 30, any strictly negative balance meets the maximum duration.

For exact numerical equalities, reproducing the implementation requires respecting the actual floating-point calculations and rounding convention.

Now we can see why the operational geometry contains more than the single level

$$D = 0 \tag{487}$$

It contains a family of negative balance levels

$$D = -h_k \tag{488}$$

We calculate their geometry.
Start with a general constant d

$$D = d \tag{489}$$

Substitute the density balance

$$\frac{f}{1+x} - \frac{nx}{1+y} = d \tag{490}$$

Multiply by the positive denominators

$$f(1+y) - nx(1+x) = d(1+x)(1+y) \tag{491}$$

Collect the terms containing 1 + y

$$\boxed{[f - d(1+x)](1+y) = nx(1+x)} \tag{492}$$

This describes any constant level of the balance.
Now take a negative level

$$d = -h, \qquad h > 0 \tag{493}$$

Then

$$[f + h(1+x)](1+y) = nx(1+x) \tag{494}$$

Divide by the strictly positive factor f + h(1 + x)

$$1 + y = \frac{nx(1+x)}{f + h(1+x)} \tag{495}$$

Subtract 1

$$\boxed{y = Y_h(x) = \frac{nx(1+x)}{f + h(1+x)} - 1} \tag{496}$$

This is a rational curve.
When h = 0 and f > 0, it reduces to the original dominance parabola.
When h > 0, the denominator also depends on x.

Now calculate its second derivative

$$\boxed{Y_h''(x) = \frac{2nf(f+h)}{[f+h(1+x)]^3} > 0} \tag{497}$$

for f, n > 0.

If f = 0, n > 0, the expression simplifies

$$Y_h(x) = \frac{nx(1+x)}{h(1+x)} - 1 \tag{498}$$

Cancel 1 + x

$$\boxed{Y_h(x) = \frac{n}{h}x - 1} \tag{499}$$

This case is a straight line.

Next, find the corresponding critical value of x.

Starting from

$$[f + h(1+x)](1+y) = nx(1+x) \tag{500}$$

expand and collect powers of x

$$\boxed{nx^2 + [n - h(1+y)]x - (f+h)(1+y) = 0} \tag{501}$$

That's a quadratic equation.

Using the quadratic formula and retaining the non negative root

$$\boxed{x_h = \frac{h(1+y) - n + \sqrt{[n-h(1+y)]^2 + 4n(f+h)(1+y)}}{2n}} \tag{502}$$

For n > 0

$$\boxed{D < -h \iff x > x_h} \tag{503}$$

A strictly negative balance requires n > 0 and x > 0.

So negative confirmation levels don't intersect the strata where both densities vanish.

On the strata they do intersect

$$\frac{\partial D}{\partial n} = -\frac{x}{1+y} < 0 \tag{504}$$

That derivative doesn't vanish.

It ensures the negative level remains regular on those strata.

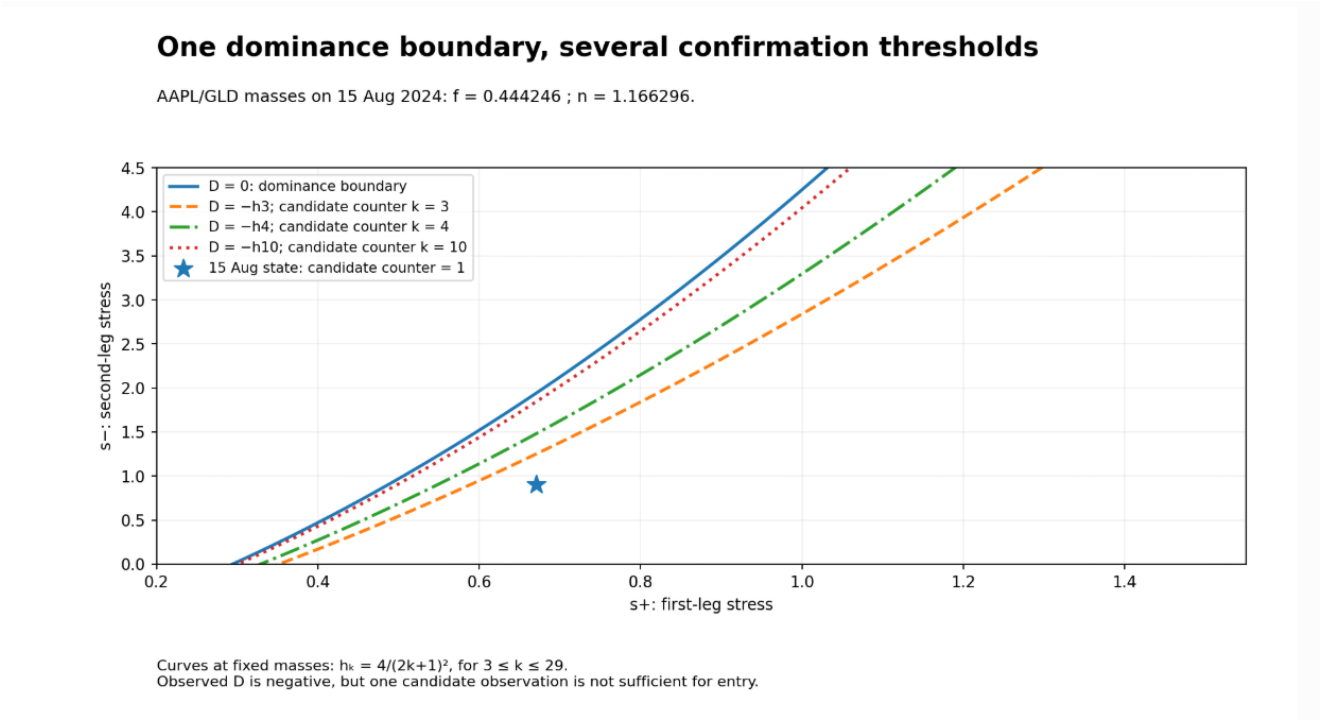


Figure 15: Required confirmation duration as a function of balance. Clipping and integer rounding create threshold regions.

Look at the observed point in this chart.

Its balance is already negative.

But its candidate counter is only 1.

It cannot enter the confirmed refuge state, even if its geometric position satisfies a threshold associated with a more advanced counter.

The geometric position isn't enough. memory matters!

45 The memory automaton and its transition families

Next, describe that memory completely.

Let G = 0 denote the growth mode and G = 1 the confirmed refuge mode, called crash in the code.

Let c be the waiting counter.

The memory state is

$$(G, c) \tag{505}$$

Outside confirmed refuge, the retained counter can take values from 0 to 29.

It cannot remain at 30 because the maximum duration is 30. When the candidate counter reaches the required duration, the model enters the confirmed state and resets the counter.

During confirmed refuge, the counter remains zero.

The formal memory state space is

$$\boxed{\mathcal{Q} = \{(0, c) : c = 0, \ldots, 29\} \cup \{(1, 0)\}} \tag{506}$$

This gives thirty waiting states and one confirmed refuge state.

Thirty-one formal memory states!

Next, examine the five transition families.

Previous state	Current condition	New state
$(0, c)$	$D \geq 0$	$(0, 0)$
$(0, c)$	$D < 0,\ c + 1 < \tau$	$(0, c + 1)$
$(0, c)$	$D < 0,\ c + 1 \geq \tau$	$(1, 0)$
$(1, 0)$	$a_+ \leq 0$	$(1, 0)$
$(1, 0)$	$a_+ > 0$	$(0, 0)$

First row: we're outside refuge, and the balance is nonnegative. The waiting counter resets.

Second: balance is negative, but the required duration hasn't been reached. The counter increases.

Third: balance is negative and the candidate counter reaches the current required duration. Entry is confirmed.

Fourth: we're already in refuge, and the exit acceleration is nonpositive. The model remains there.

Fifth: we're already in refuge, and the exit acceleration becomes strictly positive. The model exits.

These five cases cover the transitions of the main automaton.

Notice the consequences.

Outside refuge, a zero balance resets the waiting counter.

Inside refuge, a zero balance doesn't automatically cause an exit.

The exit depends on a_+.

And an observation doesn't immediately produce both an entry and an exit within the same iteration.

Now look at the historical SPY/GLD episode of February 2020.

The balances are

$$D_{24/02} \simeq +13.289481 \tag{507}$$

$$D_{25/02} \simeq -1.820929 \tag{508}$$

$$D_{26/02} \simeq -2.049789 \tag{509}$$

$$D_{27/02} \simeq -30.225347 \tag{510}$$

On 24 February, fragile dominance remains active.

On 25 February, the balance becomes negative.

The geometric boundary has been crossed!

But the waiting counter is only 1.

The confirmed state remains G = 0.

On 26 February, the balance stays negative.

The counter reaches 2.

Still no confirmation.

On 27 February, the third negative observation satisfies the duration.

The model enters

$$G = 1 \tag{511}$$

So the geometric crossing occurs on 25 February, while temporal confirmation occurs on 27 February.

And GLD's signed mass was already negative before that crossing.

Its individual response didn't suddenly become antifragile on 25 February. The pair's coupled dominance changed.

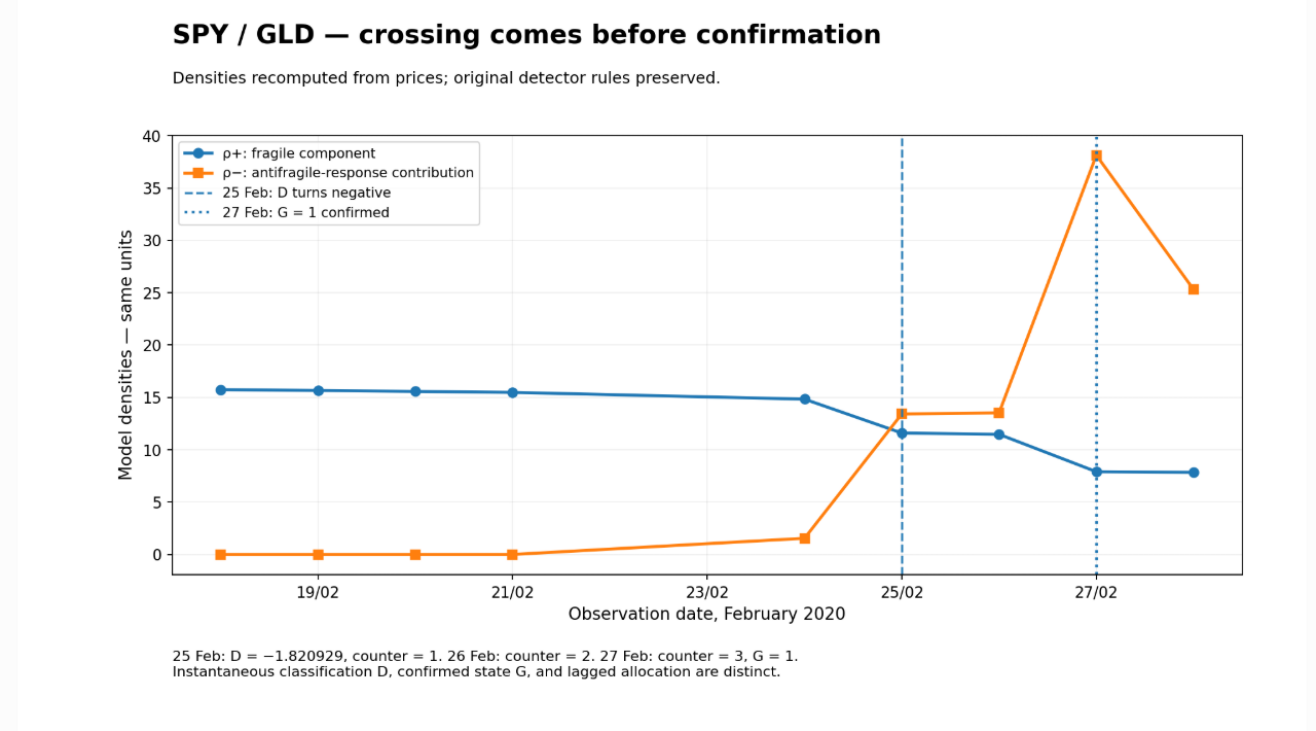


Figure 16: Instantaneous geometric crossing and subsequent confirmation in the February 2020 example.

Next, examine something even more interesting.

Imagine both densities are zero

$$u = v = D = 0 \tag{512}$$

In the density plane, our point doesn't move.

Does that mean the confirmed regime must remain unchanged?

No!

Because the exit depends on kinematics.

Look at SPY/GLD on 4, 5, and 6 April 2022.

On all three dates

$$u = v = D = 0 \tag{513}$$

But the exit signal takes these values

$$a_{+,04/04} \simeq -0.002761486 \tag{514}$$

$$a_{+,05/04} \simeq -0.002695374 \tag{515}$$

$$a_{+,06/04} \simeq +0.000439183 \tag{516}$$

On the first two dates, the acceleration is negative.

The model remains in G = 1.

On 6 April, the acceleration becomes positive.

The model exits

$$G : 1 \longrightarrow 0 \tag{517}$$

The point in the density plane didn't move. the kinematic variable crossed the exit boundary!

That's why the complete confirmed regime cannot be reconstructed from current densities alone.

We need the kinematic signal and the memory.

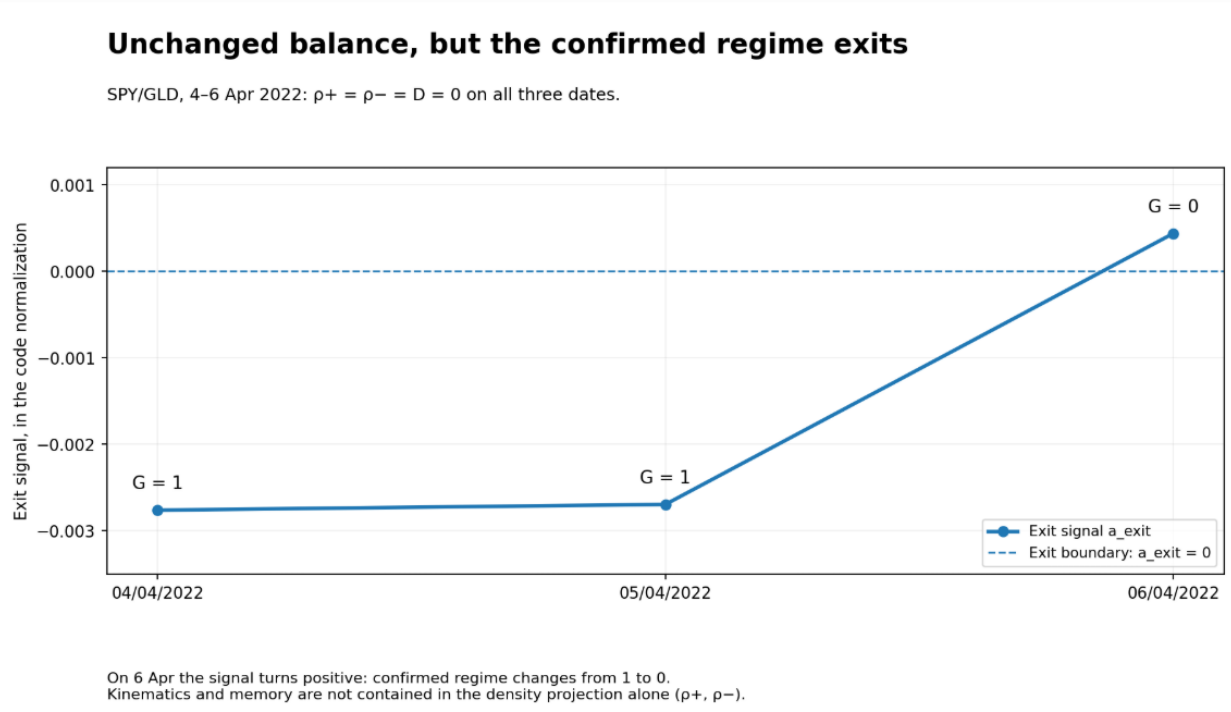


Figure 17: A zero balance does not force exit from an already confirmed state. Exit follows the specified acceleration rule.

Next, unify the regularity of the different decision boundaries.
We've encountered

$$D = 0 \tag{518}$$

for dominance

$$D = -h_k \tag{519}$$

for confirmation,
and

$$a_+ = 0 \tag{520}$$

for exit.
Do we really need to prove the regularity of each ambient surface separately?
We can use a single argument.
Consider five continuous coordinates

$$(f, n, x, y, \ell) \tag{521}$$

Define

$$\boxed{\mathcal{B}(f, n, x, y, \ell) = (D, a_+), \qquad a_+ = \ell + \frac{D}{2}} \tag{522}$$

The transformation has five inputs and two outputs.
Now examine its derivatives with respect to f and ℓ.
For the balance

$$\frac{\partial D}{\partial f} = \frac{1}{1+x} \tag{523}$$

$$\frac{\partial D}{\partial \ell} = 0 \tag{524}$$

The second derivative is zero because ℓ doesn't appear in the balance.
For the acceleration

$$\frac{\partial a_+}{\partial f} = \frac{1}{2(1+x)} \tag{525}$$

$$\frac{\partial a_+}{\partial \ell} = 1 \tag{526}$$

Put these derivatives into a 2×2 minor

$$\boxed{\frac{\partial(D, a_+)}{\partial(f, \ell)} = \begin{pmatrix} \frac{1}{1+x} & 0 \\ \frac{1}{2(1+x)} & 1 \end{pmatrix}} \tag{527}$$

Its determinant is

$$\boxed{\det \frac{\partial(D, a_+)}{\partial(f, \ell)} = \frac{1}{1+x} > 0} \tag{528}$$

The two outputs have independent variation directions everywhere in the admissible smooth ambient extension.

The map is a submersion.

What does that mean?

Its differential has the maximum possible rank for two output coordinates.

The regular-level theorem then applies to the levels of D and a_+, wherever those levels are admissible.

In a five-dimensional ambient space, one independent regular equality locally defines a four-dimensional hypersurface.

Two independent equalities locally define an intersection of dimension three when it's nonempty.

We can even write a coordinate change

$$(f, n, x, y, \ell) \longmapsto (D, a_+, n, x, y) \tag{529}$$

To invert it, solve the balance for f

$$\boxed{f = (1+x)\left(D + \frac{nx}{1+y}\right)} \tag{530}$$

And solve the acceleration for ℓ

$$\boxed{\ell = a_+ - \frac{D}{2}} \tag{531}$$

In these coordinates, the boundary levels become coordinate planes.

But remember the strata!

On a stratum where

$$D \equiv 0 \tag{532}$$

the equation D = 0 is already satisfied everywhere.

It's no longer an independent condition within that stratum.

The acceleration becomes

$$a_+ = \ell \tag{533}$$

So the exit condition is simply

$$\boxed{\ell = 0} \tag{534}$$

Regularity in the ambient space and regularity restricted to a stratum must not be confused!

Now we can describe the complete hybrid decision space of one pair

$$\boxed{\mathcal{E} = [0, +\infty)^4 \times \mathbb{R}_\ell \times \mathcal{Q}} \tag{535}$$

This combines continuous variables with discrete memory states.

That's why the detector is a hybrid system.

For the current decision, the two continuous values D, a_+, together with the memory state, are sufficient to determine the transition.

But to calculate the next state, we also need the historical windows, the polynomial filter window, the last valid mass estimate, and the initial normalization.

Under the default rolling settings, obtaining an updated estimate can involve a historical depth reaching 504 price levels.

So the reduced decision variables don't contain every piece of information needed to update the system.

We can also track several decision margins

$$x - x^\star \tag{536}$$

$$-D - h_k \tag{537}$$

$$k - \tau \tag{538}$$

$$a_+ \tag{539}$$

The first compares observed stress with its critical threshold.

The second compares the balance with a negative confirmation level.

The third compares the waiting counter with the required duration.

The fourth measures the exit condition.

These margins have different meanings and units.

We shouldn't add them together arbitrarily as if they were one universal risk measure.

The code also supports alternative configurations.

Entry can use

$$D < e \tag{540}$$

Exit can use an acceleration threshold

$$a_+ > q \tag{541}$$

or a balance recrossing

$$D > r \tag{542}$$

The main configuration studied here uses zero thresholds and acceleration-based exit.

The native parameter grid includes three smoothing settings, 11, 21, and 31; two horizons, 126 and 252; three minimum confirmations, 2, 3, and 5; and two exit modes.

That gives 36 configurations per pair, or 108 result rows for the three pairs.

Those alternative settings must remain distinct from the main configuration we're demonstrating.

46 Shared refuge coordinates and joint state spaces

Now we've completed the decision geometry of one pair.

But the Janus archive contains three pairs:

SPY/GLD, QQQ/GLD, and AAPL/GLD.

And there's something important here.

All three pairs share GLD!

So if the calendars and calculations are aligned, the variables calculated for that branch can be shared.

We shouldn't automatically treat the three pairs as completely independent systems.

For the positive legs, define

$$f_1, x_1, \qquad f_2, x_2, \qquad f_3, x_3 \tag{543}$$

Those are six coordinates.

For the common GLD leg, retain

$$n, y \tag{544}$$

So the joint primitive space is

$$\boxed{(f_1, x_1, f_2, x_2, f_3, x_3, n, y)} \tag{545}$$

That's eight coordinates, not twelve independent coordinates.

Now define the common coefficient

$$\boxed{k_c = \frac{n}{1+y}} \tag{546}$$

For pair i

$$u_i = \frac{f_i}{1+x_i} \tag{547}$$

And

$$v_i = \frac{nx_i}{1+y} \tag{548}$$

Substituting kc

$$\boxed{v_i = k_c x_i} \tag{549}$$

So for the three pairs

$$\boxed{(v_1, v_2, v_3) = k_c(x_1, x_2, x_3)} \tag{550}$$

The three antifragile contributions share the same factor k_c.

Next, examine the rank of the joint map from eight primitives to six densities.

Take the Jacobian minor corresponding to the six columns

$$(f_1, x_1, f_2, x_2, f_3, x_3) \tag{551}$$

Each pair contributes a 2×2 block

$$\begin{pmatrix} \frac{1}{1+x_i} & -\frac{f_i}{(1+x_i)^2} \\ 0 & k_c \end{pmatrix} \tag{552}$$

Its determinant is

$$\frac{k_c}{1+x_i} \tag{553}$$

The determinant of the full six-dimensional minor is the product of the three block determinants

$$\boxed{\frac{k_c^3}{(1+x_1)(1+x_2)(1+x_3)} > 0} \tag{554}$$

This is nonzero in the sector $n > 0$.

So the joint density map has rank six.

It starts with eight coordinates and produces six independent density coordinates.

The regular fibers therefore have dimension two.

We can recover them explicitly.

At fixed densities

$$x_i = \frac{v_i}{k_c} \tag{555}$$

Then

$$f_i = u_i \left(1 + \frac{v_i}{k_c}\right) \tag{556}$$

And

$$n = k_c(1 + y) \tag{557}$$

The two free parameters are kc > 0 and y ≥ 0.

Now add three kinematic quantities

$$\ell_1, \qquad \ell_2, \qquad \ell_3 \tag{558}$$

The joint decision system has eleven continuous coordinates, with discrete memory in

$$\boxed{\mathcal{Q}^3} \tag{559}$$

The power 3 indicates one memory state for each pair.

In the archived controls, the shared GLD proportionality constraints are reproduced with a maximum residual of approximately 5.7×10^{-13}.

All three pairs have both densities positive simultaneously on 317 dates.

But this shared structure depends on alignment.

If different pairs use different calendar intersections, their rolling GLD estimates may differ.

Sharing a ticker doesn't automatically guarantee identical estimates after different historical processing!

Auxiliary dynamics and invariances

47 Formal pressure and retrospective potentials

Now the complete construction also contains several auxiliary mathematical objects and an options-pricing layer.

We need to preserve them, but without confusing them with the actual density and regime-decision rules.

We start with the formal pressure terms

$$\boxed{p_{\pm} = \omega_{\pm}\rho_{\pm}} \tag{560}$$

Here p± denotes the formal pressures constructed by the program.

$\rho\pm$ are the two densities we've already studied.

And $\omega\pm$ are their pressure coefficients.

We use ω, pronounced "omega," to distinguish these coefficients from the portfolio weights used later.

The equation simply says:

"Multiply each density by its corresponding pressure coefficient."

Now, in the main execution studied here, those coefficients are zero.

Therefore

$$p_{+} = 0, \qquad p_{-} = 0 \tag{561}$$

The formal pressure terms don't drive entry and exit in that configuration.

And when we informally refer to ρ_{-} as the antifragile pressure or contribution, we mean the density used by the coupling operator.

That isn't the same mathematical object as the column p_minus.

Similar names don't make two calculations equivalent.

Now, the program also constructs a potential.

Before writing its equation, let's understand the two operations used to build it.

The first is centering.

Imagine a series

$$2, \quad 4, \quad 6 \tag{562}$$

Its mean is 4.

Subtract that mean from each value

$$-2, \quad 0, \quad 2 \tag{563}$$

The new series has a mean of zero.

That's centering.

We denote the operation by CN.

The second operation is cumulative integration.

The program uses the trapezoidal rule.

The idea is to approximate the area under a sequence by treating each interval as a trapezoid and accumulating those areas.

We denote that operation by TN.

Now, the potential is

$$\boxed{\psi_+ = \gamma C_N T_N C_N T_N C_N D, \qquad \psi_- = -\psi_+} \tag{564}$$

The operations are applied from right to left.

First center the balance D.

Then integrate.

Center again.

Integrate a second time.

Center once more.

Finally multiply by γ.

Here γ is a coefficient in the potential construction, not the path γ used earlier in the topology proof.

The second equation makes the two constructed potentials opposites.

But that doesn't mean the two operational densities are negative copies of one another.

The potential and the densities are different objects.

Now, suppose D is constant across the series.

The first centering operation removes that constant.

The result is zero.

Every subsequent integration and centering of zero remains zero.

So a constant source produces a zero potential in this construction.

But there's an especially important detail: the centering operations use the entire historical archive.

That makes the potential retrospective.

It isn't a real-time signal calculated causally from only the information available before each historical decision.

And it doesn't feed the main entry and exit detector.

We can't give an auxiliary retrospective potential a decision role that it doesn't have in the code!

48 Inertia, volatility inputs, and singular limits

Next, examine inertia.

The program includes

$$\boxed{I = \frac{\sigma^2}{\beta}} \tag{565}$$

σ is a volatility.

So σ^2 is its corresponding variance.

β is the coefficient used in this auxiliary expression.

We must identify the coefficient used by each function rather than assuming every variable named β is the complete Beta construction or the date-specific rolling Janus mass.

The expression requires:

$$\beta \neq 0 \tag{566}$$

Because division by zero is undefined.

The program also defines a drift correction

$$\boxed{\mu_c = -\beta\sigma^2} \tag{567}$$

The Greek letter μ, commonly denotes drift in financial models.

The subscript c identifies the correction.

Now multiply the drift correction by inertia

$$\mu_c I = (-\beta\sigma^2)\frac{\sigma^2}{\beta} \tag{568}$$

For $\beta \neq 0$, the coefficient cancels.

So

$$\boxed{\mu_c I = -\sigma^4} \tag{569}$$

Why the fourth power?

Because

$$\sigma^2 \times \sigma^2 = \sigma^4 \tag{570}$$

Now rearrange the inertia formula

$$I = \frac{\sigma^2}{\beta} \quad \Longrightarrow \quad \sigma^2 = \beta I \tag{571}$$

Substitute into the drift correction

$$\mu_c = -\beta(\beta I) \tag{572}$$

Therefore

$$\boxed{\mu_c = -\beta^2 I} \tag{573}$$

We have two identities

$$\boxed{\mu_c I = -\sigma^4, \qquad \mu_c = -\beta^2 I} \tag{574}$$

But now look at what happens near $\beta = 0$.

At fixed variance

$$I = \frac{\sigma^2}{\beta} \tag{575}$$

becomes singular as β approaches zero.

Its magnitude increases without bound.

Meanwhile

$$\mu_c = -\beta\sigma^2 \tag{576}$$

approaches zero.

So the singularity of inertia doesn't mean the drift correction explodes.

The two expressions behave differently near zero.

Also

$$|I| = \frac{\sigma^2}{|\beta|} \tag{577}$$

At fixed variance, increasing $|\beta|$ reduces the magnitude of inertia.

Now, in the options summary, the columns mass_corr and inertia_atm use the provider's implied volatility iv_near.

The column iv_mass_near comes from a different inversion.

Those are different volatility inputs and should not be silently treated as the same quantity.

Auxiliary mechanical constructions include the following equations

$$F = ma, \qquad a = \frac{F}{m} \tag{578}$$

An energy-like expression

$$\frac{1}{2} m\sigma^2 \tag{579}$$

Illustrative components named quarks, gluons, or nucleons.

And a reference function of the form

$$\frac{1}{\mathcal{R}^3} \tag{580}$$

But these auxiliary objects don't define the operational densities.

Those remain

$$u = \frac{f}{1+x}, \qquad v = \frac{nx}{1+y} \tag{581}$$

49 Stochastic price dynamics and pathwise comparison

Next, return to Brownian motion, which I mentioned at the beginning.

The model includes an auxiliary stochastic price simulation

$$\boxed{\frac{dS_t}{S_t} = (\mu - \beta\sigma^2)\,dt + \sigma\,dW_t} \tag{582}$$

We identify the terms.

S_t is the simulated price at time t.

dS_t/S_t is its infinitesimal relative change.

μ is the reference drift.

$-\beta\sigma^2$ is the mass-related drift correction.

dt is a small time interval.

W_t is Brownian motion.

And dW_t is its random increment.

The equation says:

"The simulated relative price change combines a drift corrected by the mass coefficient with a random component scaled by volatility."

This is an equation for an auxiliary simulation, not a proof that every observed market price follows that exact process.

At constant coefficients, its solution is

$$\boxed{S_t = S_0 \exp\left[\left(\mu - \beta\sigma^2 - \frac{1}{2}\sigma^2\right)t + \sigma W_t\right]} \tag{583}$$

Why does the additional term $-\sigma^2/2$ appear?

Because the transformation from a multiplicative stochastic differential equation to log-price coordinates requires the log-price drift correction associated with Brownian variance.

Next, compare two simulations.

The first uses a coefficient β.

The second uses $\beta = 0$.

But I want both simulations to use the same Brownian path.

Why?

Because We isolate the effect of the mass coefficient without introducing another difference coming from separate random draws.

The two solutions are

$$S_t^{\beta} = S_0 \exp\left[\left(\mu - \beta\sigma^2 - \frac{1}{2}\sigma^2\right)t + \sigma W_t\right] \tag{584}$$

$$S_t^{0} = S_0 \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W_t\right] \tag{585}$$

Take their ratio

$$\frac{S_t^{\beta}}{S_t^{0}} \tag{586}$$

The initial prices cancel.

The common drift terms cancel.

The Brownian terms cancel because the paths are identical.

And the log-price drift corrections cancel.

We're left with

$$\boxed{\frac{S_t^{\beta}}{S_t^{0}} = e^{-\beta\sigma^2 t}} \tag{587}$$

That's the exact effect of the coefficient in this comparison using the same random path.

But the two demonstration trajectories exported by the program use separate random draws.

Their visible differences therefore include the difference between those random realizations.

They cannot all be attributed to the mass correction alone.

And this stochastic simulation doesn't turn our variance map into physical walls enclosing the price.

The simulated price, its measured variance, and the geometry classifying its state remain different objects.

50 Price, return, and mass-scaling invariances

Next, examine invariances.

An invariance is a property that remains unchanged under a specified transformation.

Why is this useful?

Because it tells us what information the model actually depends on.

First, multiply every price in a series by a positive constant c

$$\widetilde{P}_t = cP_t \tag{588}$$

The logarithmic return becomes

$$\ln\left(\frac{\widetilde{P}_t}{\widetilde{P}_{t-1}}\right) = \ln\left(\frac{cP_t}{cP_{t-1}}\right) \tag{589}$$

The common factor cancels

$$\boxed{\ln\left(\frac{cP_t}{cP_{t-1}}\right) = \ln\left(\frac{P_t}{P_{t-1}}\right)} \tag{590}$$

So the logarithmic returns remain unchanged.

The normalized price series also remains unchanged

$$\frac{cP_t}{cP_0} = \frac{P_t}{P_0} \tag{591}$$

Therefore, pure positive rescaling of the nominal price level preserves the returns, masses, stresses, densities, and confirmed regime.

Now consider changing the constant reference rate used in the excess return.

That adds or subtracts the same constant from all values of E in a regression window.

But centered covariance is unchanged by adding a constant to one variable.

So the rolling mass remains unchanged.

We've also demonstrated another transformation

$$(f, n) \mapsto (\lambda f, \lambda n), \qquad \lambda > 0 \tag{592}$$

It gives

$$D \mapsto \lambda D \tag{593}$$

The zero boundary doesn't move.

But the magnitude changes, so confirmation duration and exit can change.

Now consider a different transformation of price

$$\boxed{\widetilde{P}_t = P_t e^{ct}} \tag{594}$$

where c is constant and t counts observations.

This adds a regular exponential growth component.

What happens to logarithmic returns?

Calculate

$$\widetilde{r}_t = \ln\left(\frac{P_t e^{ct}}{P_{t-1}e^{c(t-1)}}\right) \tag{595}$$

Separate the ratios

$$\widetilde{r}_t = \ln\left(\frac{P_t}{P_{t-1}}\right) + \ln\left(e^{ct-c(t-1)}\right) \tag{596}$$

Since

$$ct - c(t-1) = c \tag{597}$$

we obtain

$$\boxed{\widetilde{r}_t = r_t + c} \tag{598}$$

Every logarithmic return receives the same additive constant.

But variance depends on deviations from the mean.

Adding c to every return also adds c to their mean.

So

$$(r_i + c) - (\overline{r} + c) = r_i - \overline{r} \tag{599}$$

The deviations remain unchanged.

Therefore, U, V, their historical references, and their stresses remain unchanged.

The 252-observation excess return changes by

$$\boxed{\widetilde{E}_t = E_t + 252c} \tag{600}$$

But this constant shift doesn't change the centered covariance of U and E.

So

$$\boxed{\widetilde{m} = m} \tag{601}$$

And because the masses and stresses remain unchanged, the densities and balance remain unchanged

$$\boxed{\widetilde{D} = D} \tag{602}$$

We've changed the returns while preserving exactly the same dominance geometry!

This is an important mathematical result.

It demonstrates that the geometry of the density domains doesn't uniquely determine the asset's realized or future returns.

But this invariance concerns the densities and the balance.

It doesn't automatically preserve the fitted-price kinematics or the entire confirmed-state automaton.

Options and coordinate transformations

51 Replication, parity, and diagnostic distance

We now examine option valuation and replication.

The analogy was meant to show how a valuation framework depends on parameters that can change even when the underlying price doesn't.

But to understand the options layer scientifically, we have to use its own equations.

And we must not confuse its parity boundary or its volatility geometry with the historical Janus dominance boundary D = 0.

For this section, I'll use:

S for the underlying price.

K for the strike.

T for time to maturity.

r for the interest rate used in valuation.

C for the call price.

And Pput for the put price.

I use Pput so it isn't confused with the price series P_t from the regime calculations.

Now define the discounted strike

$$\boxed{B = Ke^{-rT}} \tag{603}$$

This is the present value of the amount K, using the continuous discounting convention of the formula.

The exponential factor e^{-rT} brings a future payment back to its present value.

Now the synthetic replication is

$$\boxed{S_{\text{synth}} = B + C - P_{\text{put}}} \tag{604}$$

This says:

"Take the discounted strike, add the call, and subtract the put."

Why this particular combination?

We prove its payoff at maturity.

The call payoff is

$$(S_T - K)^+ \tag{605}$$

The notation z^+ means

$$z^+ = \max(z, 0) \tag{606}$$

So the call payoff is the positive part of the difference between the final underlying price and the strike.
The put payoff is

$$(K - S_T)^+ \tag{607}$$

At maturity, the bond component pays K.
So the complete payoff is

$$\boxed{K + (S_T - K)^+ - (K - S_T)^+} \tag{608}$$

Now check both cases.
First

$$S_T \geq K \tag{609}$$

The call pays ST − K.
The put pays zero.
So

$$K + (S_T - K) - 0 = S_T \tag{610}$$

Second

$$S_T < K \tag{611}$$

The call pays zero.
The put pays K − ST.
So

$$K + 0 - (K - S_T) = S_T \tag{612}$$

Therefore, in both cases

$$\boxed{K + (S_T - K)^+ - (K - S_T)^+ = S_T} \tag{613}$$

The individual option payoffs have kinks at the strike.
But in this combination, those kinks cancel.
The resulting payoff is simply the underlying price at maturity.
That's the basis of the affine replication.
Under the valuation assumptions of the coded parity relation

$$\boxed{C - P_{\text{put}} = S - B} \tag{614}$$

This says:

"The difference between call and put prices equals the difference between the underlying price and the discounted strike."

At fixed S, B, this is a straight line in the call–put price plane.

If the strike also varies

$$\boxed{C - P_{\text{put}} + e^{-rT}K - S = 0} \tag{615}$$

This is a plane in the three-dimensional coordinate space (C, Pput, K), at fixed S, r, T.

Now, observed market quotes may not satisfy theoretical parity exactly.

So the program calculates a residual

$$\boxed{\epsilon_{\text{parité}} = (C - P_{\text{put}}) - (S - B)} \tag{616}$$

The Greek letter ϵ, is the discrepancy.

If it is zero, the quoted call–put difference matches theoretical parity.

If it's nonzero, there is a gap.

Next, calculate the normal distance to the parity line.

The line has normal vector

$$(1, -1) \tag{617}$$

Its length is

$$\sqrt{1^2 + (-1)^2} = \sqrt{2} \tag{618}$$

So its unit normal is

$$\frac{(1, -1)}{\sqrt{2}} \tag{619}$$

The perpendicular distance is

$$\boxed{d_{\text{parité}} = \frac{|\epsilon_{\text{parité}}|}{\sqrt{2}}} \tag{620}$$

The tangent direction is

$$\frac{(1, 1)}{\sqrt{2}} \tag{621}$$

Why?

Because increasing the call and put by the same amount leaves their difference unchanged

$$(C + \delta) - (P_{\text{put}} + \delta) = C - P_{\text{put}} \tag{622}$$

So this movement stays along the parity line.

Now real markets also have bid–ask spreads.

We cannot automatically interpret every small parity residual as an executable arbitrage opportunity.

The code therefore uses a tolerance.

When the call and put bid–ask spreads are valid

$$\boxed{\tau_{\text{marché}} = \max\left(0.05,\ 0.001S,\ \frac{\text{spread}_C + \text{spread}_P}{2}\right)} \tag{623}$$

The first term is a fixed floor.

The second is proportional to the underlying price.

The third is half the sum of the two bid–ask spreads.

The maximum determines the tolerance.

The observation passes the parity diagnostic when

$$\boxed{|\epsilon_{\text{parité}}| \leq \tau_{\text{marché}}} \tag{624}$$

The first term is a fixed floor.

The second is proportional to the underlying price.

The third is half the sum of the two bid–ask spreads.

The maximum determines the tolerance.

The observation passes the parity diagnostic when

$$\boxed{|\epsilon_{\text{parité}}| \leq \tau_{\text{marché}}} \tag{625}$$

This defines a band around the parity line.

Its perpendicular half-width is

$$\frac{\tau_{\text{marché}}}{\sqrt{2}} \tag{626}$$

That's a boundary in the options parity diagnostic. it's not the Janus fragile–antifragile boundary!

52 Corrected kernel coordinates and the shear group

Now the options kernel has another transformation.

It uses a coefficient β in a corrected rate

$$\boxed{r_\beta = r - \beta\sigma^2} \tag{627}$$

Here σ is the volatility used in the pricing formula.

And β is the coefficient used by that options layer, not automatically the date-specific rolling financial mass of the historical regime detector.

Now define

$$\boxed{w = \sigma\sqrt{T}} \tag{628}$$

and

$$\boxed{\lambda_o = \ln(S/K) + rT} \tag{629}$$

The first combines volatility and time to maturity.
The second combines log-moneyness and the rate term.
The corrected kernel coordinates are

$$\boxed{d_1^\beta = \frac{\lambda_o}{w} + \left(\frac{1}{2} - \beta\right) w} \tag{630}$$

$$\boxed{d_2^\beta = d_1^\beta - w} \tag{631}$$

Now compare them with the case $\beta = 0$.
For the first coordinate

$$d_1^\beta = d_1^0 - \beta w \tag{632}$$

And for the second

$$d_2^\beta = d_2^0 - \beta w \tag{633}$$

So both coordinates are shifted by the same quantity $-\beta$w.
Since

$$w = d_1^0 - d_2^0 \tag{634}$$

we can rewrite

$$\begin{aligned} d_1^\beta &= d_1^0 - \beta(d_1^0 - d_2^0) \\ &= (1-\beta)d_1^0 + \beta d_2^0 \end{aligned} \tag{635}$$

For the second coordinate

$$d_2^\beta = -\beta d_1^0 + (1+\beta)d_2^0 \tag{636}$$

Put both expressions into a matrix

$$\boxed{\begin{pmatrix} d_1^\beta \\ d_2^\beta \end{pmatrix} = M_\beta \begin{pmatrix} d_1^0 \\ d_2^0 \end{pmatrix}} \tag{637}$$

where

$$\boxed{M_\beta = \begin{pmatrix} 1-\beta & \beta \\ -\beta & 1+\beta \end{pmatrix}} \tag{638}$$

Now calculate its determinant

$$\det M_\beta = (1-\beta)(1+\beta) + \beta^2 \tag{639}$$

Expand

$$1 - \beta^2 + \beta^2 = 1 \tag{640}$$

Therefore

$$\boxed{\det M_\beta = 1} \tag{641}$$

The transformation is invertible and preserves oriented area.

But it doesn't generally preserve Euclidean lengths or angles.

Its inverse is

$$\boxed{M_\beta^{-1} = M_{-\beta}} \tag{642}$$

And its composition law is

$$\boxed{M_\beta M_\gamma = M_{\beta+\gamma}} \tag{643}$$

Two successive corrections combine into one correction with the sum of their parameters.

Now, to understand the geometry, define the average

$$\bar{d} = \frac{d_1 + d_2}{2} \tag{644}$$

and the difference

$$w = d_1 - d_2 \tag{645}$$

Both d_i shift by $-\beta w$.

So their average shifts by that amount.

But their difference doesn't change.

Therefore

$$\boxed{(\bar{d}, w) \longmapsto (\bar{d} - \beta w, w)} \tag{646}$$

That's a shear transformation.

Imagine a square grid.

You move its points horizontally by an amount proportional to their vertical coordinate.

The grid becomes slanted.

Areas are preserved, but lengths and angles can change.

The transported Euclidean metric is

$$\boxed{g_\beta = M_\beta^\top M_\beta} \tag{647}$$

Multiplying the matrices gives

$$\boxed{g_\beta = \begin{pmatrix} 1 - 2\beta + 2\beta^2 & -2\beta^2 \\ -2\beta^2 & 1 + 2\beta + 2\beta^2 \end{pmatrix}} \tag{648}$$

This matrix is positive definite because for any nonzero vector z

$$z^\top g_\beta z = \|M_\beta z\|^2 > 0 \tag{649}$$

Since Mβ is invertible, it cannot send a nonzero vector to zero.

The metric is flat and has determinant

$$\boxed{\det g_\beta = (\det M_\beta)^2 = 1} \tag{650}$$

Now compare this with the Janus density metric g_4.

Here, the options-kernel transformation is an invertible map from two dimensions to two dimensions.

The density map went from four dimensions to two, losing directions.

These two geometries don't have the same rank structure, because their maps are different!

53 Coordinate metrics and differential rank

Next, study the transformation from the usual option parameters (K, σ) to the kernel coordinates ($d_1{}^\beta$, $d_2{}^\beta$).

Keep S, r, T, β fixed.

The transformation is

$$(K, \sigma) \longmapsto (d_1^\beta, d_2^\beta) \tag{651}$$

Its Jacobian determinant is

$$\boxed{\det \frac{\partial(d_1^\beta, d_2^\beta)}{\partial(K, \sigma)} = \frac{1}{K\sigma} > 0} \tag{652}$$

We see why.

First

$$\lambda_o = \ln(S/K) + rT \tag{653}$$

Its derivative with respect to K is

$$\frac{\partial \lambda_o}{\partial K} = -\frac{1}{K} \tag{654}$$

Since w doesn't depend on K

$$\frac{\partial d_1^\beta}{\partial K} = -\frac{1}{Kw} \tag{655}$$

And since

$$d_2^\beta = d_1^\beta - w \tag{656}$$

we also have

$$\frac{\partial d_2^\beta}{\partial K} = -\frac{1}{Kw} \tag{657}$$

Now

$$w = \sigma\sqrt{T} \tag{658}$$

so

$$\frac{\partial w}{\partial \sigma} = \sqrt{T} \tag{659}$$

Therefore

$$\frac{\partial d_2^\beta}{\partial \sigma} = \frac{\partial d_1^\beta}{\partial \sigma} - \sqrt{T} \tag{660}$$

When we calculate the 2×2 determinant, the common derivative terms cancel, leaving

$$\frac{\sqrt{T}}{Kw} \tag{661}$$

And since w = $\sigma\sqrt{\text{T}}$

$$\frac{\sqrt{T}}{Kw} = \frac{1}{K\sigma} \tag{662}$$

That's why the determinant is strictly positive.

We can also recover the inverse.

From

$$d_2 = d_1 - w \tag{663}$$

we obtain

$$w = d_1 - d_2 \tag{664}$$

Then

$$\boxed{\sigma = \frac{w}{\sqrt{T}}} \tag{665}$$

Now start with

$$d_1 = \frac{\lambda_o}{w} + \left(\frac{1}{2} - \beta\right) w \tag{666}$$

Multiply by w

$$wd_1 = \lambda_o + \left(\frac{1}{2} - \beta\right) w^2 \tag{667}$$

Isolate λo

$$\lambda_o = wd_1 - \left(\frac{1}{2} - \beta\right) w^2 \tag{668}$$

But

$$\lambda_o = \ln(S/K) + rT \tag{669}$$

Rearrange

$$\ln(K/S) = rT - wd_1 + \left(\frac{1}{2} - \beta\right) w^2 \tag{670}$$

Apply the exponential and multiply by S

$$\boxed{K = S \exp\left[rT - wd_1 + \left(\frac{1}{2} - \beta\right) w^2\right]} \tag{671}$$

We've recovered both K and σ from the kernel coordinates.
Since w = $\sigma\sqrt{\mathrm{T}} > 0$

$$d_1 > d_2 \tag{672}$$

The positive parameter quadrant maps onto the half-plane $\mathrm{d}_1 > \mathrm{d}_2$.

The determinant of the transported metric is

$$\boxed{\det g_{K,\sigma} = \frac{1}{K^2\sigma^2}} \tag{673}$$

Now suppose maturity T varies too.

Then we have three parameters K, σ, T but only two kernel coordinates.

Different parameter combinations can produce the same outputs.

At fixed λ_o, w, the fibers are

$$\boxed{\sigma(T) = \frac{w}{\sqrt{T}}} \tag{674}$$

$$\boxed{K(T) = Se^{rT-\lambda_o}} \tag{675}$$

Along these fibers

$$Ke^{-rT} = Se^{-\lambda_o} \tag{676}$$

remains constant.

And because the kernel coordinates remain fixed, the generated kernel prices remain fixed too.

54 Generated option prices and parity-preserving shifts

Now we have to distinguish the map toward kernel coordinates from the map toward actual calculated option prices.

They are different transformations, and they don't necessarily have the same rank properties.

The option prices generated by the corrected kernel are

$$\boxed{C_\beta = SN(d_1^\beta) - BN(d_2^\beta)} \tag{677}$$

$$\boxed{P_\beta = BN(-d_2^\beta) - SN(-d_1^\beta)} \tag{678}$$

Cβ is the generated call price.

Pβ is the generated put price.

S is the underlying price.

B is the discounted strike.

And $d_1{}^\beta$, $d_2{}^\beta$ are the corrected kernel coordinates we just constructed.

Now, what is N?

Here, N denotes the standard normal cumulative distribution function.

It returns the theoretical probability that a standard normally distributed variable is less than or equal to its argument.

For example

$$N(0) = 0.5 \tag{679}$$

The standard normal distribution is symmetric around zero.
Therefore

$$\boxed{N(z) + N(-z) = 1} \tag{680}$$

That symmetry will allow us to verify call–put parity.
We subtract the generated put from the generated call

$$C_\beta - P_\beta \tag{681}$$

Substitute both formulas

$$\begin{aligned} C_\beta - P_\beta = SN(d_1^\beta) - BN(d_2^\beta) \\ - \left[BN(-d_2^\beta) - SN(-d_1^\beta)\right] \end{aligned} \tag{682}$$

Remove the parentheses

$$\begin{aligned} C_\beta - P_\beta = SN(d_1^\beta) - BN(d_2^\beta) \\ - BN(-d_2^\beta) + SN(-d_1^\beta) \end{aligned} \tag{683}$$

Group the terms multiplied by S, then those multiplied by B

$$\begin{aligned} C_\beta - P_\beta = S\left[N(d_1^\beta) + N(-d_1^\beta)\right] \\ - B\left[N(d_2^\beta) + N(-d_2^\beta)\right] \end{aligned} \tag{684}$$

Both brackets equal 1.
So

$$\boxed{C_\beta - P_\beta = S - B} \tag{685}$$

We've proved that the generated prices preserve the parity identity.
Since the uncorrected kernel also satisfies it

$$C_0 - P_0 = S - B \tag{686}$$

we can subtract the two equalities

$$(C_\beta - P_\beta) - (C_0 - P_0) = 0 \tag{687}$$

Rearrange

$$\boxed{C_\beta - C_0 = P_\beta - P_0} \tag{688}$$

The correction changes the call and put by the same amount.

In the call–put plane, that's a displacement parallel to (1, 1), the tangent direction of the parity line.

The kernel correction moves the generated prices along the call–put parity relationship!

But this is an identity of the generated prices.

It doesn't imply that every observed market quote has zero parity residual.

55 Exact vega of the corrected kernel

Next, examine what happens when implied volatility changes.

The sensitivity of an option price to volatility is called vega.

For the generated call

$$\mathcal{V}_\beta = \frac{\partial C_\beta}{\partial \sigma} \tag{689}$$

That's the partial derivative of the call price with respect to volatility, holding the other specified parameters fixed.

But the exact vega of our corrected kernel isn't simply the uncorrected uncorrected lognormal-kernel expression.

We need an additional factor.

First define

$$\boxed{e_\beta = e^{-\beta w^2}} \tag{690}$$

Now, the standard normal probability density is

$$\boxed{\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}} \tag{691}$$

The symbol φ, is the density of the standard normal distribution.

It is also the derivative of the normal cumulative distribution function

$$N'(z) = \varphi(z) \tag{692}$$

Now use

$$d_2^\beta = d_1^\beta - w \tag{693}$$

The ratio between the two normal densities is

$$\frac{\varphi(d_2^\beta)}{\varphi(d_1^\beta)} = \exp\left[-\frac{(d_2^\beta)^2}{2} + \frac{(d_1^\beta)^2}{2}\right] \tag{694}$$

Substitute

$$d_2^\beta = d_1^\beta - w \tag{695}$$

Expand the square

$$(d_1^\beta - w)^2 = (d_1^\beta)^2 - 2wd_1^\beta + w^2 \tag{696}$$

The two terms involving $(\mathrm{d_1}^\beta)^2$ cancel.
So

$$\frac{\varphi(d_2^\beta)}{\varphi(d_1^\beta)} = e^{wd_1^\beta - w^2/2} \tag{697}$$

Now, from the definition of $\mathrm{d_1}^\beta$:

$$wd_1^\beta - \frac{w^2}{2} = \lambda_o - \beta w^2 \tag{698}$$

And

$$e^{\lambda_o} = e^{\ln(S/K)+rT} = \frac{S}{B} \tag{699}$$

because

$$B = Ke^{-rT} \tag{700}$$

Therefore

$$\frac{\varphi(d_2^\beta)}{\varphi(d_1^\beta)} = \frac{S}{B}e^{-\beta w^2} \tag{701}$$

So

$$\boxed{B\varphi(d_2^\beta) = S\varphi(d_1^\beta)e_\beta} \tag{702}$$

This identity will simplify the derivatives of the generated option prices.
Now differentiate the call

$$C_\beta = SN(d_1^\beta) - BN(d_2^\beta) \tag{703}$$

with respect to σ.

We obtain

$$\mathcal{V}_\beta = S\varphi(d_1^\beta)\frac{\partial d_1^\beta}{\partial \sigma} - B\varphi(d_2^\beta)\frac{\partial d_2^\beta}{\partial \sigma} \tag{704}$$

Replace the second normal-density term using the identity we just proved

$$\mathcal{V}_\beta = S\varphi(d_1^\beta)\left[\frac{\partial d_1^\beta}{\partial \sigma} - e_\beta \frac{\partial d_2^\beta}{\partial \sigma}\right] \tag{705}$$

Since

$$d_2^\beta = d_1^\beta - w \tag{706}$$

and

$$\frac{\partial w}{\partial \sigma} = \sqrt{T} \tag{707}$$

we have

$$\frac{\partial d_2^\beta}{\partial \sigma} = \frac{\partial d_1^\beta}{\partial \sigma} - \sqrt{T} \tag{708}$$

Substitute

$$\mathcal{V}_\beta = S\varphi(d_1^\beta)\left[(1 - e_\beta)\frac{\partial d_1^\beta}{\partial \sigma} + e_\beta\sqrt{T}\right] \tag{709}$$

Now differentiate

$$d_1^\beta = \frac{\lambda_o}{w} + \left(\frac{1}{2} - \beta\right) w \tag{710}$$

With λo fixed and w = $\sigma\sqrt{\mathrm{T}}$

$$\frac{\partial d_1^\beta}{\partial \sigma} = \sqrt{T}\left[-\frac{\lambda_o}{w^2} + \frac{1}{2} - \beta\right] \tag{711}$$

Use the definition of $\mathrm{d_1}^\beta$ to rewrite it

$$\boxed{\frac{\partial d_1^\beta}{\partial \sigma} = \sqrt{T}\left(1 - 2\beta - \frac{d_1^\beta}{w}\right)} \tag{712}$$

Substitute that expression into the vega calculation.

After factoring out $S\varphi(d_1{}^{\beta})\sqrt{T}$, define

$$\boxed{R_\beta = 1 - (1 - e_\beta)\left(2\beta + \frac{d_1^\beta}{w}\right)} \tag{713}$$

The exact vega becomes

$$\boxed{\mathcal{V}_\beta = S\varphi(d_1^\beta)\sqrt{T}\,R_\beta} \tag{714}$$

The extra factor $R\beta$ comes from the dependence of the corrected coordinates on volatility.
Now the kernel also uses

$$\boxed{Q_\beta = 1 - (1 - e_\beta)\frac{d_1^\beta}{w}} \tag{715}$$

Why introduce another factor?
Because it appears in the local second derivatives of the generated call price.
At fixed volatility

$$\boxed{C_{SS} = \frac{\varphi(d_1^\beta)}{Sw}Q_\beta} \tag{716}$$

and

$$\boxed{C_{KK} = \frac{S\varphi(d_1^\beta)}{K^2 w}Q_\beta} \tag{717}$$

CSS is the second derivative of the call with respect to the underlying price.
CKK is the second derivative with respect to the strike.
These derivatives describe local curvature of the price in the corresponding fixed-volatility slices.
The factors multiplying $Q\beta$ are positive in the admissible parameter region.
So the sign of the local curvature is controlled by $Q\beta$.
Now compare the definitions of $R\beta$ and $Q\beta$.
We obtain

$$\boxed{R_\beta = Q_\beta - 2\beta(1 - e_\beta)} \tag{718}$$

And

$$\boxed{\beta(1 - e_\beta) \geq 0} \tag{719}$$

Why?

If $\beta > 0$, then e$\beta < 1$.

Both factors are positive.

If $\beta < 0$, then e$\beta > 1$.

Both factors are negative, so their product is positive.

If $\beta = 0$, the product is zero.

Therefore

$$R_\beta \leq Q_\beta \tag{720}$$

If exact vega is strictly positive, then R$\beta > 0$.

That implies

$$Q_\beta > 0 \tag{721}$$

Positive exact vega implies positive local curvature in these fixed-volatility slices of the coded kernel!

But this does not automatically establish convexity along a complete strike-dependent implied-volatility surface.

That's a different derivative, which we'll examine shortly.

56 Curvature, local invertibility, and numerical inversion

Next, study the rank of the transformation from option parameters to option prices

$$(K, \sigma) \longmapsto (C_\beta, P_\beta) \tag{722}$$

Its Jacobian is

$$\begin{pmatrix} C_K C_\sigma \\ P_K P_\sigma \end{pmatrix} \tag{723}$$

Its determinant is

$$C_K P_\sigma - C_\sigma P_K \tag{724}$$

Use the generated parity identity

$$C_\beta - P_\beta = S - Ke^{-rT} \tag{725}$$

Differentiate with respect to σ.

The right-hand side doesn't depend on σ, so

$$C_\sigma - P_\sigma = 0 \tag{726}$$

Therefore

$$P_\sigma = C_\sigma \tag{727}$$

Now differentiate parity with respect to K

$$C_K - P_K = -e^{-rT} \tag{728}$$

Rearrange

$$P_K = C_K + e^{-rT} \tag{729}$$

Substitute both identities into the determinant

$$\begin{aligned} \det &= C_K C_\sigma - C_\sigma(C_K + e^{-rT}) \\ &= -e^{-rT} C_\sigma \end{aligned} \tag{730}$$

Since C$\sigma = \mathcal{V}\beta$

$$\boxed{\det \frac{\partial(C_\beta, P_\beta)}{\partial(K, \sigma)} = -e^{-rT}\mathcal{V}_\beta} \tag{731}$$

So the price map is locally invertible when exact vega is nonzero.

When the vega vanishes, the determinant vanishes, and the map has rank one in this construction.

The metric transported from the price plane has

$$\boxed{\det g_{\text{prix}} = e^{-2rT}\mathcal{V}_\beta^2} \tag{732}$$

Notice the distinction.

The transformation from (K, σ) to ($d_1{}^\beta$, $d_2{}^\beta$) is regular throughout the positive parameter domain.

But the transformation from (K, σ) to (Cβ, Pβ) can lose invertibility when exact vega vanishes.

Two successive transformations do not necessarily have the same rank properties!

Now consider the direct effect of β on the generated call price.

The derivative is

$$\boxed{\partial_\beta C_\beta = Sw\varphi(d_1^\beta)(e_\beta - 1)} \tag{733}$$

The factors S, w, $\varphi(d_1{}^\beta)$ are positive.

So the sign depends on e_$\beta - 1$.

If $\beta < 0$, then e$\beta > 1$, and the derivative is positive.

If $\beta > 0$, then e$\beta < 1$, and the derivative is negative.

So, with all other parameters fixed, the generated call price reaches its maximum at $\beta = 0$.

Therefore

$$\boxed{C_\beta \le C_0, \qquad P_\beta \le P_0} \tag{734}$$

This is a property of the coded pricing kernel.

It isn't the same thing as the sign of an asset's historical return response to variance.

Next, examine the numerical volatility solver.

Usually, we start with volatility and calculate an option price.

Here, the solver starts with a quoted call price and searches for a volatility that reproduces it.

That's an implied-volatility inversion.

But the denominator used in the numerical iteration is

$$\boxed{\mathcal{V}_{\text{utilisé}} = S\varphi(d_1^{\beta})\sqrt{T}} \tag{735}$$

Compare it with the exact vega

$$\mathcal{V}_{\beta} = S\varphi(d_1^{\beta})\sqrt{T}\, R_{\beta} \tag{736}$$

The iteration denominator doesn't include Rβ.

So it's not the exact derivative of the corrected kernel unless R$\beta = 1$.

But that doesn't automatically mean the solver's accepted solutions are wrong.

Why?

Because the solver also checks the final price residual.

A finite solution is accepted when

$$\boxed{|C_{\beta}(\sigma) - C_{\text{coté}}| < 10^{-6}} \tag{737}$$

This means the absolute difference between the generated call price and the quoted call price must be less than one millionth in the price units used.

The solver also imposes limits.

Volatility is bounded between 0.01 and 3.

The numerical step is limited to 0.5.

And the iteration count is limited to 100.

In the four near-the-money observations studied, exact vega and local convexity are positive.

The ratio of exact vega to the denominator used by the solver is approximately:

Asset	Exact vega / solver denominator
SPY	1.000185
QQQ	1.000110
AAPL	0.994633
GLD	0.978389

These numbers measure the discrepancy between the two derivative expressions on the selected observations.

They don't prove that accepted solutions fail the final price-residual check.

57 Coordinate checks and the geometry of fitted smiles

Now, for the 6 July 2026 expiry valued on 5 July

$$T = \frac{1}{365} \tag{738}$$

The recorded coordinate shifts

$$-\beta\sigma\sqrt{T} \tag{739}$$

are approximately:

Asset	Coordinate shift
SPY	−0.008017367
QQQ	−0.016849174
AAPL	+0.021045127
GLD	+0.162885607

The matrix transformation reproduces the directly calculated coordinates with a maximum discrepancy around

$$2.8 \times 10^{-17} \tag{740}$$

The recalculated calls reproduce those used for inversion within approximately

$$1.31 \times 10^{-7} \tag{741}$$

And the generated call–put parity identity agrees within approximately

$$5.7 \times 10^{-14} \tag{742}$$

These are very small numerical differences.

But remember: the solver fits the call.

It doesn't guarantee that the generated put exactly matches the put quoted in the market.

Generated parity and observed-quote agreement are different checks.

Next, finish the options geometry with the implied-volatility smile.

The smile describes how implied volatility varies with strikes for a given expiry.

The program uses a coordinate called log-moneyness

$$\boxed{k = \ln(K/S)} \tag{743}$$

If K = S, then k = 0.

If K > S, then k > 0.

If K < S, then k < 0.

For each expiry, the program fits

$$\boxed{\widehat{\sigma}_T(k) = p_T + q_T k} \tag{744}$$

pT is the fitted intercept.

qT is the fitted slope.

And k is the log-moneyness.

That's a straight line in the log-moneyness–volatility plane.

But now substitute

$$k = \ln(K/S) \tag{745}$$

We obtain

$$\boxed{\widehat{\sigma}_T(K) = p_T + q_T \ln(K/S)} \tag{746}$$

In strike coordinates, this is a logarithmic curve.

Its first derivative is

$$\boxed{\partial_K \widehat{\sigma}_T = \frac{q_T}{K}} \tag{747}$$

Its second derivative is

$$\boxed{\partial_{KK} \widehat{\sigma}_T = -\frac{q_T}{K^2}} \tag{748}$$

The slope now varies with the strike.

A straight line in one coordinate system can become a curve in another!

This is conceptually similar to the dominance boundary being diagonal in density coordinates and parabolic in stress coordinates.

But the mathematical objects remain different.

The numerical example contains 16 fitted smiles from 1,022 accepted points.

The export retains fit coefficients and statistics, but not every individual corrected data point.

And it doesn't construct a complete continuous implied-volatility surface $\Sigma(\mathrm{K}, \mathrm{T})$.

58 Total derivatives and conditional reconstruction

Now suppose we had a calibrated curve

$$\sigma = \Sigma(K) \tag{749}$$

The call price would depend on K directly and indirectly through this volatility function.

Its first total derivative would be

$$\frac{dC}{dK} = C_K + C_\sigma \Sigma' \tag{750}$$

Here $\Sigma\prime$ is the derivative of the volatility function with respect to the strike.

Differentiate again, applying the chain rule to every term.

We obtain

$$\boxed{\frac{d^2C}{dK^2} = C_{KK} + 2C_{K\sigma}\Sigma' + C_{\sigma\sigma}(\Sigma')^2 + C_\sigma \Sigma''} \tag{751}$$

We identify the terms.

CKK is the direct strike curvature at fixed volatility.

CKσ is the cross-sensitivity between strike and volatility.

$\Sigma\prime$ describes the slope of the volatility curve.

C$\sigma\sigma$ is the second derivative of the option price with respect to volatility.

And $\Sigma\prime\prime$ is the curvature of the volatility curve itself.

Fixed-volatility gamma does not replace the total second derivative along a strike-dependent volatility curve!

Next, put the data in its correct context.

The options snapshot is dated 5 July 2026.

It contains 2,613 contract rows and 891 call–put pairs.

Of those, 845 pass the selected parity tolerance.

But the Janus historical regime extract ends on 17 April 2026.

These are distinct datasets.

The options snapshot doesn't contain a complete sequence of option chains aligned with every date needed to reconstruct the historical regime series.

We can still write a formal conditional connection.

Define the replication operator

$$\boxed{\mathcal{R}(B, C, P) = B + C - P} \tag{752}$$

Let $\mathcal{J}$ denote the historical Janus detector.

Then

$$\boxed{\mathcal{T} = \mathcal{J} \circ (\mathcal{R}_+, \mathcal{R}_-)} \tag{753}$$

The symbol $\circ$ means function composition.

First, reconstruct the two price series using the replication operators.

Then feed those reconstructed series into Janus.

If the reconstructed prices equal the original prices at every necessary historical date

$$B_{\pm,t} + C_{\pm,t} - P_{\pm,t} = S_{\pm,t} \tag{754}$$

then every subsequent deterministic calculation receives the same input.

With the same initial state and calculation conventions, the detector reconstructs the same returns, masses, variances, stress values, densities, confirmation states, and allocations.

But the available options snapshot doesn't provide the complete historical series required to verify this empirically.

It's a conditional mathematical equivalence, not a completed historical reconstruction from option chains.

The options snapshot does not create the historical D = 0 boundary used by the Janus density detector!

59 Mathematical distinctions between auxiliary layers

Next, place the physical Janus analogy in its proper mathematical context.

The project draws structural inspiration from constructions involving two branches, coupled fields, scale factors, opposing contributions, and acceleration terms.

But a structural analogy isn't an identification of mathematical objects.

The physical model uses spacetime metrics.

The financial program uses normalized price series and density fields built from statistically estimated masses and stresses.

Those are different objects.

And we've already demonstrated

$$\boxed{g_4 = J^\top J} \tag{755}$$

This transported metric is degenerate.

Its rank is at most 2 in a four-dimensional primitive space.

An invertible coordinate transformation cannot change that rank.

So this financial metric cannot simply be identified with a nondegenerate four-dimensional spacetime metric.

To construct another four-dimensional metric, we would need to define it explicitly and prove its properties.

Now look at the fields named H2plus and H2minus in the program.

They satisfy

$$\boxed{H_-^2 = -H_+^2} \tag{756}$$

But the column names don't automatically mean we're dealing with two real squares of independent kinematic quantities.

A real square is nonnegative.

Two nonzero real squares cannot be negatives of one another.

So these columns must be interpreted according to their actual coded definitions: opposing signed balances.

Similarly, the potential relation

$$\psi_- = -\psi_+ \tag{757}$$

shows an antisymmetric construction.

It doesn't prove that the financial potential satisfies the same spatial field equations as the physical potential.

Our financial potential uses temporal centering and integration over an archive.

And the confirmation dependence on an inverse square root, while resembling other mathematical timescales, isn't automatically the same mechanism as a physical Jeans time.

Finally, the drift correction

$$-\beta\sigma^2 \tag{758}$$

appears both in the auxiliary simulation and in the corrected options kernel.

But the simulation and pricing formula are not automatically equivalent.

A discounted expectation under a drift r $-$ $\beta\sigma^2$ would introduce a factor

$$Se^{-\beta\sigma^2 T} \tag{759}$$

in the first call term.

The coded options formula retains S.

So we must distinguish the simulated stochastic dynamics from the pricing kernel actually implemented.

The physical analogy can inspire the model. but the financial equations, their definitions, and their proofs determine what has actually been established!

Allocation and reconstruction

60 Lagged allocation on the portfolio simplex

Now one final operational transformation remains.

We've measured the responses, constructed the domains, studied their geometry and topology, and described their confirmation rules.

But how does a confirmed regime become a portfolio position?

That's the geometry of allocation.

The confirmed regime is represented by

$$G_t \in \{0, 1\} \tag{760}$$

When $G_t = 0$, the detector is in its growth mode.

When $G_t = 1$, it's in the confirmed refuge mode.

But this state isn't yet the return of the trading strategy.

We still need to determine which leg is held and when the allocation takes effect.

The rule is

$$\boxed{w_t = (1 - G_{t-1}, G_{t-1})} \tag{761}$$

Here w_t is the vector of portfolio weights at date t.

Its first coordinate is the allocation to the positive leg.

The second is the allocation to the refuge leg.

And notice the index:

$t - 1$.

The portfolio uses the previously confirmed state.

It doesn't pretend to use a regime calculated at the current close for a position that would already have been taken before that close.

Now, if

$$G_{t-1} = 0 \tag{762}$$

then

$$w_t = (1 - 0, 0) \tag{763}$$

So

$$\boxed{w_t = (1, 0)} \tag{764}$$

The portfolio is allocated to the first leg.

If

$$G_{t-1} = 1 \tag{765}$$

then

$$w_t = (1 - 1, 1) \tag{766}$$

So

$$\boxed{w_t = (0, 1)} \tag{767}$$

The portfolio is allocated to the second leg.

These two points are the vertices of the allocation simplex.

For two nonnegative portfolio weights summing to one, the simplex is simply the line segment joining (1, 0) and (0, 1).

The program uses its two endpoints.

It doesn't construct a continuous mixture such as 30% in one leg and 70% in the other under this rule.

Next, calculate the portfolio return.

Earlier, for masses and variances, we used logarithmic returns.

For allocation, the code uses simple returns.

Define

$$\boxed{q_t^{\pm} = \frac{P_t^{\pm}}{P_{t-1}^{\pm}} - 1} \tag{768}$$

If a price rises from 100 to 105

$$q = \frac{105}{100} - 1 = 0.05 \tag{769}$$

The simple return is 5%.

The corresponding logarithmic return is

$$r = \ln(1.05) \tag{770}$$

They're close for small changes but not identical.

The executed strategy return is

$$\boxed{q_t^{\text{stratégie}} = (1 - G_{t-1})q_t^{+} + G_{t-1}q_t^{-}} \tag{771}$$

This equation says:

"Multiply each leg's simple return by its portfolio weight determined from the previous confirmed regime, then add the two contributions."

If the previous regime is zero

$$q_t^{\text{stratégie}} = q_t^+ \tag{772}$$

If it's one

$$q_t^{\text{stratégie}} = q_t^- \tag{773}$$

That's how the confirmed regime becomes an allocation.

But switching between legs can create costs.

So the program also measures portfolio turnover.

It uses the L^1 norm

$$\boxed{\|w_t - w_{t-1}\|_1} \tag{774}$$

What does L^1 mean?

We take the absolute change in each portfolio weight and add those absolute values.

Now substitute the two allocation vectors.

At date t

$$w_t = (1 - G_{t-1}, G_{t-1}) \tag{775}$$

At the previous date

$$w_{t-1} = (1 - G_{t-2}, G_{t-2}) \tag{776}$$

The difference in the first coordinate is

$$-G_{t-1} + G_{t-2} \tag{777}$$

The difference in the second is

$$G_{t-1} - G_{t-2} \tag{778}$$

They are opposites.

But their absolute values are equal.

So

$$\begin{aligned} \|w_t - w_{t-1}\|_1 = |G_{t-2} - G_{t-1}| \\ + |G_{t-1} - G_{t-2}| \end{aligned} \tag{779}$$

Therefore

$$\boxed{\|w_t - w_{t-1}\|_1 = 2|G_{t-1} - G_{t-2}|} \tag{780}$$

Now imagine a complete switch

$$(1,0) \longrightarrow (0,1) \tag{781}$$

The first leg loses one unit of weight.

The second gains one unit.

The sum of absolute changes is

$$1 + 1 = 2 \tag{782}$$

So the L^1 turnover is 2.

This is another geometry: the geometry of portfolio allocations.

And it provides a basis for calculating the cost of rotating from one leg to the other.

The last step of the chain is now clear: a geometric dominance, after temporal confirmation, becomes a movement between two portfolio allocations!

61 Historical reconstruction and numerical consistency

Now we've described the mathematical mechanism.

But here's another question.

Do these equations actually reproduce the historical states recorded by Janus?

Because you can write a perfectly consistent mathematical demonstration of a function and later discover that the program executes something slightly different.

So we have to check.

The retained Janus historical extraction covers 2 January 2015 to 17 April 2026.

It contains three pairs:

SPY/GLD, QQQ/GLD, and AAPL/GLD.

Each pair has 2,839 observations.

So:

$$2,839 \times 3 = 8,517 \tag{783}$$

But the first 503 observations for each pair belong to the initialization period, before the two rolling masses are estimable.

Across the three pairs

$$503 \times 3 = 1,509 \tag{784}$$

So after excluding initialization

$$8,517 - 1,509 = \boxed{7,008} \tag{785}$$

That gives 2,336 estimable observations per pair.

Why make this distinction?

Because an initialization zero isn't the same thing as a genuinely estimated mass that was subsequently clipped to zero.

Mixing them would distort the counts of zero densities, strata, and confirmed states.

Now, earlier checks of individual sensitivities reconstructed 13,505 valid coefficients across five series, including four Janus asset-response series and the fragility coordinate of the Beta residual.

They identified 41 sign changes and independently verified their 82 endpoints using ordinary least squares.

Those 41 mass-sign changes aren't 41 changes in pair dominance.

And they're not the same as the 25 confirmed entries found in the three Janus pairs.

The sign of an individual mass, the sign of the pair balance, and the confirmed regime are three different objects!

Next, compare the balances reconstructed from the extracted prices with the archived balances.

The horizontal axis contains the archived balance.

The vertical axis contains the recalculated balance.

If both calculations agree, the points fall on the equality diagonal.

The observations agree with that diagonal to small numerical differences.

Here are the results:

Pair	Estimable	$D > 0$	$D < 0$	$D = 0$	G differences
SPY/GLD	2,336	1,695	531	110	0
QQQ/GLD	2,336	1,505	607	224	0
AAPL/GLD	2,336	1,418	541	377	0
Total	7,008	4,618	1,679	711	0

We verify the total: 4,618+1,679+711=7,008 Every estimable observation is accounted for.

Now if we add the 503 initialization zeros to each pair, the total recorded zero-balance counts become:

613 for SPY/GLD.

727 for QQQ/GLD.

880 for AAPL/GLD.

But the initialization zeros must remain separate from genuinely estimated zero-mass states.

Now, what's the numerical accuracy?

The maximum absolute error in the balance reconstructed from prices is approximately

$$9.34 \times 10^{-10} \tag{786}$$

for SPY/GLD,

$$1.24 \times 10^{-9} \tag{787}$$

for QQQ/GLD,

and

$$1.49 \times 10^{-9} \tag{788}$$

for AAPL/GLD.

The algebraic checks performed directly on previously exported variables produce smaller errors, around 10^{-13}.

Those are different numerical operations.

Rebuilding from prices and checking formulas using already calculated columns are not the same test.

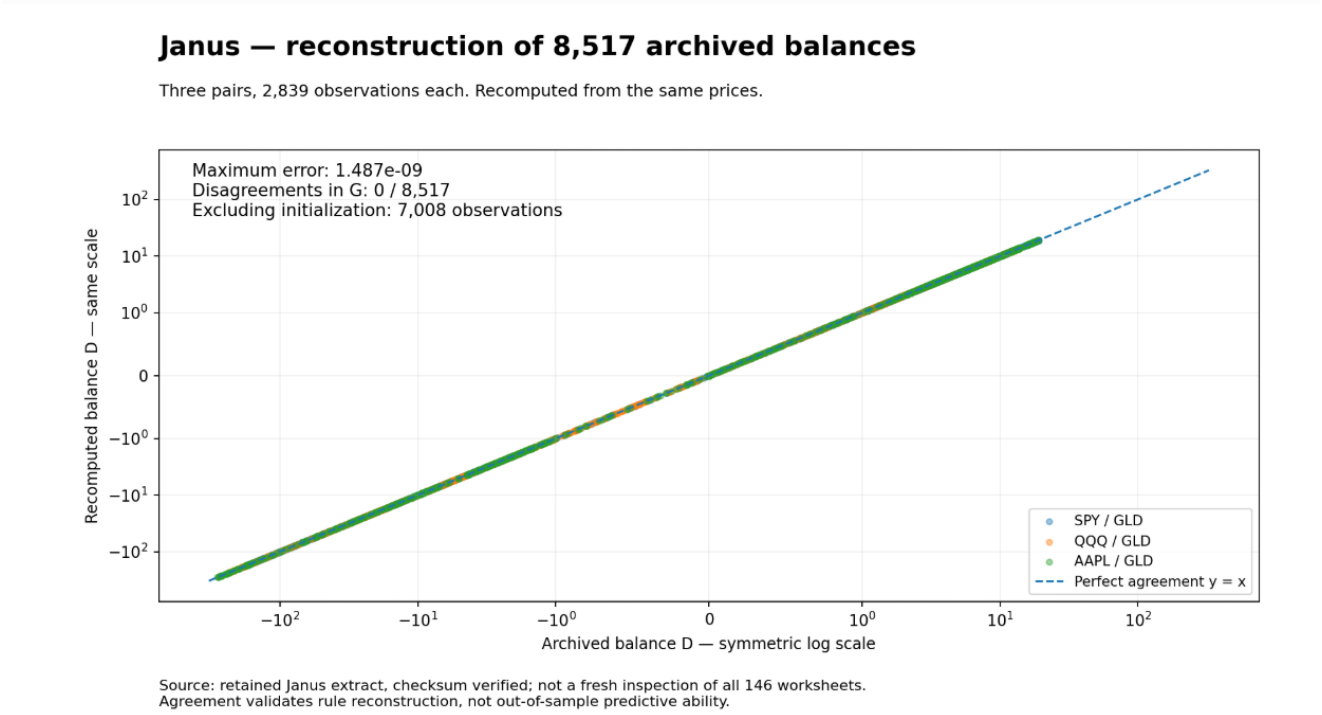


Figure 18: Reconstructed and archived balances for 8,517 recorded pair observations. The 7,008 estimable observations are analyzed separately from initialization; the reconstructed confirmed states have no recorded disagreements.

62 Unified conclusions

The construction links a sequence of distinct mathematical maps. Returns determine rolling dispersion and response estimates; clipped masses and stresses determine the two density contributions; their difference determines instantaneous dominance. Positive-mass boundary normalization establishes the topology of the regular domains. The density map then supplies a separate differential geometry, with invisible primitive directions, a degenerate pullback metric, and rank changes on specified strata.

Temporal memory augments the current geometric state. Confirmation and exit rules therefore require their own state-transition equations. Executed allocations use the confirmed state with the stated lag and produce returns through the corresponding portfolio weights. The option constructions provide additional replication, coordinate, and pricing maps whose parity identities and sensitivities must be assessed on their own parameter domains.

A moving observation and a changing framework can each alter classification. Boundary deformation, channel opening, temporal confirmation, and portfolio rotation are separate mechanisms. The numerical reconstruction checks their agreement with the recorded quantities while preserving initialization, clipping, timing, and calculation conventions. These identities and geometric properties do not by themselves establish predictive accuracy or investment performance.